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Sohn et al.

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(54) **NON-TOXIC PROTEASE HAVING
IMPROVED PRODUCTIVITY**

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A61P 21/00 (2006.01)

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CPC **C12N 9/52** (2013.01); **A61P 21/00**
(2018.01)

(58) **Field of Classification Search**

CPC **C12N 9/52**; **A61P 21/00**

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2006/0258847 A1 11/2006 Johnson et al.
2012/0128649 A1 5/2012 Chaddock et al.

FOREIGN PATENT DOCUMENTS

EP 2524963 A1 11/2012
KR 1019940701061 A 4/1994
KR 1020100088683 A 8/2010

(Continued)

OTHER PUBLICATIONS

Cabrita, L.D., et al., "Protein expression and refolding—a practical
guide to getting the most out of inclusion bodies", *Biotechnology
Annual Review*, 2004, pp. 31-50; DOI:10.1016/S1387-2656(04)10002-
1, vol. 10, Publisher: Elsevier.

(Continued)

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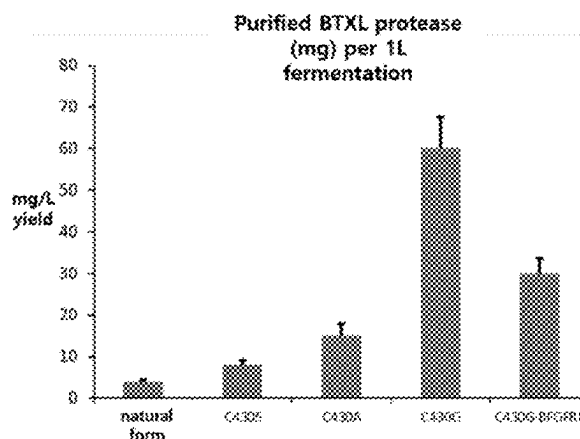
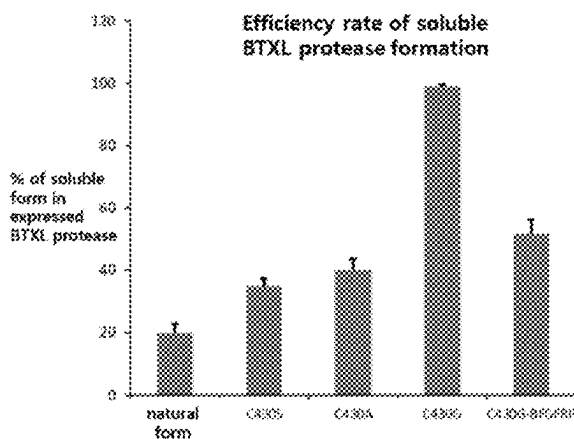
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(57) **ABSTRACT**

The present invention relates to a mutated non-toxic pro-
tease in which the amino acid cysteine (Cys) at position 430
of a non-toxic protease represented by the amino acid
sequence of SEQ ID NO: 1 is substituted with an amino acid
other than cysteine. According to the present invention, it is
possible to recover a refolded non-toxic protease from an
insoluble fraction from which the non-toxic protease was
almost impossible to recover in the prior art, and thus it is
possible to produce the non-toxic protease with significantly
improved productivity.

19 Claims, 21 Drawing Sheets

Specification includes a Sequence Listing.



(56)

References Cited

FOREIGN PATENT DOCUMENTS

KR	1020120034927 A	4/2012
KR	2020120034927 A	4/2012
KR	1020140036239 A	3/2014
KR	101636846 B1	6/2016

OTHER PUBLICATIONS

Clark, E. De B., "Refolding of recombinant proteins", *Current Opinion in Biotechnology*, 1998, pp. 157-163, vol. 9, Publisher: Current Biology.

Dhaked, R.K., et al., *Indian J Med Res*, 2010, pp. 489-503, vol. 132.

Fonfria, E., et al., "Engineering Botulinum Toxins to Improve and Expand Targeting and Snare Cleavage Activity", *Toxins*, 2018, pp. 278; doi:10.3390/toxins10070278, vol. 10, No. 7, Publisher: www.mdpi.com/journal/toxins.

Masuyer, G., et al., "Engineered Botulinum Neurotoxins as New Therapeutics", *Annual Review of Pharmacology and Toxicology*, 2013, pp. 27-51, vol. 53, No. 3.

Mauskop, A., "The Use of Botulinum Toxin in the Treatment of Headaches", *Current Pain and Headache Reports*, 2002, pp. 320-323, vol. 6, Publisher: Current Science Inc.

Mauskop, A., "Botulinum toxin in headache treatment: the end of the road?", *Cephalgia*, , pp. 769-771, vol. 26, Publisher: Blackwell Publishing Ltd.

Saffarian, P., et al., "Expression and purification of recombinant TAT-BoNT/A (1-448) under denaturing and native conditions", *Bioengineered*, 2016, pp. 478-483, vol. 7, No. 6.

Turton, K. et al., "Botulinum and tetanus neurotoxins: structure, function and therapeutic utility", *Trends in Biochemical Sciences*, 2002, pp. 552-558, vol. 27, No. 11.

Verheyden, J., et al., "Other Noncosmetic Uses of Botox", *Dis Mon*, 2002, pp. 357-366; doi:10.1053/mda.2002.25136, vol. 48.

Yamaguchi, H., et al., "Refolding Techniques for Recovering Biologically Active Recombinant Proteins from Inclusion Bodies", *biomolecules*, 2014, pp. 235-251; doi:10.3390/biom4010235, vol. 4, Publisher: www.mdpi.com/journal/biomolecules/.

Avcil, M., et al., "Efficacy of bioactive peptides loaded on hyaluronic acid microneedle patches: A monocentric clinical study", *J Cosmet Dermatol*, 2019, DOI:10.1111/jocd.13009, Publisher: Wiley.

Issue of PESR in European Patent Application No. 23154792.8 on Jun. 7, 2023.

FIG. 1

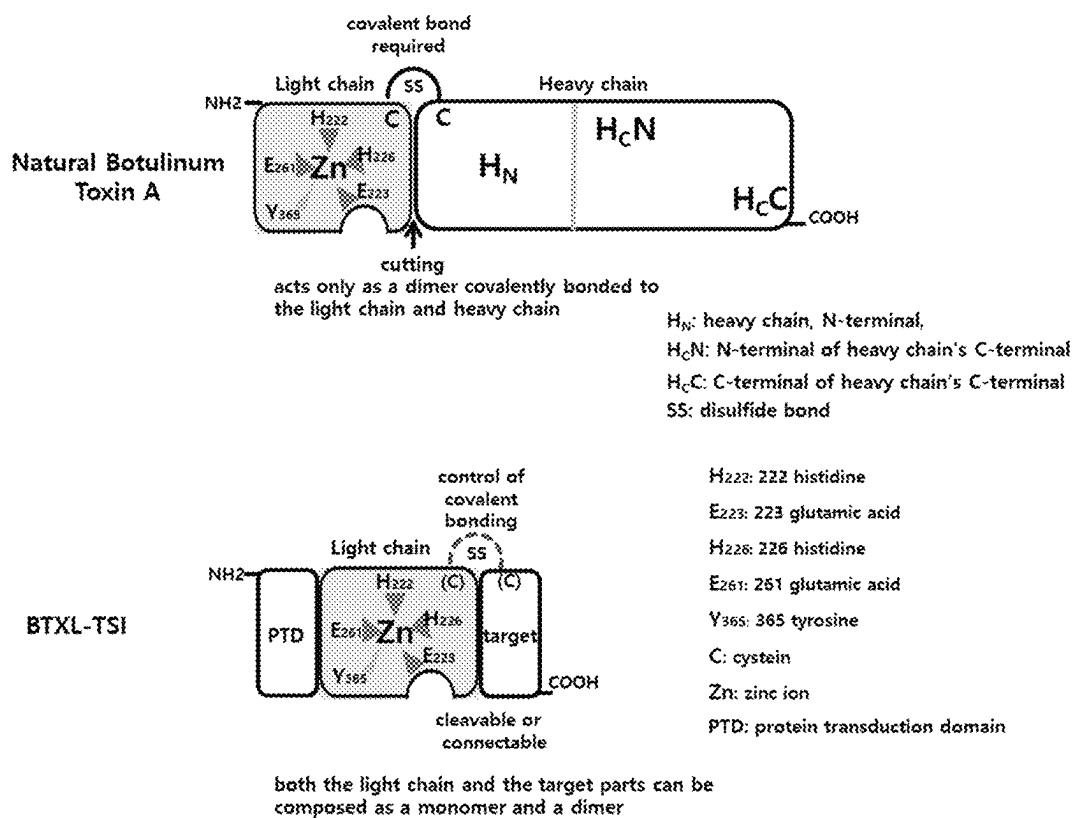


FIG. 2

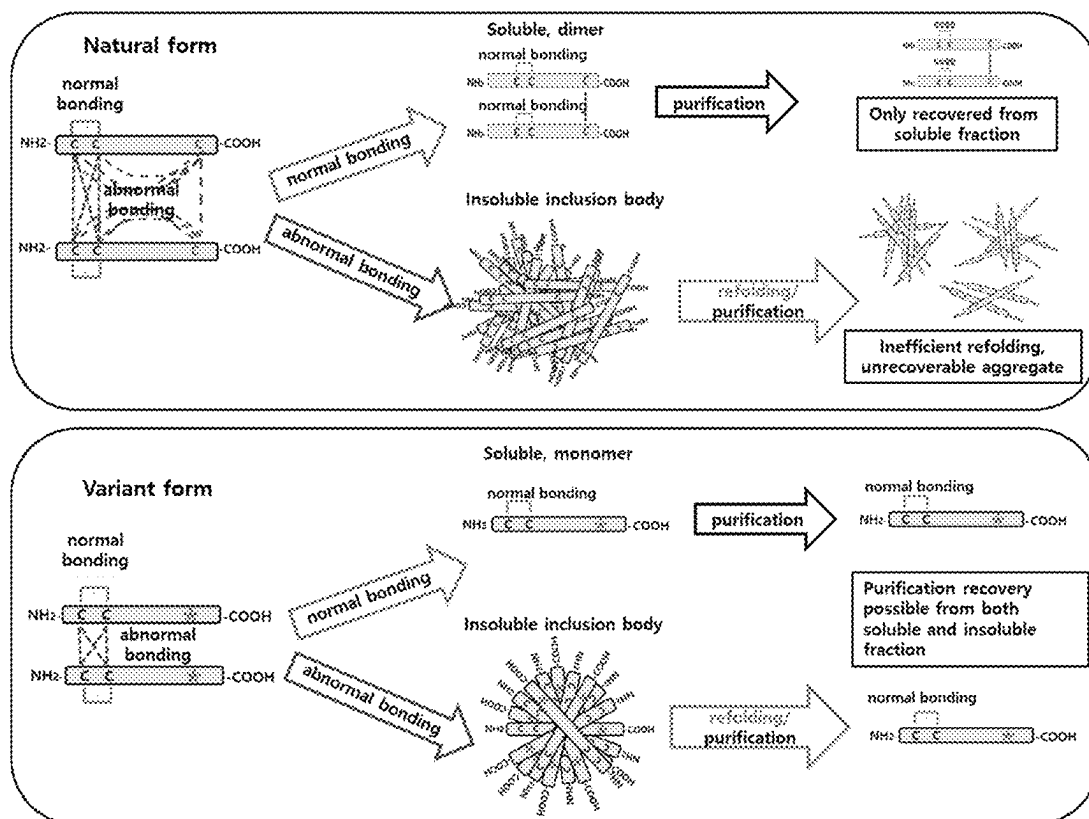


FIG. 3

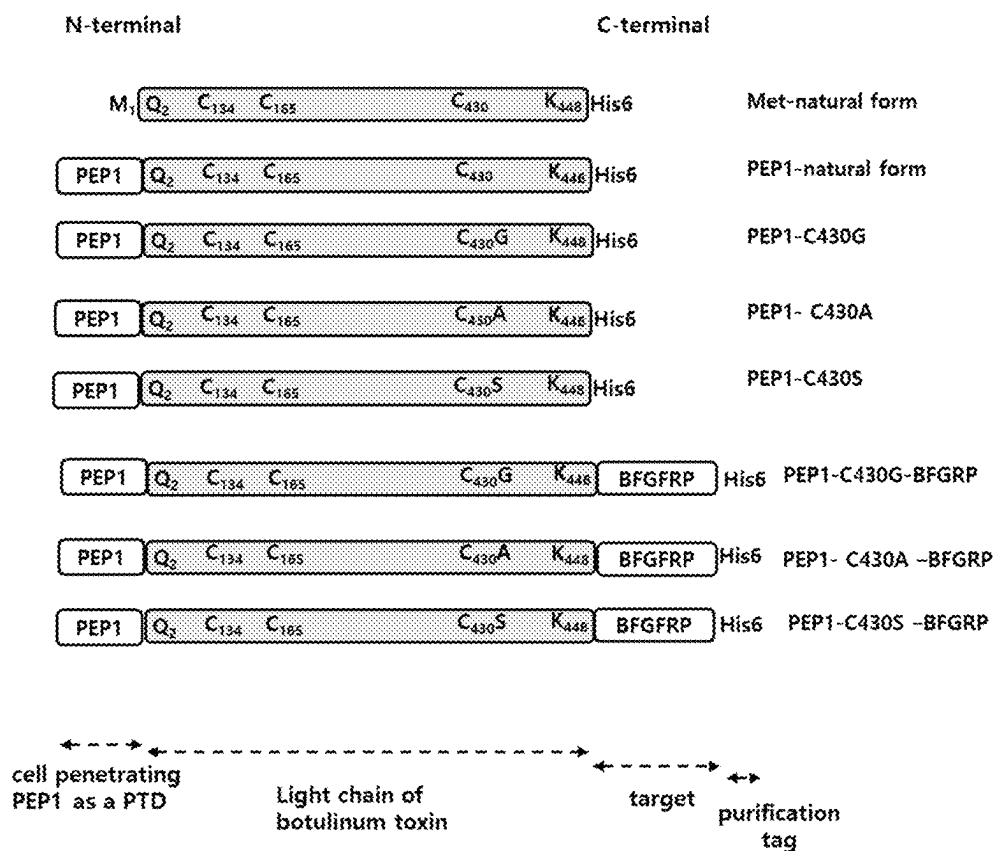


FIG. 4

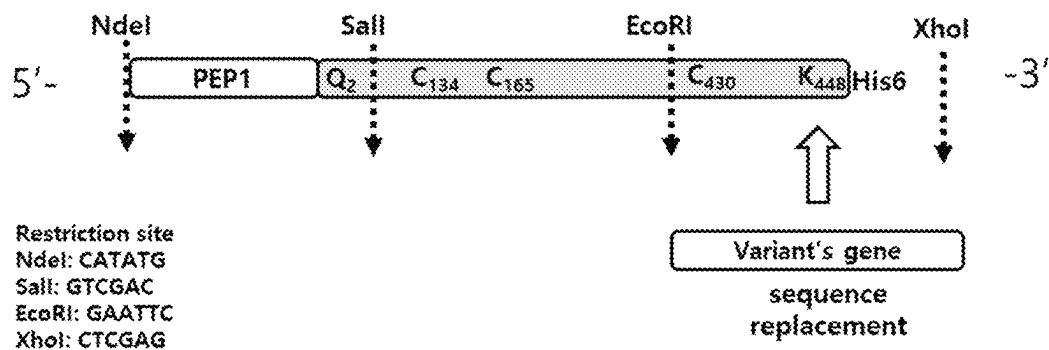


FIG. 5

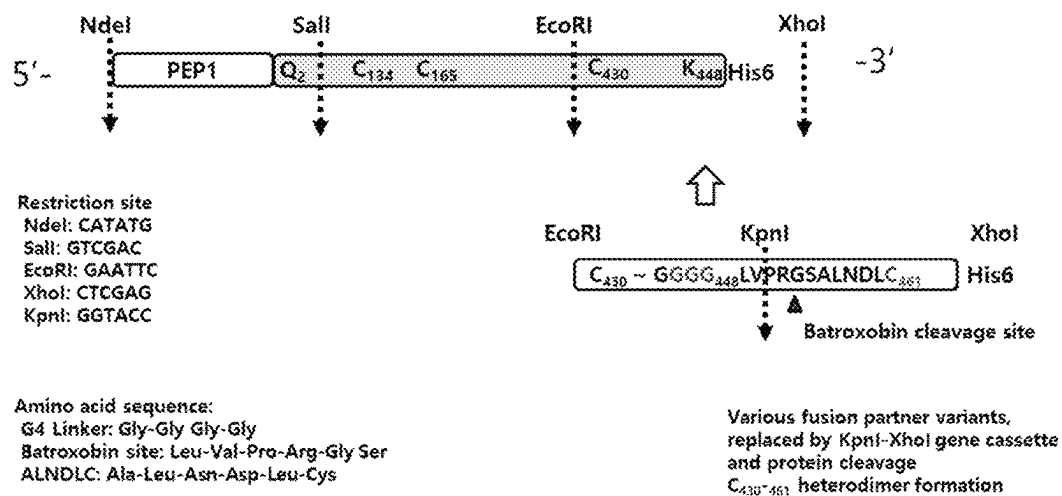


FIG. 6

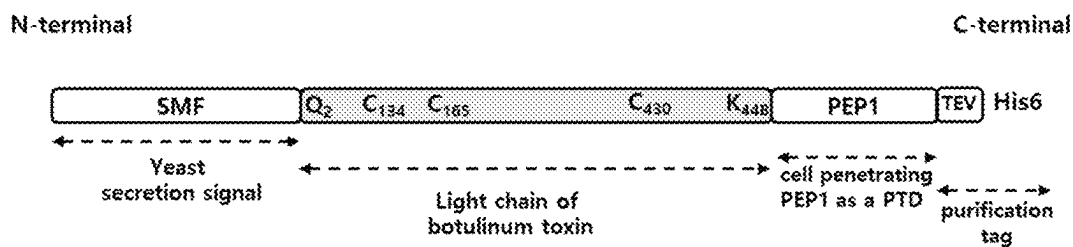


FIG. 7

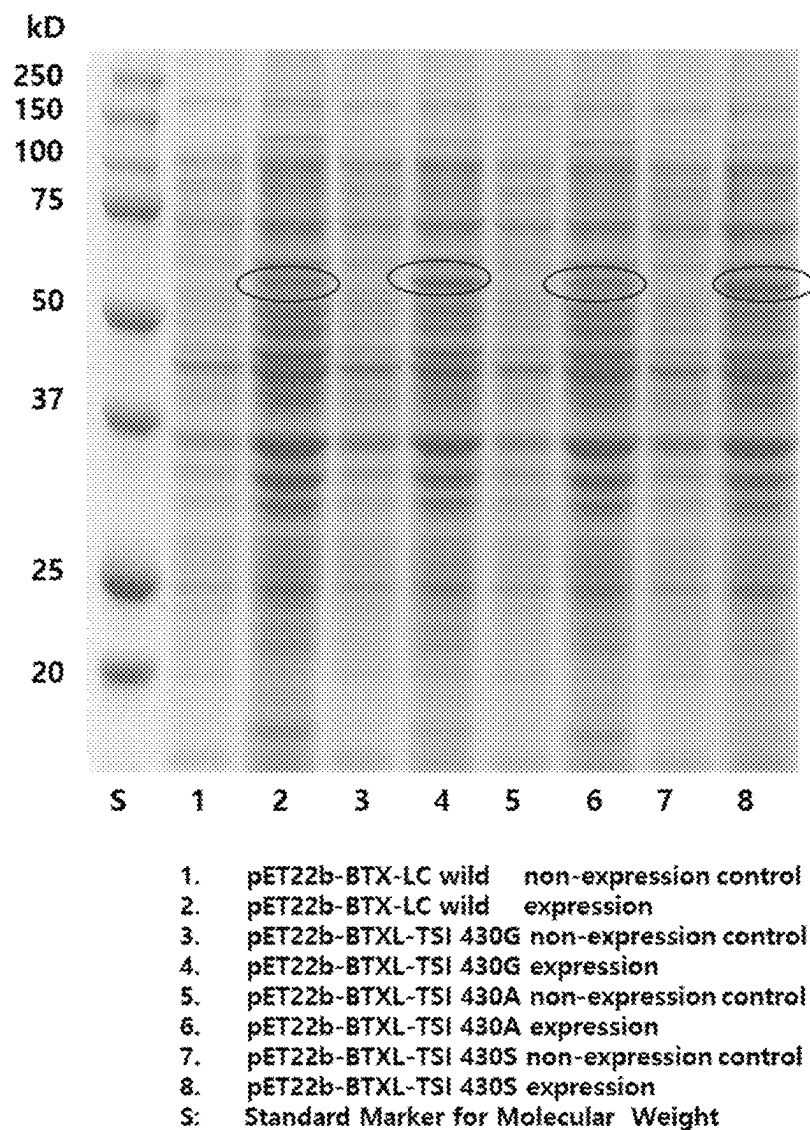


FIG. 8

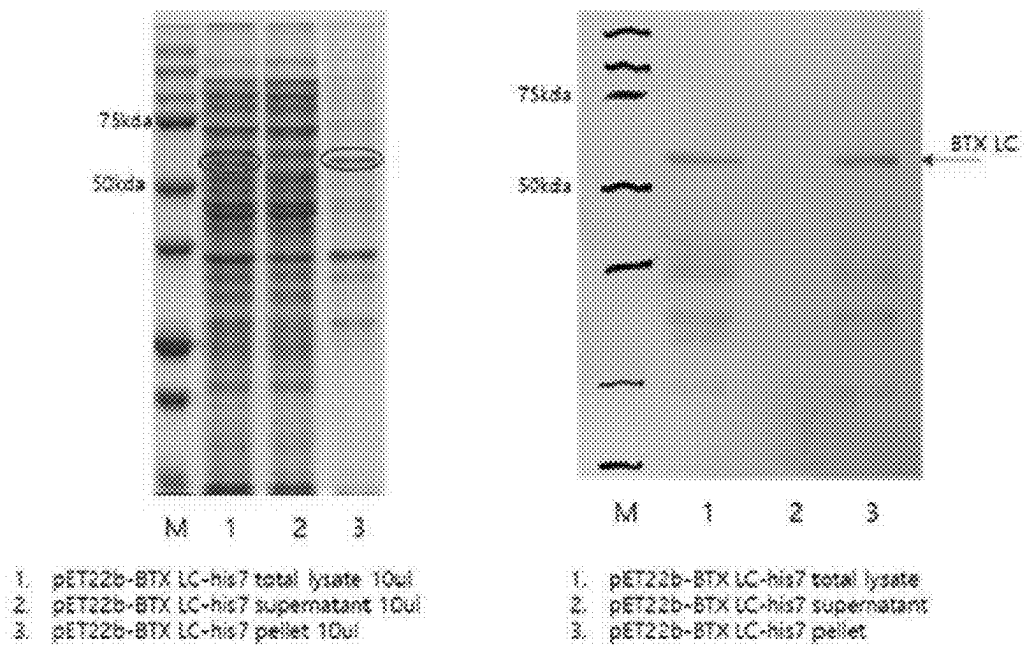


FIG. 9

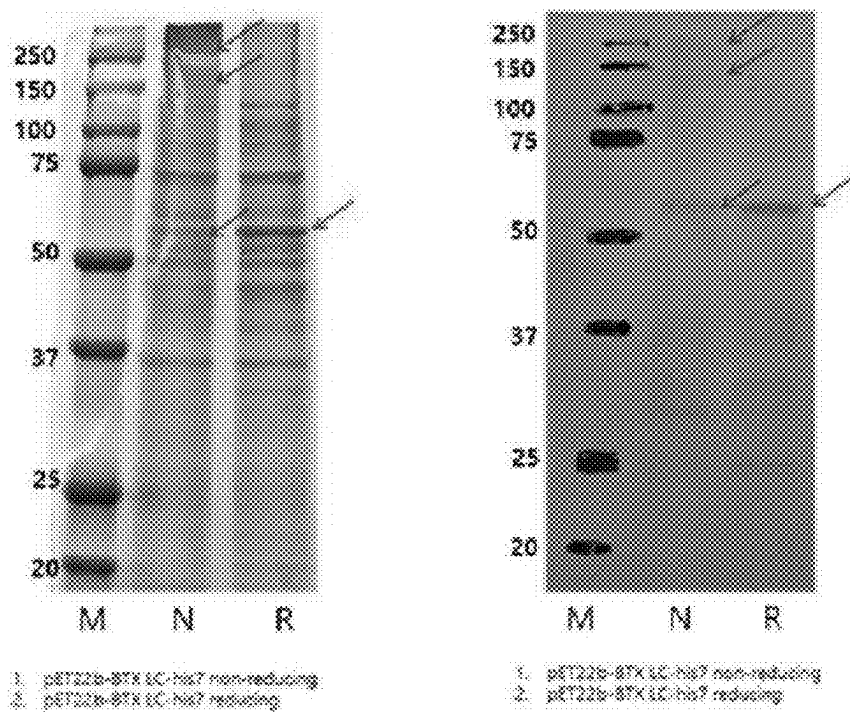
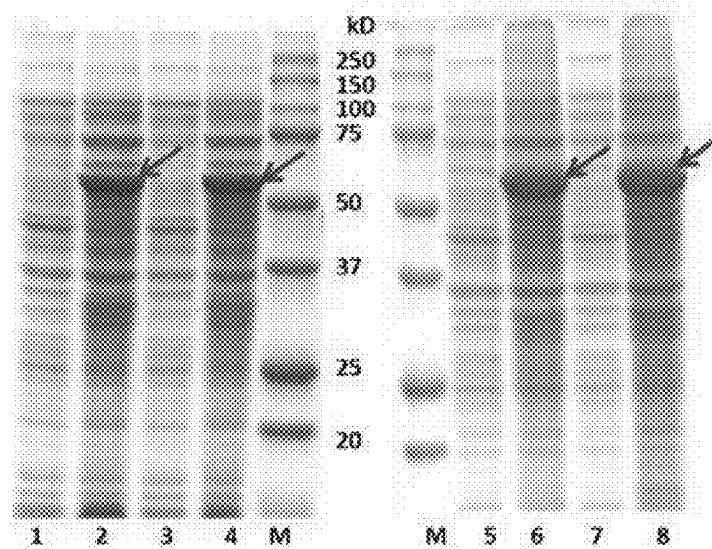
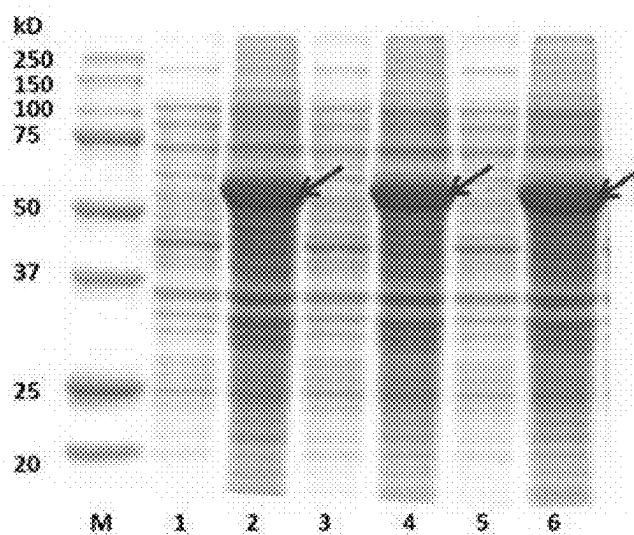


FIG. 10



1. pET22b-BTX-LC wild opti-codon non-expression control
 2. pET22b-BTX-LC wild opti-codon expression
 3. pET22b-BTX-TSI wild opti-codon 430G non-expression control
 4. pET22b-BTX-TSI wild opti-codon 430G expression
 5. pET22b-BTX-TSI wild opti-codon 430A non-expression control
 6. pET22b-BTX-TSI wild opti-codon 430A expression
 7. pET22b-BTX-TSI wild opti-codon 430S non-expression control
 8. pET22b-BTX-TSI wild opti-codon 430S expression
- M: Molecular weight standard marker

FIG. 11



1. pET22b-BTXL-TSI wild opti-codon 430G-BFGFRP non-expression control
 2. pET22b-BTXL-TSI wild opti-codon 430G-BFGFRP expression
 3. pET22b-BTXL-TSI wild opti-codon 430A-BFGFRP non-expression control
 4. pET22b-BTXL-TSI wild opti-codon 430A-BFGFRP expression
 5. pET22b-BTXL-TSI wild opti-codon 430S-BFGFRP non-expression control
 6. pET22b-BTXL-TSI wild opti-codon 430S-BFGFRP expression
- M: Molecular weight standard marker

FIG. 12

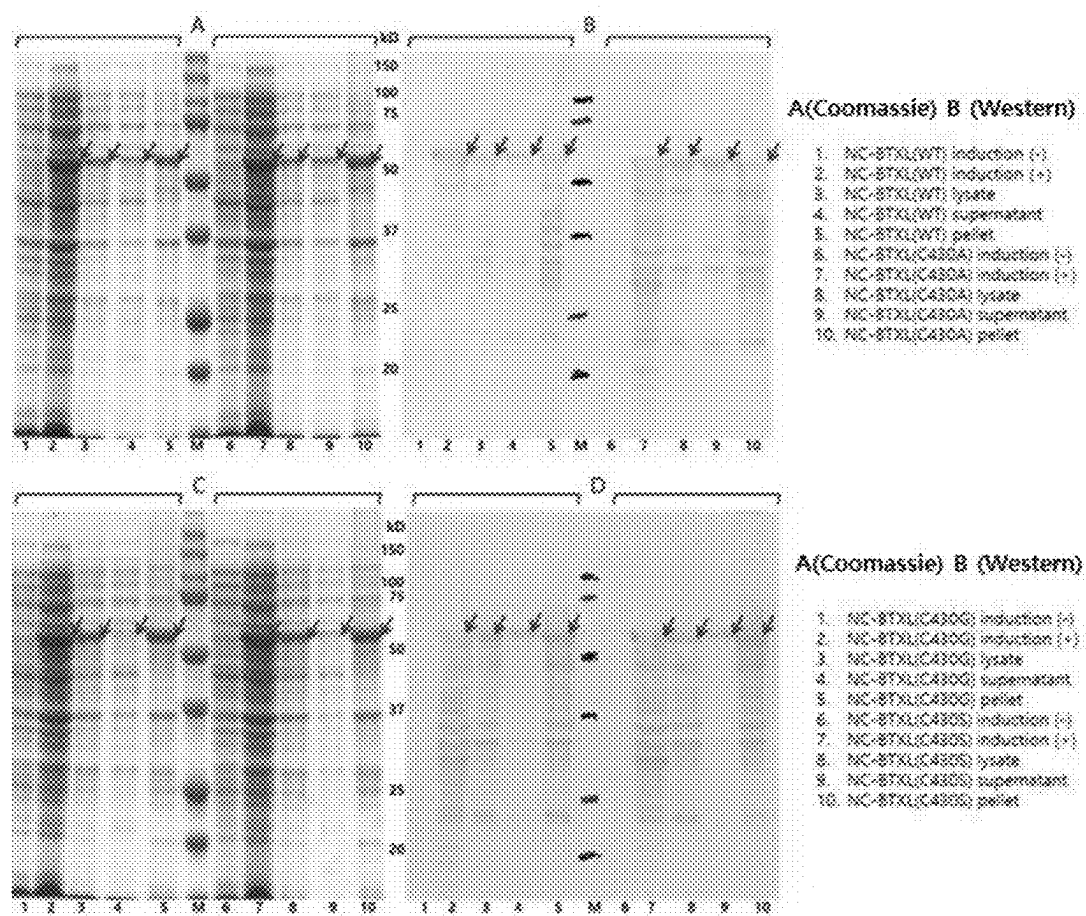


FIG. 13

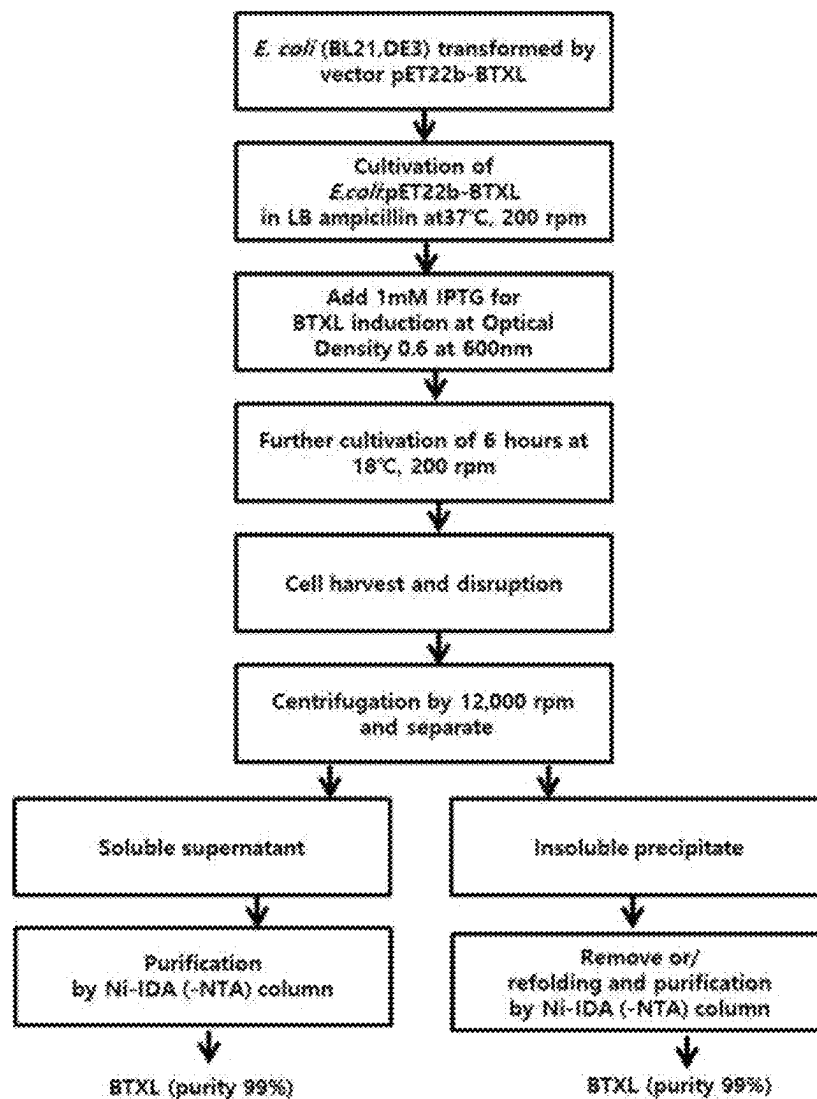


FIG. 14

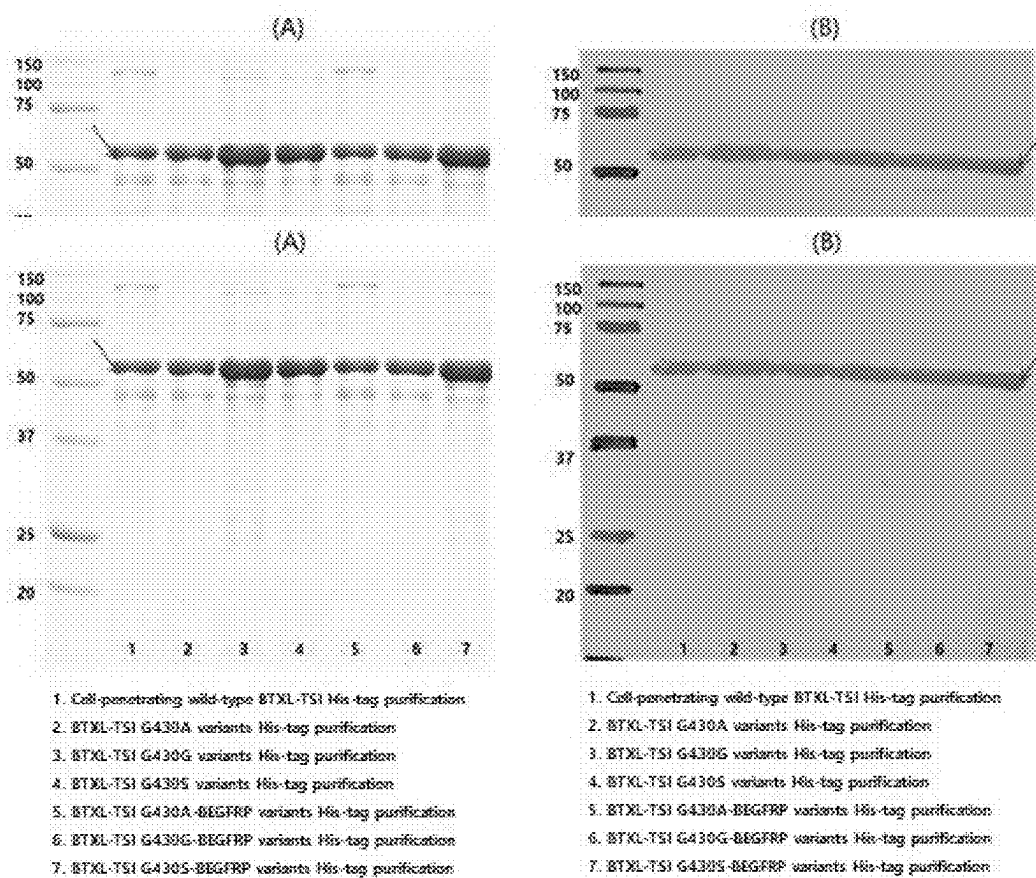


FIG. 15

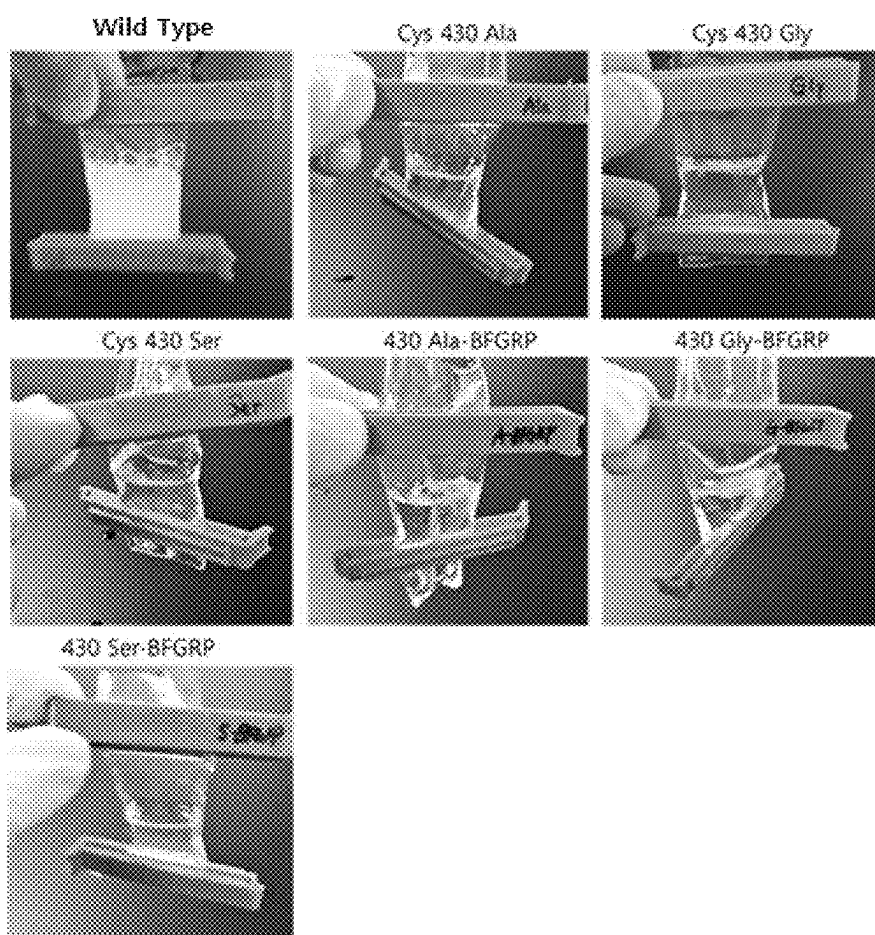


FIG. 16

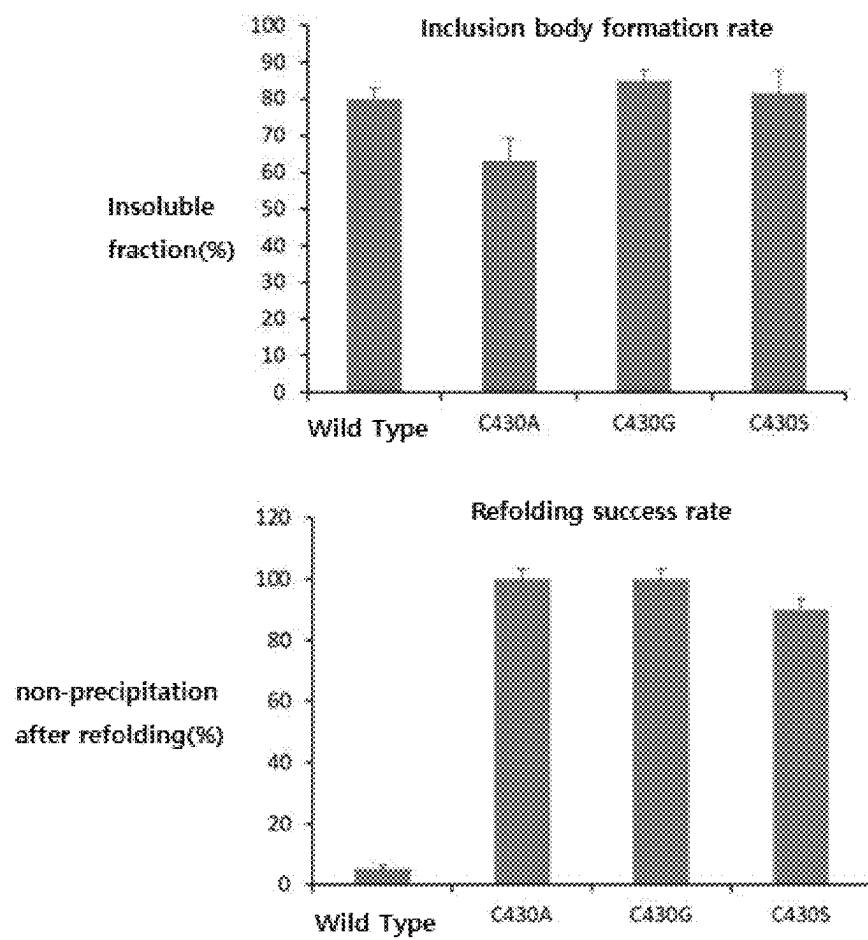
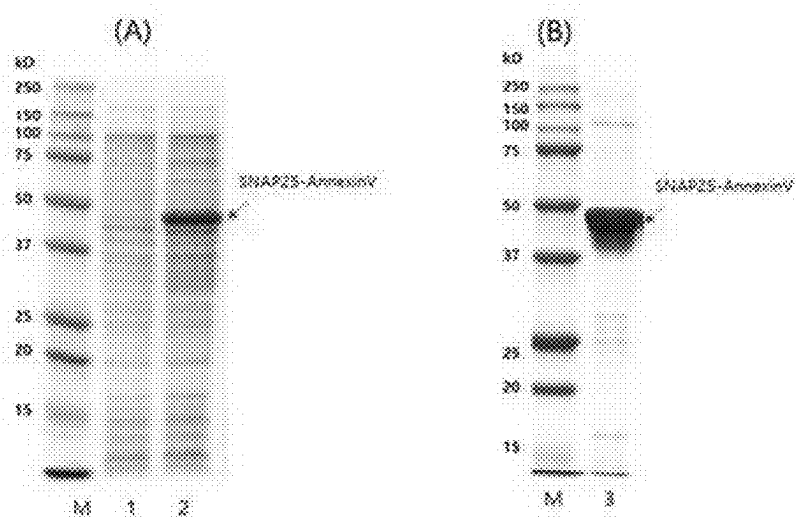


FIG. 17



1. pET22b-SNAP25-Annexin V non-expression control

2. pET22b-SNAP25-Annexin V expression

3. purified SNAP25-Annexin V protein

M: Molecular weight standard marker

FIG. 18

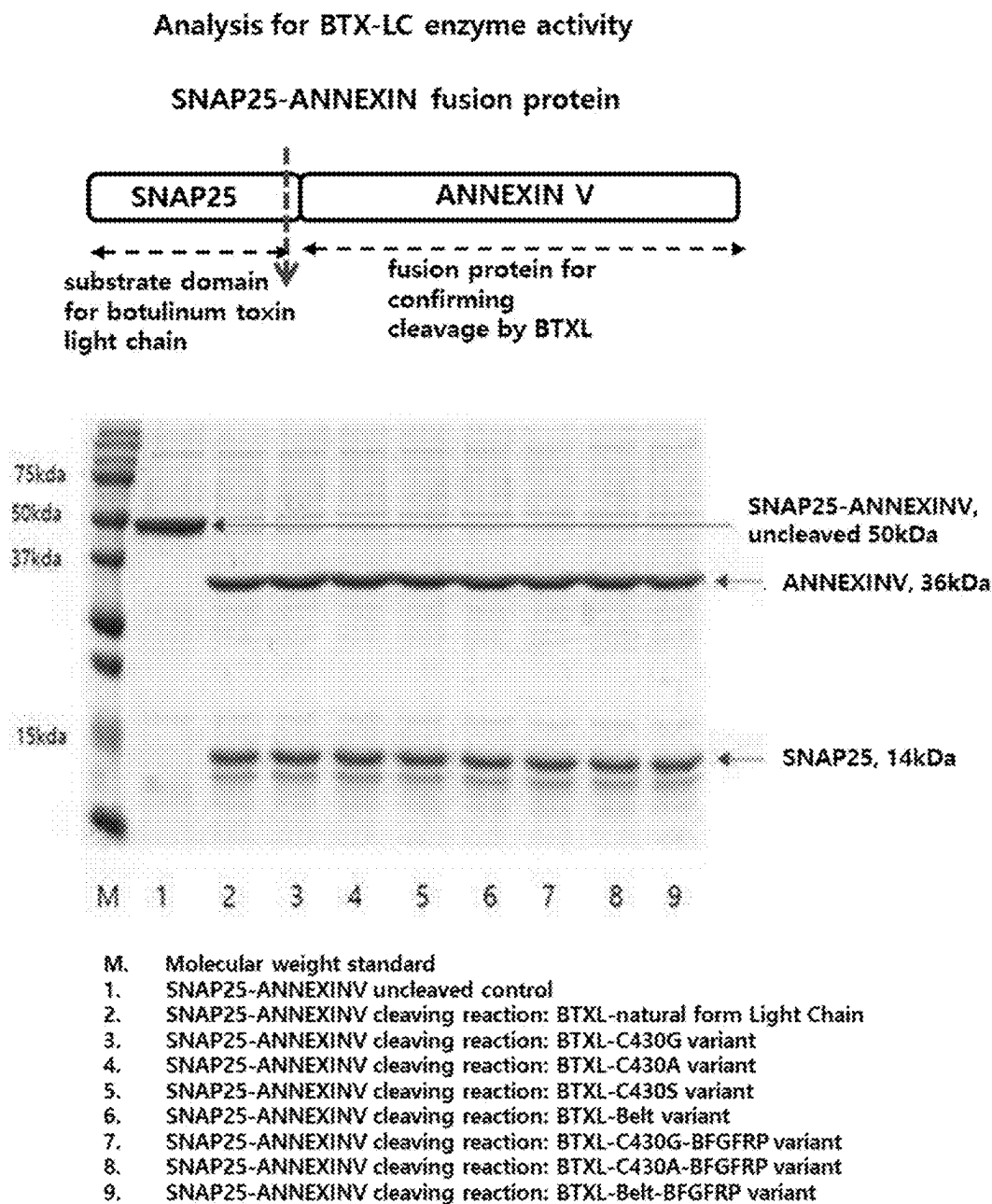


FIG. 19

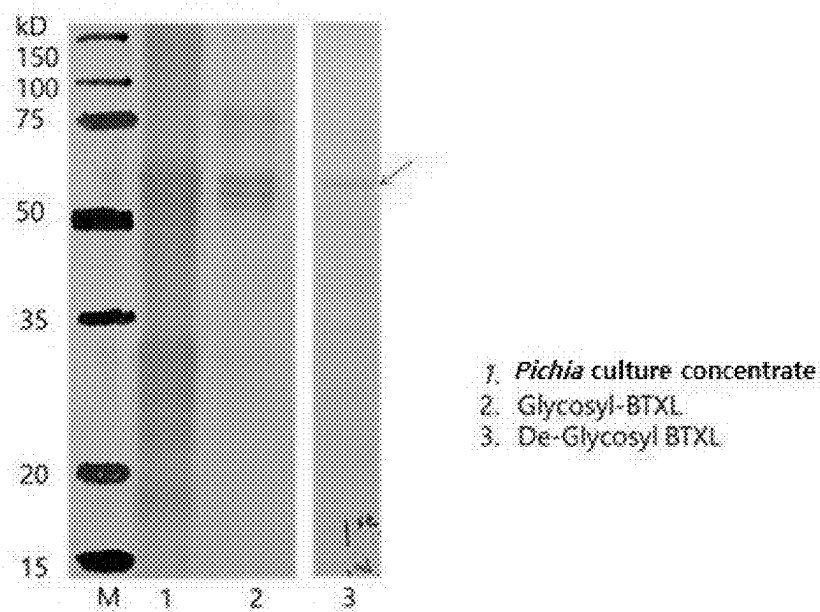


FIG. 20

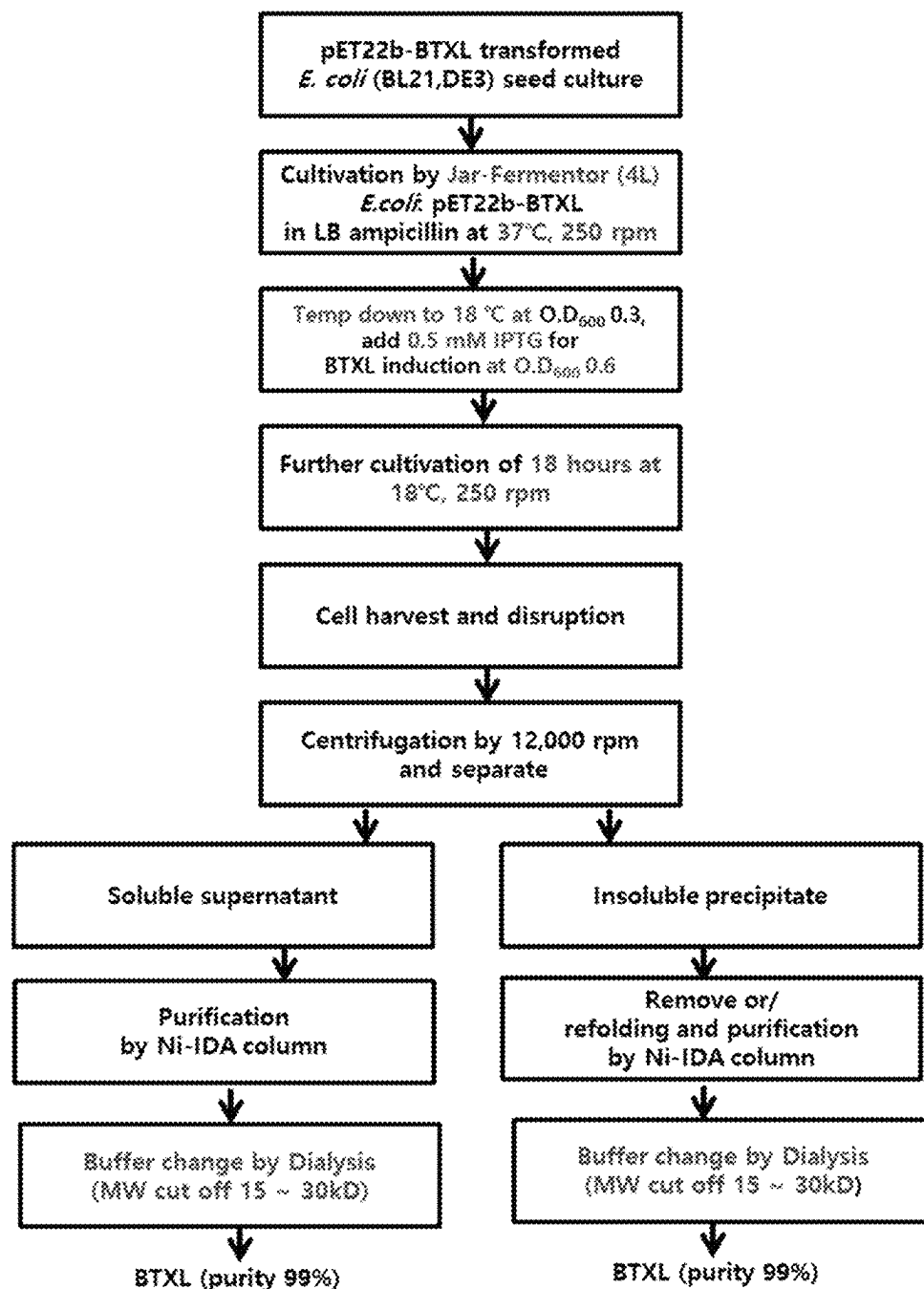


FIG. 21

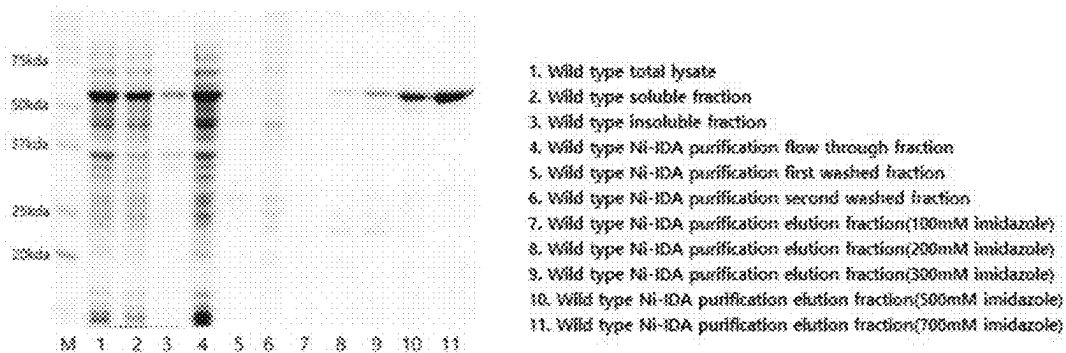


FIG. 22

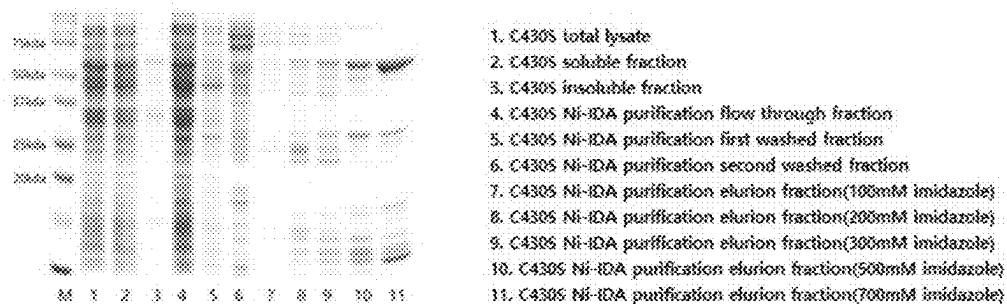


FIG. 23

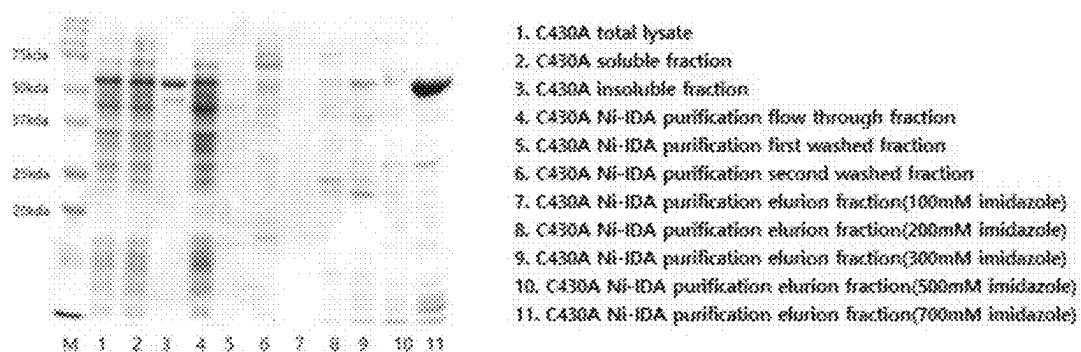


FIG. 24

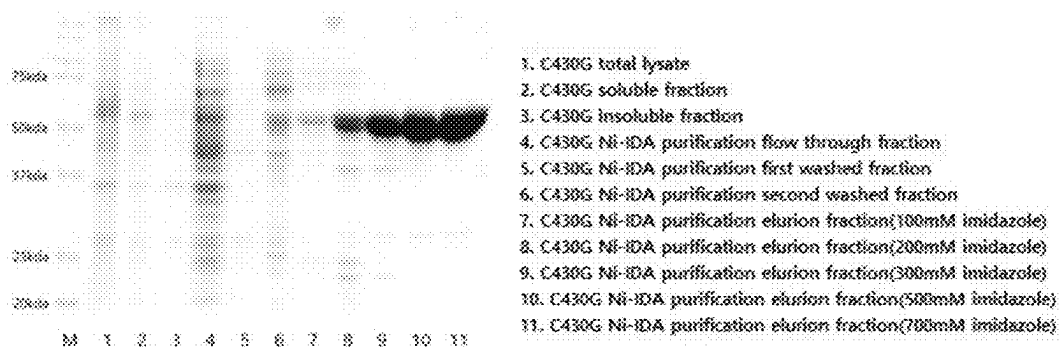


FIG. 25

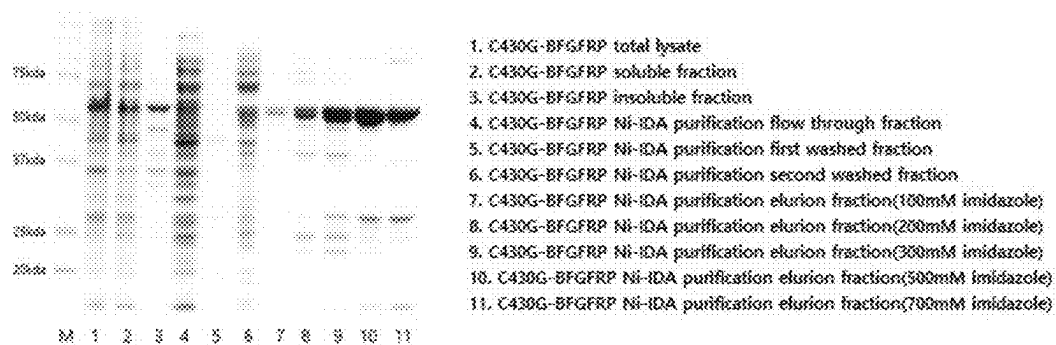
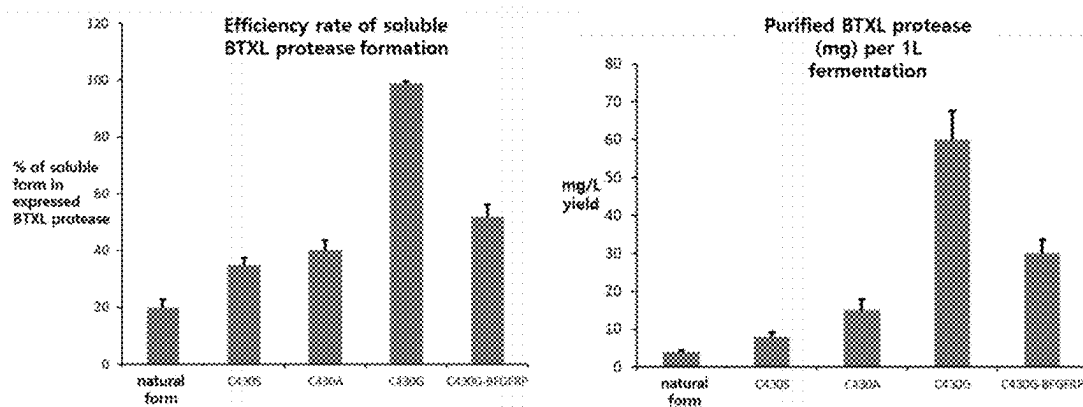


FIG. 26



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**NON-TOXIC PROTEASE HAVING
IMPROVED PRODUCTIVITY****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a United States national phase under 35 USC § 371 of International Patent Application No. PCT/KR2020/005623 filed Apr. 28, 2020, which in turn claims priority under 35 USC § 119 of Korean Patent Application No. 10-2019-0069259 filed Jun. 12, 2019. The disclosures of all such applications are hereby incorporated herein by reference in their respective entireties, for all purposes.

**REFERENCE TO SEQUENCE LISTING
SUBMITTED VIA EFS-WEB**

This application includes an electronically submitted sequence listing in .txt format. The .txt file contains a sequence listing entitled "607_SeqListing_ST25.txt" created on Dec. 7, 2021 and is 96,478 bytes in size. The sequence listing contained in this .txt file is part of the specification and is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

The present invention relates to a non-toxic protease with improved productivity, and more particularly, to a non-toxic protease which is produced with improved productivity as a result of inhibiting irreversible protein aggregation and improving refolding efficiency through introduction of a point mutation.

BACKGROUND ART

Some of the best known examples of substances that act on nerve cells by incapacitating the secretory function of cellular physiological substances in target cells include clostridial neurotoxins (e.g., botulinum neurotoxins commercially available under trade names such as Dysport™, Neurobloc™, and Botox™). The non-toxic protease domains constituting these substances are zinc-endopeptidases which are a well-known group of proteases that act on target cells by incapacitating the secretory function of cellular physiological substances. Interestingly, the non-toxic protease domains alone do not kill the target cells on which they act.

The non-toxic protease domains act by proteolytically cleaving SNARE proteins (e.g., intracellular transport proteins known as SNAP-25, VAMP, or syntaxin) (see Gerald K (2002) "Cell and Molecular Biology" (4th edition) John Wiley & Sons, Inc.). The acronym "SNARE" derives from Soluble NSF Attachment Receptor, where NSF means N-ethylmaleimide—Sensitive Factor. SNARE proteins are integral to intracellular vesicle formation, and thus to secretion of molecules through vesicle transport from a cell. Accordingly, once delivered to a desired target cell, the non-toxic protease is capable of inhibiting cellular secretion from the target cell.

Clostridial neurotoxins represent a major group of non-toxic toxin molecules and comprise two polypeptide chains joined together by a disulfide bond. The two chains are termed the heavy chain (H-chain) having a molecular weight of about 100 kDa and the light chain (L-chain) having a molecular weight of about 50 kDa. It is the L-chain, which

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possesses a protease function and exhibits a high substrate specificity for vesicle and/or plasma membrane associated (SNARE) proteins involved in the exocytic process (e.g., synaptobrevin, syntaxin or SNAP-25). These substrates are important components of the neurosecretory machinery (Korean Patent Application Publication No. 10-2014-0036239).

Non-toxic proteases can temporarily inhibit muscle contraction *in vivo*, and thus they are used as bio-drugs for the treatment of some Dystonia or muscle contraction disorders. That is, non-toxic proteases inhibit muscle contraction by temporarily inhibiting the release of acetylcholine. Recently, non-toxic proteases have been successfully used as injectable biological drugs in the treatment of various muscle contraction disorders, such as blepharospasm, migraine headache, reduction of facial wrinkles, etc. (Mauskop A. The use of Botulinum Toxin in the treatment of headaches. *Curr Pain Headache Rep* 2002; 6:320-3; Mauskop A. Botulinum toxin in headache treatment: The end of the road? *Cephalalgia* 2007; 27:468; Verheyden J, Blitzer A. Other noncosmetic uses of Botox. *Dis Mon* 2002; 48:357-66; Dhaked R K, Singh M K, Singh P, Gupta P. Botulinum toxin: Bioweapon & magic drug. *Indian J Med Res* 2010; 132: 489-503; Turton K, Chaddock J A, Acharya K R. Botulinum and tetanus neurotoxins: structure, function and therapeutic utility. *Trends Biochem Sci* 2002; 27:552-8; Masuyer G, Chaddock J A, Foster K A, Acharya K R. Engineered botulinum neurotoxins as new therapeutics. *Annu Rev Pharmacol Toxicol* 2014; 54:27-51).

Non-toxic proteases can be used for various purposes as described above and have high commercial potential, and thus attempts have been made to mass-produce non-toxic proteases using recombinant microorganisms. However, a significant portion of a recombinant non-toxic protease expressed in a recombinant microorganism transformed with a recombinant expression vector tends to aggregate in the form of inclusion bodies in an insoluble fraction depending on the composition and structure of the protein. In this case, the protein is dissolved as much as possible in a solution of urea, guanidium chloride, a surfactant or a reducing agent, and then purified through centrifugation, dialysis, ion exchange, hydrophobic interaction, reverse phase and gel filtration HPLC, various chromatography columns packed with an appropriate type of resin, etc. The solution used in this complex purification process changes the structure of the protein, resulting in loss of the activity of the non-toxic protease, and even when a complex additional process of refolding the purified protein to restore the protease activity is introduced, a problem arises in that a problem that most of the inclusion bodies are not recovered due to low restoration efficiency (Saffarian, *BIOENGINEERED* 2016, VOL. 7, NO. 6, 478-483). This refolding process is generally performed by diluting or dialyzing the protein in a buffer solution free of a reducing/denaturing agent depending on the size of the protein to induce disulfide bonds between cysteine amino acids inside the protein through oxidation, and then separating soluble and insoluble fractions by filtration or centrifugation, thereby recovering only a protein whose unique three-dimensional structure has been restored. Even when this complicated process is performed, the recovery rate of the refolded non-toxic protease from the insoluble fraction is quite low (Cabrita, 2004, *Biotechnology Annual Review*, 10:31-50; Clark, *Current Opinion in Biotechnology* 1998, 9:157-163; Yamaguchi, *Biomolecules* 2014, 4, 235-251).

Accordingly, the present inventors have made extensive efforts to effectively recover a refolded non-toxic protease

even from insoluble fractions in which most of non-toxic proteases are present but which are almost impossible to recover due to their irreversible aggregation into inclusion bodies, in a process of producing a non-toxic protease using a recombinant microorganism. As a result, the present inventors have found that, when a point mutation is induced in some amino acids of a non-toxic protease, it is possible to recover a non-toxic protease that reversibly aggregates into inclusion bodies, is effectively refolded even in insoluble fractions, and has activity, and the production of the non-toxic protease may be maximized, thereby completing the present invention.

SUMMARY OF THE INVENTION

An object of the present invention is to provide a mutated non-toxic protease which is produced in improved yield using a recombinant microorganism, and a method for producing the same.

To achieve the above object, the present invention provides a mutated non-toxic protease in which the amino acid cysteine at position 430 of a non-toxic protease represented by the amino acid sequence of SEQ ID NO: 1 is substituted with an amino acid other than cysteine.

The present invention also provides a fusion non-toxic proteinase in which the mutated non-toxic protease is fused to any one or more peptides selected from the group consisting of:

- i) a cell penetrating peptide;
- ii) a belt domain fragment peptide; and
- iii) a cell targeting peptide.

The present invention also provides a fusion non-toxic proteinase in which the mutated non-toxic protease is fused with any one or more peptides selected from the group consisting of:

- i) a yeast secretion signal peptide;
- ii) a cell penetrating peptide; and
- iii) a cell targeting peptide.

The present invention also provides a mutant gene encoding the mutated non-toxic protease.

The present invention also provides a gene construct comprising: the mutant gene; and any one or more nucleic acids selected from the group consisting of

- i) a nucleic acid encoding a cell penetrating peptide;
- ii) a nucleic acid encoding a belt domain fragment peptide; and
- iii) a nucleic acid encoding a cell targeting peptide.

The present invention also provides a gene construct comprising: the mutant gene; and any one or more nucleic acids selected from the group consisting of

- i) a nucleic acid encoding a yeast secretion signal peptide;
- ii) a nucleic acid encoding a cell penetrating peptide; and
- iii) a nucleic acid encoding a cell targeting peptide.

The present invention also provides a recombinant vector into which the mutant gene has been introduced.

The present invention also provides a recombinant microorganism into which the recombinant vector has been introduced.

The present invention also provides a method for producing a mutated non-toxic protease or a fusion non-toxic protease,

the method comprising steps of:

- (a) producing the mutated non-toxic protease or the fusion non-toxic protease by culturing the recombinant microorganism;
- (b) disrupting the recombinant microorganism; and

(c) purifying the mutated non-toxic protease or the fusion non-toxic protease from the disrupted recombinant microorganism.

The present invention also provides a pharmaceutical composition for treating Dystonia containing the mutated or fusion non-toxic protease as an active ingredient.

The present invention also provides a hyaluronic acid microneedle patch containing the mutated or fusion non-toxic protease.

The present invention also provides a method for alleviating or treating symptoms of Dystonia, the method comprising a step of administering the mutated or fusion non-toxic protease to a subject in need of alleviation or treatment of symptoms of Dystonia.

The present invention also provides the use of the mutated or fusion non-toxic protease for the manufacture of a medicament for use in the alleviation or treatment of symptoms of Dystonia.

The present invention also provides the above mutated or fusion non-toxic protease for use in a method for alleviating or treating symptoms of Dystonia.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 schematically shows the structural difference between a mutated non-toxic protease according to the present invention and a wild-type non-toxic protease.

FIG. 2 schematically shows the principle that irreversible insoluble inclusion bodies are formed as a result of expression of the wild-type non-toxic protease and this formation is inhibited in the mutated non-toxic protease according to the present invention.

FIG. 3 schematically shows the structures of a mutated non-toxic protease and a fusion non-toxic protease according to various embodiments of the present invention.

FIG. 4 schematically shows the restriction enzyme positions used in recombinant vector cloning to produce various mutated non-toxic proteases and fusion mutated non-toxic proteases according to the present invention.

FIG. 5 schematically shows the structure of a fusion non-cytotoxic protease into which (Gly)₄ capable of imparting structural flexibility to a fusion site between a mutated non-cytotoxic protease and a peptide (fusion partner) has been introduced and a cleavage amino acid sequence (LVPRGS) of the thrombin-like enzyme batroxobin, a protein cleavage enzyme, has been introduced so that it can cleave between the mutated non-toxic protease and the peptide (fusion partner) to form heterodimers linked by a disulfide bond.

FIG. 6 schematically shows the structure of a fusion non-toxic protease for expression in the yeast *Pichia pastoris*.

FIG. 7 shows the results of analyzing the expression patterns of mutated non-toxic proteases according to various embodiments of the present invention in *E. coli* by Coomassie blue staining.

FIG. 8 shows the results of expressing a wild-type non-toxic protease in *E. coli* and analyzing the expression patterns thereof in the supernatant and precipitate by Coomassie blue staining and Western blotting.

FIG. 9 shows the results of expressing a wild-type non-toxic protease in *E. coli* and analyzing the aggregation patterns of the wild-type non-toxic protease under non-reducing and reducing conditions by Coomassie blue staining and Western blotting.

FIG. 10 shows the results of codon-optimizing the mutated non-toxic proteases according to various embodiments of the present invention.

ments of the present invention so as to be suitable for expression in *E. coli*, and then analyzing the expression patterns thereof in *E. coli* by Coomassie blue staining.

FIG. 11 shows the results of codon-optimizing the fusion non-toxic proteases according to various embodiments of the present invention so as to be suitable for expression in *E. coli*, and then analyzing the expression patterns thereof in *E. coli* by Coomassie blue staining.

FIG. 12 shows the results of lysing *E. coli* cells expressing the mutated non-toxic proteases according to various embodiments of the present invention, separating the cell lysate into a soluble fraction and an insoluble fraction, and then analyzing the amount of a non-toxic protease in each fraction by Coomassie blue staining and Western blotting.

FIG. 13 is a schematic diagram showing a production process according to the present invention.

FIG. 14 shows the results of purifying the mutated non-toxic protease and fusion non-toxic protease according to various embodiments of the present invention, and then analyzing the non-toxic proteases by Coomassie blue staining and Western blotting.

FIG. 15 shows the results of confirming the excellent refolding effects of the mutated non-toxic proteases and fusion non-toxic proteases according to various embodiments of the present invention.

FIG. 16 shows the results of comparing the amounts of wild-type and mutated non-toxic proteases present in insoluble fractions (A) with the amounts of wild-type and mutant non-toxic proteases recoverable by refolding (B).

FIG. 17 shows the results of fusing an enzyme active substrate of a non-toxic protease with ANNEXIN V, expressing and purifying the fusion non-toxic protease in *E. coli*, and performing Coomassie blue staining, in order to analyze the cleavage activities of the mutated non-toxic protease and fusion non-toxic protease according to the present invention.

FIG. 18 shows the results of analyzing the cleavage activities of the mutated non-toxic protease and fusion non-toxic protease according to the present invention by Coomassie blue staining using the SNAP25-ANNEXIN V fusion protein.

FIG. 19 shows the results of expressing and purifying the fusion non-toxic protease in the recombinant yeast *Pichia pastoris* transformed with the fusion non-toxic protease according to the present invention, and analyzing the fusion non-toxic protease by Western blotting.

FIG. 20 is a schematic diagram showing a process for fermenting and purifying a large amount of a non-toxic protease according to the present invention.

FIG. 21 shows the results of performing SDS-PAGE electrophoresis and Coomassie staining after fermenting and purifying a large amount of a wild-type non-toxic protease.

FIG. 22 shows the results of performing SDS-PAGE electrophoresis and Coomassie staining after fermenting and purifying a large amount of a mutated non-toxic protease (C430S) according to the present invention.

FIG. 23 shows the results of SDS-PAGE electrophoresis and Coomassie staining after fermenting and purifying a large amount of a mutated non-toxic protease (C430A) according to the present invention.

FIG. 24 shows the results of SDS-PAGE electrophoresis and Coomassie staining after fermenting and purifying a large amount of a mutated non-toxic protease (C430G) according to the present invention.

FIG. 25 shows the results of SDS-PAGE electrophoresis and Coomassie staining after fermenting and purifying a

large amount of a fusion non-toxic protease (C430G-BFG-FRP) according to the present invention.

FIG. 26 is a graph comparing the water-soluble protease formation rate and purification yield of each of the wild-type non-toxic protease and the mutant non-toxic protease according to the present invention.

DETAILED DESCRIPTION AND PREFERRED EMBODIMENTS OF THE INVENTION

Unless otherwise defined, all technical and scientific terms used in the present specification have the same meanings as commonly understood by those skilled in the art to which the present disclosure pertains. In general, the nomenclature used in the present specification is well known and commonly used in the art.

In the present invention, it has been found that, when a mutation at the amino acid cysteine at position 430 of a wild-type protease is induced in order to solve the problem associated with the decrease in productivity resulting from protein aggregation which frequently occurs during purification of the wild-type non-toxic protease in a process of producing the non-toxic protease using a recombinant microorganism, protein aggregation resulting from inclusion body formation is inhibited while the protease activity is maintained, and thus the non-toxic protease contained in an insoluble fraction can be recovered even by a simple refolding process alone, and even when various functional peptides are fused to the mutated non-toxic protease, the fusion mutated non-toxic protease can be produced with high productivity.

Therefore, in one aspect, the present invention is directed to a mutated non-toxic protease in which the amino acid cysteine at position 430 of a non-toxic protease represented by the amino acid sequence of SEQ ID NO: 1 is substituted with an amino acid other than cysteine.

In the present invention, the amino acid cysteine at position 430 of the non-toxic protease is substituted with any amino acid other than cysteine so as not to form an inappropriate disulfide bond with another cysteine. The amino acid substituting for the cysteine at position 430 is not limited, but the cysteine (Cys) may preferably be substituted with glycine (Gly), alanine (Ala) or serine (Ser).

In another aspect, the present invention is directed to a fusion non-toxic proteinase in which the mutated non-toxic protease is fused with any one or more peptides selected from the group consisting of:

- i) a cell penetrating peptide;
- ii) a belt domain fragment peptide; and
- iii) a cell targeting peptide.

In the present invention, the cell penetrating peptide may be represented by the amino acid sequence of SEQ ID NO: 3, the belt domain fragment peptide may be represented by the amino acid sequence of SEQ ID NO: 5, and the cell targeting peptide may be represented by the amino acid sequence of SEQ ID NO: 7, but the present invention is not limited thereto. Preferably, the fusion non-toxic protease is for expression in *E. coli*.

In still another aspect, the present invention is directed to a fusion non-toxic proteinase in which the mutated non-toxic protease is fused with any one or more peptides selected from the group consisting of:

- i) a yeast secretion signal peptide;
- ii) a cell penetrating peptide; and
- iii) a cell targeting peptide. Preferably, the fusion non-toxic protease is for expression in yeast.

In the present invention, the yeast secretion signal peptide may be represented by the amino acid sequence of SEQ ID NO: 46, the cell penetrating peptide may be represented by the amino acid sequence of SEQ ID NO: 3, and the cell targeting peptide may be represented by the amino acid sequence of SEQ ID NO: 7, but the present invention is not limited thereto.

In the present invention, when the non-toxic protease constituting the fusion non-toxic protease is fused with another peptide at its N-terminus, the fusion may be performed after removal of the amino acid methionine at position 1.

In the present invention, a tag peptide for purification may further be fused to the mutated non-toxic protease or the fusion non-toxic protease. The tag peptide for purification may be selected from the group consisting of glutathione-S-transferase (GST), C-myc tag, a chitin-binding domain, streptavidin binding protein (SBP), a cellulose-binding domain, a calmodulin-binding peptide, S-tag, Strep-tag II, FLA, protein A, protein G, histidine affinity tag (HAT), poly-His, thioredoxin, pelB leader, and maltose binding protein (MBP), but is not limited thereto. It is possible to use any tag peptide for purification that is used in a process of expressing and purifying a protein in a microorganism in the art. It will be apparent to those skilled in the art that the peptide for purification may be cleaved according to a method known in the art after the completion of purification, and a mutated non-toxic protease or fusion non-toxic protease from which the tag peptide for purification has been detached may be used for its original purpose.

In the present invention, any one of the peptides (i.e., a peptide for fusion or purification) may be fused directly or via a linker to the non-toxic protease, and the linker sequence comprises about 3 to 20 amino acids, more preferably about 3 to 10 amino acids. The linker sequence is preferably flexible so that the non-toxic protease is not maintained in an undesirable conformation. The linker sequence may be used, for example, to space the non-toxic protease moiety apart from the fused peptide. Specifically, when the non-toxic protease is fused with two or more peptides selected from the group consisting of a cell penetrating peptide, a belt domain fragment peptide, a cell targeting peptide, a yeast secretion signal peptide, and a tag peptide for purification, the peptide linker sequence may be appropriately placed as needed between the non-toxic protease and the peptide and/or between any two or more selected peptides to provide molecular flexibility. In order to provide flexibility, the linker preferably predominantly comprises amino acids having small side chains, such as glycine, alanine and serine. Preferably, at least about 80 or 90 percent of the linker sequence comprises a glycine, alanine or serine residue, in particular a glycine or serine residue. One suitable linker sequence may be a peptide linker, preferably represented by (Gly)_N (wherein N is an integer ranging from 3 to 10), but is not limited thereto.

The present invention may further comprise a peptide sequence cleavable by a protease for cleavage between the C-terminus of the non-toxic protease and the N-terminus of a peptide selected from the group consisting of a cell penetrating peptide, a belt domain fragment peptide, a cell targeting peptide, a yeast secretion signal peptide, and a tag peptide for purification.

In the present invention, the cleavable peptide sequence may be LVPRGS, which is one of the cleavage sequences of the thrombin-like enzyme batroxobin, but is not limited thereto. In one embodiment, the cleavable peptide sequence is derived from the cleavage sequence of human fibrinogen

alpha-chain, which is one of the cleavage sequences of the thrombin-like enzyme batroxobin, which is a protein cleaving enzyme. When the peptide fused with a non-toxic protease is cleaved by the cleavage sequence, the non-toxic protease may be linked to the cleaved peptide by a disulfide bond to form a heterodimer.

However, any proteases that may cleave the peptide fused with the non-toxic protease include, without limitation, trypsin, pepsin, Lys-C endoprotease, Lys-N endoprotease, arginyl, endopeptidase, plasmin, omptin and clostridial proteases, as described in EP2524963. In one embodiment, the protease for cleavage is trypsin or Lys-C endoprotease. In one embodiment, the protease is a protease that cleaves a non-toxic protease non-native (i.e. exogenous) cleavage site. Non-native proteases that may be utilized include, but are not limited to, enterokinase (DDDDK↓) (SEQ ID NO: 50), factor Xa (IEGR↓ (SEQ ID NO: 51)/IDGR↓) (SEQ ID NO: 52), TEV (Tobacco Etch Virus) (ENLYFQ↓G) (SEQ ID NO: 53), thrombin (LVPR↓GS) (SEQ ID NO: 54), and precleavage (LEVLFQ↓GP) (SEQ ID NO: 55), (↓ indicates a cleavage site). In the present invention, the cell penetrating peptide may be selected from the group consisting of Protein-derived Penetration (RQIKIWQNRRMKWKK) (SEQ ID NO: 56), Tat peptide (GRKKRRQRRPPQ) (SEQ ID NO: 57), pVEC (LLILRRRIRKQAHASK) (SEQ ID NO: 58), chimeric transportan (GWTLSAGYLLGKINLKALAALAKKIL) (SEQ ID NO: 59), MPG (GALFLGFLGAAGSTMGAWSQPQKKRKV) (SEQ ID NO: 60), Pep-1 (KETWWETWWTEWSQPQKKRKV) (SEQ ID NO: 61), synthetic Polyarginines ((R)_n; 6<n<12), MAP (KLALKLALKALKALKLA) (SEQ ID NO: 62), and R₆W₃ (RRWRRWR) (SEQ ID NO: 63), but is not limited thereto (see C. Bechara et al. FEBS Letter 587 (2013) 1693-1702).

In one embodiment of the present invention, the cell targeting peptide may be NH₂-Thr-Tyr-Arg-Ser-Arg-Lys-Tyr-(Thror Ser)-Ser-Trp-Tyr-COOH [TYRSRKY(S/T)SWY], which is a peptide derived from the amino acids at positions 105 to 115 of human basic fibroblast growth factor, but is not limited thereto. It will be obvious to those skilled in the art that any peptides used as targeting moieties for conventional non-toxic proteases, such as lectin wheat germ agglutinin, nerve growth factor (NGF), epidermal growth factor, an antibody fragment, or a growth hormone releasing hormone (GHRH) ligand (see Elena Fonfria et al., Toxins (Basel). 2018 July; 10(7): 278.) may be used in fusion with the non-toxic protease.

In still another aspect, the present invention is directed to a mutant gene encoding the mutated non-toxic protease.

In the present invention, the mutant gene may be represented by a nucleotide sequence selected from the group consisting of SEQ ID NOs: 30, 32 and 34.

In yet another aspect, the present invention is directed to a gene construct comprising: the mutant gene; and any one or more nucleic acids selected from the group consisting of

- i) a nucleic acid encoding a cell penetrating peptide;
- ii) a nucleic acid encoding a belt domain fragment peptide; and
- iii) a nucleic acid encoding a cell targeting peptide.

In the present invention, the nucleic acid encoding the cell penetrating peptide may be represented by the nucleotide sequence of SEQ ID NO: 4, the nucleic acid encoding the belt domain fragment peptide may be represented by the nucleotide sequence of SEQ ID NO: 6, and the nucleic acid encoding the cell targeting peptide may be represented by the nucleotide sequence of SEQ ID NO: 8, but the present invention is not limited thereto.

In another aspect, the present invention is directed to a gene construct comprising: the mutant gene; and any one or more nucleic acids selected from the group consisting of

- i) a nucleic acid encoding a yeast secretion signal peptide;
- ii) a nucleic acid encoding a cell penetrating peptide; and
- iii) a nucleic acid encoding a cell targeting peptide.

In the present invention, the nucleic acid encoding the yeast secretion signal peptide may be represented by the nucleotide sequence of SEQ ID NO: 47, the nucleic acid encoding the cell penetrating peptide may be represented by the nucleotide sequence of SEQ ID NO: 4, and the nucleic acid encoding the cell targeting peptide may be represented by the nucleotide sequence of SEQ ID NO: 8, but the present invention is not limited thereto.

In the present invention, the mutant gene or gene construct may further comprise a nucleic acid encoding a tag peptide for purification. The tag peptide for purification may be selected from the group consisting of glutathione-S-transferase (GST), C-myc tag, a chitin-binding domain, streptavidin binding protein (SBP), a cellulose-binding domain, a calmodulin-binding peptide, S-tag, Strep-tag II, FLA, protein A, protein G, histidine affinity tag (HAT), poly-His, thioredoxin, pelB leader, and maltose binding protein (MBP), but is not limited thereto. It is possible to use any tag peptide for purification that is used in a process of expressing and purifying a protein in a microorganism in the art.

In the present invention, the gene construct may further comprise a nucleic acid encoding the linker peptide that connects the peptide to the non-toxic protease, wherein the nucleic acid encoding the linker peptide may be represented by the nucleotide sequence of SEQ ID NO: 26.

The present invention may further comprise a nucleic acid encoding a peptide sequence cleavable by a cleavage protease between the C-terminus of the non-toxic protease and the N-terminus of a peptide selected from the group consisting of a cell penetrating peptide, a belt domain fragment peptide, a cell targeting peptide, a yeast secretion signal peptide, and a tag peptide for purification. The nucleic acid encoding the cleavable peptide sequence may be represented by the nucleotide sequence of SEQ ID NO: 28, but is not limited thereto.

The present invention encompasses fragments and variants of polypeptides (proteins) and genes (nucleic acids) provided as specific sequences as long as they maintain their original structural and/or functional characteristics. That is, these fragments and variants may have a sequence homology of at least 90%, more preferably at least 95%, at least 97% or at least 99% to the sequences provided in the present invention.

In another aspect, the present invention is directed to a recombinant vector having the mutant gene introduced therein.

In the present invention, as the recombinant vector, there may be used without limitation any vector known in the art that may be introduced into bacterial cells, yeast cells, mammalian cells, insect cells, plant cells, or amphibian cells and used for overexpression of recombinant proteins. Preferably, there may be used a recombinant vector that may be introduced into bacterial cells or yeast cells and used for overexpression of recombinant proteins. Preferably, the pET22b(+) (Novagen, Merk Millipore) vector may be used for expression in *E. coli* and the pPIC9 vector may be used for expression in yeast. However, it will be obvious to those skilled in the art that vectors usable in the present invention are not limited to these vectors, and vectors known in the art

may be applied for the expression of the mutant non-toxic protease or fusion non-toxic protease of the present invention.

In another aspect, the present invention is directed to a recombinant microorganism having the recombinant vector introduced therein.

In the present invention, the recombinant microorganism may be bacteria or yeast, but is not limited thereto.

In another aspect, the present invention is directed to a method for producing a mutated non-toxic protease or a fusion non-toxic protease, the method comprising steps of:

- (a) producing the mutated non-toxic protease or the fusion non-toxic protease by culturing the recombinant microorganism;
- (b) disrupting the recombinant microorganism; and
- (c) purifying the mutated non-toxic protease or the fusion non-toxic protease from the disrupted recombinant microorganism.

In the present invention, step (a) may be performed in two steps: seed culture followed by main culture (i.e., culture for fermentation).

In the present invention, in step (a), the recombinant microorganism may be cultured at 30 to 38° C., and then when the OD₆₀₀ of the recombinant microorganism reaches 0.2 to 0.4, the culture temperature may be lowered to 16 to 20° C. and the recombinant microorganism may be further cultured. Preferably, in the main culture step in step (a), the recombinant microorganism may be cultured at 36 to 38° C., more preferably 36.5 to 37.5° C., and then when the OD₆₀₀ of the recombinant microorganism reaches 0.25 to 0.35, the culture temperature may be 17 to 19° C. and the recombinant microorganism may be further cultured. In this case, protein expression may be induced by adding 0.4 to 0.6 mM IPTG to the culture medium when the OD₆₀₀ reaches 0.5 to 0.7. Through this process, recovery of the mutated non-toxic protease or the fusion non-toxic protease according to the present invention from a soluble fraction is significantly increased. That is, in the initial stage, culture is performed at a high temperature in order to rapidly increase the amount of the recombinant microorganism, and then when the recombinant microorganism reaches a certain amount, the culture temperature of the recombinant microorganism is lowered to slow the metabolism of the recombinant microorganism. In addition, expression of a non-toxic protease is gradually induced by adding a small amount of IPTG, so that the refolding efficiency of the non-toxic protease is increased, and thus most of the non-toxic protease may be included in the soluble fraction. This is demonstrated by the results obtained in the present invention.

The effect of increasing the expression level of the protease in the soluble fraction is particularly pronounced in the case of the mutated protease C430G derived in the present invention. This is believed to be because, in the case of a wild-type mutant protease, a disulfide bond between cysteines is induced, and in the case of C430S and C430A, which are other mutated proteases, some non-specific disulfide bonds are induced.

In the present invention, step (c) may comprise separating the disrupted recombinant microorganism into a soluble fraction and an insoluble fraction by centrifugation, and independently purifying the mutated non-toxic protease or the fusion non-toxic protease from each fraction.

In the present invention, step (c) may comprise separating the disrupted recombinant microorganism into a soluble fraction and an insoluble fraction by centrifugation, and purifying the mutated non-toxic protease or the fusion non-toxic protease from the soluble fraction.

In the present invention, suitable medium and culture conditions that are used to culture the recombinant microorganism may be those known in the art. For example, the recombinant microorganism may be inoculated and cultured in a suitable medium for transformed *E. coli* or yeast, and then expression of the protein may be induced under suitable conditions. For example, expression of the protein in *E. coli* may be induced by supplying isopropyl- β -D-thiogalactoside, and expression of the protein in isopropyl- β or methanol-assimilating yeast may be induced by supplying methyl alcohol. After completion of the step of inducing protein expression by culture, the culture or culture medium may be recovered, and "pure recombinant protein" may be recovered therefrom. As used herein, the term "pure recombinant protein" means that the recombinant protein does not contain 5% or more, preferably 1% or more of any other proteins derived from the host cell, other than the recombinant protein of the present invention and a protein derived from the DNA sequence encoding the same.

Purification of the recombinant protein expressed in the transformed microorganism may be performed by various methods known in the art. Usually, the cell lysate or culture may be centrifuged to remove cell debris, culture impurities, etc., and then subjected to precipitation, for example, salting out (ammonium sulfate precipitation and sodium phosphate precipitation), solvent precipitation (protein fraction precipitation using acetone, ethanol, isopropyl alcohol, etc.) and the like, and subjected to dialysis, electrophoresis, and various types of column chromatography. As the types of chromatography, techniques such as ion exchange chromatography, gel-filtration chromatography, HPLC, reverse phase HPLC, adsorption chromatography, affinity column chromatography and ultrafiltration may be used alone or in combination.

In the present invention, the term "non-toxic protease" is meant to include a peptide that has significantly reduced toxicity due to removal of the heavy chain domain from the natural botulinum toxin type A and has the natural botulinum toxin type A light-chain domain having protease activity, as well as a variant of the peptide.

In another aspect, the present invention is directed to a pharmaceutical composition for treating Dystonia containing the mutated or fusion non-toxic protease as an active ingredient.

In the present invention, the pharmaceutical composition may be for transdermal administration, but is not limited thereto.

In another aspect, the present invention is directed to a method for alleviating or treating symptoms of Dystonia, the method comprising a step of administering the mutated or fusion non-toxic protease to a subject in need of alleviation or treatment of symptoms of Dystonia.

In another aspect, the present invention is directed to the use of the mutated or fusion non-toxic protease for the manufacture of a medicament for use in the alleviation or treatment of symptoms of Dystonia.

In another aspect, the present invention is directed to the use of the above-described mutated or fusion non-toxic protease for use in a method for alleviating or treating symptoms of Dystonia.

In the present invention, the muscular dystonia may be selected from the group consisting of facial spasm, eyelid spasm, torticollis, blepharospasm, spasmodic torticollis, cervical dystonia, oromandibular dystonia, spasmodic dysphonia, migraine, anal pruritus, and hyperhidrosis, but is not limited thereto.

In another aspect, the present invention is directed to a hyaluronic acid microneedle patch containing the mutated or fusion non-toxic protease.

In the present invention, the hyaluronic acid microneedle patch may be used for the treatment or alleviation of a symptom selected from the group consisting of facial spasm, eyelid spasm, torticollis, blepharospasm, spasmodic torticollis, cervical dystonia, oromandibular dystonia, spasmodic dysphonia, migraine, anal pruritus, and hyperhidrosis, but is not limited thereto.

As used herein, the term "treating" means reversing, alleviating, inhibiting the progression of, or preventing a disorder or disease to which the term applies, or one or more symptoms of the disorder or disease, unless otherwise stated. As used herein, the term "treatment" refers to the act of treating when "treating" is defined as above. As used herein, the term "treatment" refers to the act of treating when the term "treating" is defined as above. Accordingly, treatment or therapy of the disease in a mammal may include one or more of the following:

- (1) inhibiting the development of the disease;
- (2) preventing the spread of the disease;
- (3) relieving the disease;
- (4) preventing the recurrence of the disease; and
- (5) palliating symptoms of the disease.

As used herein, the term "effective amount" means an amount that is high enough to deliver the desired benefit, but low enough to avoid serious side effects within the scope of medical judgment. The amount of the non-toxic protease that is administered into the body by the composition of the present invention may be appropriately adjusted in consideration of the route of administration and the for administration.

The composition of the present invention may be administered to the subject once daily, once every several days, once every several months, or once or more every several years. Unit dosage means physically discrete units suitable for unit administration to human subjects and other mammals, each unit comprising a suitable pharmaceutical carrier and a predetermined amount of the protease of the present invention that exhibits a therapeutic effect. However, the dosage may vary depending on the severity of the patient's disease and the active ingredient and auxiliary active ingredient used. In addition, the total daily dosage may be divided into several times and continuously administered as needed. Accordingly, the above dosage range is not intended to limit the scope of the present invention in any way.

As used herein, the term "pharmaceutically acceptable" refers to a composition which is physiologically acceptable and, when administered to humans, does not cause allergic reactions such as gastrointestinal disorders and dizziness, or similar reactions.

The pharmaceutical composition of the present invention may be formulated using a method known in the art so as to provide quick, sustained or delayed release of the active ingredient after administration to mammals. The dosage forms may be powders, granules, tablets, emulsions, syrups, aerosols, soft or hard gelatin capsules, sterile injection solutions, sterile powders, or patches. In addition, the composition for preventing or treating a disease according to the present invention may be administered through several routes, including oral, transdermal, subcutaneous, intravenous and intramuscular routes. The dosage of the active ingredient may be suitably selected depending on various factors such as the route of administration, and patient's age, sex, weight and severity, and the active ingredient may be

administered in combination with a known compound having the effect of preventing, alleviating or treating the symptoms of the disease.

Hereinafter, the present invention will be described in more detail with reference to examples. It will be obvious to those skilled in the art that these examples serve only to illustrate the present invention, and the scope of the present invention is not limited by these examples.

Example 1. Construction of Vectors for Expression of Mutated Non-Toxic Proteases and Fusion Non-Toxic Proteases

As shown in FIG. 3, expression vectors for producing a mutated non-toxic protease and a fusion non-toxic protease were constructed. To this end, the following nucleotide sequences were synthesized: a nucleotide sequence (SEQ ID NO: 2) encoding a wild-type non-toxic protease (BTX-LC) represented by the amino acid sequence of SEQ ID NO: 1; a nucleotide sequence (SEQ ID NO: 4) encoding a cell penetrating peptide (PEP1); a nucleotide sequence (SEQ ID

NO: 6) encoding a belt domain peptide fragment (belt'); and a nucleotide sequence (SEQ ID NO: 8) encoding a targeting peptide.

Based on the synthesized nucleotide sequences, BTX-LC and PEP1-(ΔM)BTX-L-His were cloned by restriction enzymes (NdeI and XhoI) into the multiple cloning site (MCS) downstream of the T7 promoter-lac operator of the expression vector pET22b for *E. coli*. Thereafter, using the restriction enzymes EcoRI and XhoI, recombinant vectors capable of expressing various non-toxic proteases shown in Table 1 below, including the following non-toxic proteases, were constructed: PEP1-(Δ)BTX-L-His (*E. coli* codon optimization; theoretical pI/Mw: 8.66/55003.59 Da) capable of improving expression efficiency in *E. coli*; a mutated non-toxic protease in which the amino acid cysteine (Cys) at position 430 of the wind-type non-toxic protease is substituted with each of glycine (Gly), alanine (Ala) and serine (Ser); and a PEP1-(ΔM)BTX-L-BFGFRP-His (theoretical pI/Mw: 8.95/56436.13) fusion non-toxic protease. Table 2 below shows the sequences (gene cassettes for cloning of mutated non-toxic proteases) used in cloning for a point mutation at the amino acid at position 430 of the wild-type protease.

TABLE 1

	Amino acid sequence	Nucleotide sequence
BTX-LC	MQFVNKQFNYKDPVNGVDIAYIKIPNVG QMOPVKAFKIHNNKIWVIPERDTFTNPEEG DLNPPPEAKQVPVSYDSTYLSTDNEKDN YLKGVTKLFERIYSTDLGRMLLTSIVRGIP FWGGSTIDTELKVIDTNCINVIQPDGSYRS EELNLVIIGPSADI IQFECKSFGHEVLNLTNR NGYGSTQYIRFSPDFTFGFEESLEVDTNPL LGAGKFATDPAVTLAHLIHAGHRLYGIA INPNRVFKVNTNAYYEMSGLEVSFEELRT FGGHDAKFIDSLQENEFRLYYNKFKDIA STLNKAKSIVGTTASLQYMNKVFKEKYL SEDTSKGFSVDKLFKDKLYKMLTEIYTED NFVKFFKVLNRTNLAANFNGQNTTEINN MNFTKLKNFTGLFEFYKLLCVRGIITSKTK SLDKGYNK (SEQ ID NO: 1)	5' - atgcaatttggttaataaacaatttaattataaagatcctgtaaatgggtgtgatat tgcttatataaaaaattccaaatgtaggacaaatgcaaccagtaaaagctttta aaattcataataaaaatattgggttatccagaagagatacatttacaaatcct gaagaaggagatttaaattccaccaccagaagcaaaacaagttccagtttca tattatgattcaacatatttaagtacagataatgaaaagataattatttaaagg gagttacaaaattatttgagagaatttattcaactgatcttggaagaatgttgtt aacatcaatagtaagggaataaccattttggggtggaagacaatagatagac agaataaaagtatttgatactaatgttattaatgtgatacaaccagatggttag ttatagatcagaagaacttaacttagtaataataggacctcagctgatattat acagtttgaatgtaaaagctttggacatgaagttttgaatcttacgcgaaatg gttatggctctactcaatcacattagatttagccagattttacatttggtttgag gagtcacttgaaagtgtacaaatcctcttttaggtgcaggcaaatgtgtac agatccagcagtaacattagcacatgaacttatacatgctggacatagattat atggaatagcaattaatccaaatagggtttttaaagtaaatactaatgcctatt atgaaatgagtggttgaagtaagctttgaggaacttagaacatttggtggg gacatgagcaaaagtttatagatagtttacaggaacgaatttcgtctatatt attataaataagtttaaagatataagcaagtacacttaataaagctaaatcaatag taggtactactgctctcattacagtatatgaaaaatggttttaaagagaaatattc cctattcgaagatacatctggaaaattttcggtagataaaattaaaatttgataa gttatacaaaatgttaacagagatttacacagagataattttgttaagttttt aaagtacttaacagaaaaacatattgaattttgataaagccgtatttaagata aatatagttacctaaggtaaaattacacaatatatgatggatttaatttaagaaat acaaatttagcagcaaaacttttaaggtcaaaatcacagaaatttaataatagaa ttttactaaactaaaaaattttactggattgtttgaattttataagttgctatgtgt aagagggaataaacttctaaaactaaatcattagataaaggatacaataaag- 3' (SEQ ID NO: 2)
PEP1	MKETWWETWTEWSQPKKKRKV (SEQ ID NO: 3)	5' - atgaaggaaacttggtgggaaacttggtggactgaatggtctcaaccaaaag aagaagagaaagggtt-3' (SEQ ID NO: 4)
Belt	ALNDLC (SEQ ID NO: 5)	Gcgctgaacgatctgtgc (SEQ ID NO: 6)
BFGFRP	TYRSRKYSWY (SEQ ID NO: 7) (X stands for S or T.)	5' -ACCTATCGCAGCCGCAATATASC AGCTGGTAT-3' (SEQ ID NO: 8) (ASC stands for AGC or ACC.)
PEP1-(ΔM) BTX-L-His	MKETWWETWTEWSQPKKKRKVQFVN KQFNYKDPVNGVDIAYIKIPNVGMQPV KAFKIHNNKIWVIPERDTFTNPEEGDLNPPP EAKQVPVSYDSTYLSTDNEKDNLYLKV TKLFERIYSTDLGRMLLTSIVRGIPFWGGS TIDTELKVIDTNCINVIQPDGSYRSEELNL VIIGPSADIIQFECKSFGHEVLNLTNRNGYS TQYIRFSPDFTFGFEESLEVDTNPLLGAGK FATDPAVTLAHLIHAGHRLYGIAINPNR VFKVNTNAYYEMSGLEVSFEELRTFGGH DAKFIDSLQENEFRLYYNKFKDIASTLN	5' - atgaaggaaacttggtgggaaacttggtggactgaatggtctcaaccaaaag aagaagagaaagggttcaattttgtaataaacaatttaattataaagatcctgta aatggtgttgatattgcttatataaaaaattccaaatgtaggacaaatgcaacc agtaaaagcttttaaaattcataataaaatattgggttatccagaagagata catttacaaatcctgaagaaggagatttaaatccaccaccagaagcaaaac aagttccagtttcatattatgattcaacatatttaagtacagataatgaaaaag ataattatttaaggaggattacaaaattatttgagagaattattcaactgatctt ggaagaatgttgttaacatcaatagtaagggaataccattttggggtggaa gtcataatagatacagaataaaaagttattgatactaatgttattaatgtgatac aaccagatggttagttatagatcagaagaacttaacttagtaataataggacc

TABLE 1-continued

Amino acid sequence		Nucleotide sequence
	KAKSIVGTTASLQYMKNVFKEKYLLSEDT SGKESVDKLEKDKLYKMLTEIYTEDNEVK FEKVLNRKTYLNEKAVFKINIVPKVNYTI YDGENLRNTNLAANENGQNTNINNMFT KLKNFTGLFEFYKLLCVRGIIITSKTSLDK GYNKHHHHHH (SEQ ID NO: 9)	ctcagctgatattatcacagtttgaatgtaaaagccttggacatgaagttttgaa tcttacgcgaatgggttatggctctactcaatacattagatttagccagatttt acatttgggttttgaggagtcacttgaagttgatatacaaatcctcttttaggtgca ggcaaattttctacagatccagcagtaacattagcacatgaacttatacatg ctggacatagattataggaatagcaatttaatacaaatagggtttttaaagta aataactaatgcctattatgaaatgagtggttagaagtaagctttgaggaaact tagaactatttgggggacatgatgcaaaagtttatagatagtttacaggaaaaac gaatttcgtctattattataataagtttaagatatagcaagtagacttaata aagctaaatcaatagtaggtactactgcttcattacagtataatgaaaaatggtt ttaagagaaaatattcctctatctgaagatacatctggaaaaatttcggtagata aattaaaaatttgataagttatacaaaatgttaacagagatttacacagaggat aattttgttaagttttttaagtagttaaagaaacacatttgaatttttgataaa gccgtattttaagataaaatagtagcttaaggttaattacacaaatattatgatg atttaatttaagaaatacaaaatttagcagcaaaacttttaaggtcaaaatcacaga aatttaataatagaaattttactaaactaaaaaatttactggattgtttgaaattta taagttgctatgtgtaagaggataataaacttctaaaactaaatcattagataa aggatacaataagcatcaccatcaccatcactaa-3' (SEQ ID NO: 10)
PEP1- (AM) BTX-L- His (<i>E. coli</i> codon opti- mization)	MKETWWETWTEWSQPKKRKVQFVN KQFNYKDPVNGVDIAYIKIPNVGQMOPV KAFKIHNKIWIPIPERDTFTNPEEGDLNPPP EAKQVPVSYDYSTYSLTDNEKDNYLKGV TKLFERIYSTDLGRMLLTSIVRGIPFWGGS TIDTELKVIDTNCINVIQPDGYSRSEELNL VIIGPSADIIQFECKSFGHEVLNLTNRNGYGS TQYIRFSPDFTFGFEESLEVDNPLLGAGK FATDPAVTLAHLIHAHRLYGIINPNR VFKVNTNAYEYEMSGLEVSFEELRTFGGH DAKFIDSLQENEFRLYYYNKFKDIASLTN KAKSIVGTTASLQYMKNVFKEKYLLSEDT SGKFSVDKLEKDKLYKMLTEIYTEDNFVK FFKVLNRKTYLNFDAVFKINIVPKVNYTI YDGFNLRNTNLAANFNGQNTNINNMFT KLKNFTGLFEFYKLLCVRGHTSKTSKSLDK GYNKHHHHHH (SEQ ID NO: 11)	5'- catatgaaggaacttggtgggaaacttggtggaccgaatgggtctcaacca aagaagaagcgcaaggttcaatttggttaataaacaatttaattataaagatcc agtaaatgggtgctgcacattgcttatatacaaaattccaatgtaggccaaatgc aaccagtaaaagcttttaaaattcataaaaaatctgggttatccagaacgc gatacctttaccaatccggaagaagtgatctgaatccaccaccagaagca aaacaagttccagttagctattatgatagcacctatctgagcaccgataatga aaaagataaattatctgaagggcggttaccaaaactggttgagcgcatttatagca ctgatctgggtcgcatgctgctgaccagcatcgtagcgggtatcccattttg gggtggtagcaccatcgataccgaactgaaagtatttgataactaattgtatta atgtgatccaaccagatggttagctatcgtagcgaagaactgaatctggtaa tcatcggtccgagcgctgatattatccagtttgaatgtaaaagctttgggtcat gaagttctgaatctgaccgtaaatgggtatggcagcaccacatcacatcgctt tagcccagattttacctttgggttagaggagagcctggaagtgtgataccaatc cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggga catgaactgatacatgctggccatcgctgtatggtagcgaattaatccaa atcgcggtttttaagtaaaatccaatgcctattatgaaatgagcgggtctggaa gtaagctttgaggaaactgcgcacctttgggtggtcatgatgcaaaagtttatcga tagcctgcaggaaaaacgaatttcgctctgtattatataataagtttaagatat cgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc tgtagtatatgaaaaatggttttaagagaaaatattctgctgctgtagaatatcct ctggcaaattagcgtagataaaactgaaatttgataagctgtacaaaatgctg accgagatttacaccgaggataaatttggttaagttttttaagtagtgaaccgc aaaacctatctgaattttgataaaagcgtatttaagatcaatatcgtaccgaa ggtaaaattacaccatctatgatgggttttaactcgcgcaatccaatctggcag caaaacttaatgggtcaaaataccgaaataataaatgaattttaccaaaactga aaaattttacgggtctggtttgaattctataagctgctgTGTgtacgggtat catcaccagcaaaacaaaagcctggataaaggctacaataagcatcacc atcaccatcactaataactcgag-3' (SEQ ID NO: 12)
PEP1- (AM) BTX-L (C430G) - His	MKETWWETWTEWSQPKKRKVQFVN KQFNYKDPVNGVDIAYIKIPNVGQMOPV KAFKIHNKIWIPIPERDTFTNPEEGDLNPPP EAKQVPVSYDYSTYSLTDNEKDNYLKGV TKLFERIYSTDLGRMLLTSIVRGIPFWGGS TIDTELKVIDTNCINVIQPDGYSRSEELNL VIIGPSADIIQFECKSFGHEVLNLTNRNGYGS TQYIRFSPDFTFGFEESLEVDNPLLGAGK FATDPAVTLAHLIHAHRLYGIINPNR VFKVNTNAYEYEMSGLEVSFEELRTFGGH DAKFIDSLQENEFRLYYYNKFKDIASLTN KAKSIVGTTASLQYMKNVFKEKYLLSEDT SGKFSVDKLEKDKLYKMLTEIYTEDNFVK FFKVLNRKTYLNFDAVFKINIVPKVNYTI YDGFNLRNTNLAANFNGQNTNINNMFT KLKNFTGLFEFYKLLCVRGIIITSKTSKSLDK GYNKHHHHHH (SEQ ID NO: 13)	5'- catatgaaggaacttggtgggaaacttggtggaccgaatgggtctcaacca aagaagaagcgcaaggttcaatttggttaataaacaatttaattataaagatcc agtaaatgggtgctgcacattgcttatatacaaaattccaatgtaggccaaatgc aaccagtaaaagcttttaaaattcataaaaaatctgggttatccagaacgc gatacctttaccaatccggaagaagtgatctgaatccaccaccagaagca aaacaagttccagttagctattatgatagcacctatctgagcaccgataatga aaaagataaattatctgaagggcggttaccaaaactggttgagcgcatttatagca ctgatctgggtcgcatgctgctgaccagcatcgtagcgggtatcccattttg gggtggtagcaccatcgataccgaactgaaagtatttgataactaattgtatta atgtgatccaaccagatggttagctatcgtagcgaagaactgaatctggtaa tcatcggtccgagcgctgatattatccagtagaatgtaaaagctttgggtcat gaagttctgaatctgaccgtaaatgggtatggcagcaccacatcacatcgctt tagcccagatatacctaggattgaggagagcctggaagtgtatccaatc cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggga catgaactgatacatgctggccatcgctgtatggtagcgaattaatccaa atcgcggtttttaagtaaaatccaatgcctattatgaaatgagcgggtctggaa gtaagctttgaggaaactgcgcacctttgggtggtcatgatgcaaaagtttatcga tagcctgcaggaaaaacgaatttcgctctgtattatataataagtttaagatat cgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc tgtagtatatgaaaaatggttttaagagaaaatattctgctgctgtagaatatcct ctggcaaattagcgtagataaaactgaaatttgataagctgtacaaaatgctg accgagatttacaccgaggataaatttggttaagttttttaagtagtgaaccgc aaaacctatctgaattttgataaaagcgtatttaagatcaatatcgtaccgaa

TABLE 1-continued

Amino acid sequence	Nucleotide sequence
PEP1- (AM) BTX-L (C430A) - His MKETWWETWWTEWSQPKKKRVQFVN KQFNYKDPVNGVDIAYIKIPNVGQMOPV KAFKIHNKIWIPIERDTFTNPEEGDLNPPP EAKQVPVSYDSTYSLTDNEKDNYLKGV TKLFERIYSTDLGRMLLTSIVRGIPFWGGS TIDTELKVIDTNCINVIQPDGSYRSEELNL VIIGPSADIIQFECKSFGHEVLNLTNRNGYGS TQYIRFSPDFTGFEESLEVDTNPLLGAGK FATDPAVTLAHELIIHAGHRLYGIANPNR VPKVNTNAYYEMSGLEVSFEELRTFGGH DAKFIDSLQENEFRLYYNKFKDIASTLN KAKSIVGTASLQYMKNVFKEKYLLSEDT SGKESVDKLKEDKLYKMLTEIYTEDNEVK FEKVLNRKTYLNEDEKAVFKINIVPKVNYTI YDGENLRNTNLAANENGQNTENNMMFT KLNFTGLPEFYKLLAVRGHTSKTKSLDK GYNKHHHHHH (SEQ ID NO: 15)	ggtaaattacaccatctatgatgggttttaatctgcgcaataccaatctggcag caaactttaatgggtcaaaataccgaaattataaatatgaattttaccaaaactga aaaattttaccgggtctgtttgaattctataagctgctgGCGgtacgcgggtat catcaccagcaaaacaaaagcctggataaaggctacaataagcatcacc atcaccatcactaataaactcgag-3' (SEQ ID NO: 14) 5' - catatgaaggaaacttggtgggaaacttggtggaccgaatgggtctcaacca aagaagaagcgcaaggttcaatttggttaataaacaattttaattataaagatcc agtaaatgggtgctgcacattgcttatatcaaaattccaaatgtaggccaaatgc aaccagtaaaagctttttaaattcataataaaaactctgggttatccagaacgc gatacctttaccaatccggaagaaggtgatctgaatccaccaccagaagca aaacaagttccagttagctattatgatagcacctatctgagcaccgataatga aaaagataaattatctgaaggcggttaccaaaactggttgagcgcatttatagca ctgatctgggtcgcatgctgctgaccagcatcgtacgcgggtatcccattttg ggggtggtagcaccatcgataccgaactgaaagttattgatactaatttgattat atgtgatcccaaccagatggttagctatcgagcgaagaactgaaatctgggtaa tcatcggtccgagcgctgatattatccagtagaatgtaaaagctttgggtcat gaagttctgaatctgaccgtaaatgggttatggcagcaccgaatcacatctcgctt tagccagatatacctaggattgaggagagcctggaagtgtgataccaatc cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggca catgaactgatacatgctggccatcgctgtatgggtatcgcaattaatccaa atcgcgctttttaagtaataatccaatgcctattatgaatgagcgggtctggaa gtaagctttgaggaaactgcgcacctttgggtggatgatgcaaaagtttatcgat tagcctgcaggaaaaacgaattttcgctgctgattatataataagtttaaaagat atcgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc tgcagtatatgaaaaatggttttaagagaaaatatactgctgctggaagatcacct ctggcaaattagcgtagataaaactgaaatttgataagctgtacaaaatgctg accgagattacaccgaggataaattttggttaagtttttaagtagtaaccaccgc aaaacctatctgaattttgataaagcggtaatttaagatcaatatcgtagccgaa ggtaaattacaccatctatgatgggttttaatctgcgcaataccaatctggcag caaactttaatgggtcaaaataccgaaattataaatatgaattttaccaaaactga aaaattttaccgggtctgtttgaattctataagctgctgGCGgtacgcgggtat catcaccagcaaaacaaaagcctggataaaggctacaataagcatcacc atcaccatcactaataaactcgag-3' (SEQ ID NO: 16)
PEP1- (AM) BTX-L (C430S) - His MKETWWETWWTEWSQPKKKRVQFVN KQFNYKDPVNGVDIAYIKIPNVGQMOPV KAFKIHNKIWIPIERDTFTNPEEGDLNPPP EAKQVPVSYDSTYSLTDNEKDNYLKGV TKLFERIYSTDLGRMLLTSIVRGIPFWGGS TIDTELKVIDTNCINVIQPDGSYRSEELNL VIIGPSADIIQFECKSFGHEVLNLTNRNGYGS TQYIRFSPDFTGFEESLEVDTNPLLGAGK FATDPAVTLAHELIIHAGHRLYGIANPNR VPKVNTNAYYEMSGLEVSFEELRTFGGH DAKFIDSLQENEFRLYYNKFKDIASTLN KAKSIVGTASLQYMKNVFKEKYLLSEDT SGKESVDKLKEDKLYKMLTEIYTEDNEVK FEKVLNRKTYLNEDEKAVFKINIVPKVNYTI YDGENLRNTNLAANENGQNTENNMMFT KLNFTGLPEFYKLLAVRGHTSKTKSLDK GYNKHHHHHH (SEQ ID NO: 17)	5' - catatgaaggaaacttggtgggaaacttggtggaccgaatgggtctcaacca aagaagaagcgcaaggttcaatttggttaataaacaattttaattataaagatcc agtaaatgggtgctgcacattgcttatatcaaaattccaaatgtaggccaaatgc aaccagtaaaagctttttaaattcataataaaaactctgggttatccagaacgc gatacctttaccaatccggaagaaggtgatctgaatccaccaccagaagca aaacaagttccagttagctattatgatagcacctatctgagcaccgataatga aaaagataaattatctgaaggcggttaccaaaactggttgagcgcatttatagca ctgatctgggtcgcatgctgctgaccagcatcgtacgcgggtatcccattttg ggggtggtagcaccatcgataccgaactgaaagttattgatactaatttgattat atgtgatcccaaccagatggttagctatcgagcgaagaactgaaatctgggtaa tcatcggtccgagcgctgatattatccagtagaatgtaaaagctttgggtcat gaagttctgaatctgaccgtaaatgggttatggcagcaccgaatcacatctcgctt tagccagattttaccctaggattgaggagagcctggaagtgtgataccaatc cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggca catgaactgatacatgctggccatcgctgtatgggtatcgcaattaatccaa atcgcgctttttaagtaataatccaatgcctattatgaatgagcgggtctggaa gtaagctttgaggaaactgcgcacctttgggtggatgatgcaaaagtttatcgat tagcctgcaggaaaaacgaattttcgctgctgattatataataagttttaaagat atcgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc tgcagtatatgaaaaatggttttaagagaaaatatactgctgctggaagatcacct ctggcaaattagcgtagataaaactgaaatttgataagctgtacaaaatgctg accgagattacaccgaggataaattttggttaagtttttaagtagtaaccaccgc aaaacctatctgaattttgataaagcggtaatttaagatcaatatcgtagccgaa ggtaaattacaccatctatgatgggttttaatctgcgcaataccaatctggcag caaactttaatgggtcaaaataccgaaattataaatatgaattttaccaaaactga aaaattttaccgggtctgtttgaattctataagctgctgAGCgtacgcgggtat catcaccagcaaaacaaaagcctggataaaggctacaataagcatcacc atcaccatcactaataaactcgag-3' (SEQ ID NO: 18)
PEP1- (AM) BTX-L (C430G) - BFGFRP- His MKETWWETWWTEWSQPKKKRVQFVN KQFNYKDPVNGVDIAYIKIPNVGQMOPV KAFKIHNKIWIPIERDTFTNPEEGDLNPPP EAKQVPVSYDSTYSLTDNEKDNYLKGV TKLFERIYSTDLGRMLLTSIVRGIPFWGGS TIDTELKVIDTNCINVIQPDGSYRSEELNL VIIGPSADIIQFECKSFGHEVLNLTNRNGYGS TQYIRFSPDFTGFEESLEVDTNPLLGAGK FATDPAVTLAHELIIHAGHRLYGIANPNR VPKVNTNAYYEMSGLEVSFEELRTFGGH	5' - catatgaaggaaacttggtgggaaacttggtggaccgaatgggtctcaacca aagaagaagcgcaaggttcaatttggttaataaacaattttaattataaagatcc agtaaatgggtgctgcacattgcttatatcaaaattccaaatgtaggccaaatgc aaccagtaaaagctttttaaattcataataaaaactctgggttatccagaacgc gatacctttaccaatccggaagaaggtgatctgaatccaccaccagaagca aaacaagttccagttagctattatgatagcacctatctgagcaccgataatga aaaagataaattatctgaaggcggttaccaaaactggttgagcgcatttatagca ctgatctgggtcgcatgctgctgaccagcatcgtacgcgggtatcccattttg ggggtggtagcaccatcgataccgaactgaaagttattgatactaatttgattat

TABLE 1-continued

Amino acid sequence	Nucleotide sequence
DAKFIDSLQENEFRLYYYNKFKDIASLTN KAKSIVGTTASLQYMKNVFKEKYLSEDT SGKFSVDKLFKDKLYKMLTEIYTEDNFVK FFKVLNRKTYLNFDAVKINIVPKVNYTI YDGFNLRLNTLANFNGQNTTEINNMFNT KLNFTGLFEFYKLLGVRGIITSKTKSLDK GYNKYTRSRKYXSWYHHHHHH (SEQ ID NO: 19) (X stands for S or T.)	atgtgatccaaccagatggttagctatcgagcgaagaactgaatctggtaa tcatcggtccgagcgctgatattatccagtttgaatgtaaaagattgggtcat gaagttctgaatctgaccctaatggttatggcagcaccacatcatcgctt tagcccagattttacctttggattgaggagagcctggaagtgtgataccaatc cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggca catgaactgatacatgctggccatcgctgtatggtatcgcaattaatccaa atcgcggtttttaagtaaataccaatgcctattatgaaatgagcggtctggaa gtaagctttgaggaaactgcgcacctttggtggtcatgatgcaaaagtattatcga tagcctgcaggaaaacgaatttcgctctgtattattataataagtttaagatat cgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc tgcaagtatatgaaaaatgtttttaagagaaaatctgctgtctgaagatacct ctggcaaattagcgtagataaaactgaaatttgataagctgtacaaaatgctg accgagatttacaccgaggataaattttgttaagtttttaaaagtactgaaccgc aaaacctatctgaaattttgataaagccgtatttaagatcaatcatcgtaaccga ggtaaaattacaccatctatgatggtttaactctgcgcaataccaatctggcag caaactttaatgggtcaaaataccgaaattataaatatgaattttaccaaaactga aaaattttaccggtctggtttgaattctataagctgctg GCGg TACCGG GTATCATCACCAGCAAACCAAAGCTGGA TAAAGGCTACAATAAGACCTATCGCAGCCGC AAATAT ASC AGCTGGTATCATCACCATCACCA TCACTAATAACTCGAG-3' (SEQ ID NO: 20) (ASC stands for <u>AGC</u> or <u>ACC</u> .)
PEP1- MKETWWETWWTEWSQPKKKRKVQFVN (AM) KQFNKYKDPVNGVDIAYIKIPNVGMQPV BTX-L KAFKIHNKIWIPIERDTFTNPEEGLNPP (C430A) - EAKQVPVSYDYSTYLSSTDNEKDNLYKGV BFGFRP- TKLFERIYSTDLGRMLLTSIVRGIPFWGGS His TIDTELKVIDTNCINVIQPDGYSRSEELNL VIIGPSADIIQFECKSFGHEVLNLTRNGYGS TQYIRFSPDFTFGFEESLEVDNPLLGAGK FATDPAVTLAHELIIHAGHRLYGIAINPNR VKVNTNAYYEMSGLEVSFEELRTFGGH DAKFIDSLQENEFRLYYYNKFKDIASLTN KAKSIVGTTASLQYMKNVFKEKYLSEDT SGKFSVDKLFKDKLYKMLTEIYTEDNFVK FFKVLNRKTYLNFDAVKINIVPKVNYTI YDGFNLRLNTLANFNGQNTTEINNMFNT KLNFTGLFEFYKLLAVRGIITSKTKSLDK GYNKYTRSRKYXSWYHHHHHH (SEQ ID NO: 21) (X stands for S or T.)	5'- catatgaaggaaacttggtgggaaacttggtggaccgaatggtctcaacca aagaagaagcgcaaggttcaatttggttaataaacaatttaattataaagatcc agtaaatggtgtcgacattgcttatatacaaaatccaaatgtaggccaaatgc aaccagtaaaagcttttaaaattcataaaaaatctgggttatccagaacgc gatacctttaccaatccggaagaaggtgatctgaatccaccaccagaagca aaacaagtccagttagctattatgatagcacctatctgagcaccgataatga aaaagataaattatctgaagggcggttaccaaactggttgagcgcatattatagca ctgatctgggtcgcatgctgctgaccagcatcgtagcggtatccattttg gggtggttagcaccatcgataccgaactgaaagtatttgataactaattgtatta atgtgatccaaccagatggttagctatcgagcgaagaactgaatctggtaa tcatcggtccgagcgctgatattatccagtttgaatgtaaaagattgggtcat gaagttctgaatctgaccctaatggttatggcagcaccacatcatcgctt tagcccagattttacctttggattgaggagagcctggaagtgtgataccaatc cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggca catgaactgatacatgctggccatcgctgtatggtatcgcaattaatccaa atcgcggtttttaagtaaataccaatgcctattatgaaatgagcggtctggaa gtaagctttgaggaaactgcgcacctttggtggtcatgatgcaaaagtattatcga tagcctgcaggaaaacgaatttcgctctgtattattataataagtttaagatat cgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc tgcaagtatatgaaaaatgtttttaagagaaaatctgctgtctgaagatacct ctggcaaattagcgtagataaaactgaaatttgataagctgtacaaaatgctg accgagatttacaccgaggataaattttgttaagtttttaaaagtactgaaccgc aaaacctatctgaaattttgataaagccgtatttaagatcaatcatcgtaaccga ggtaaaattacaccatctatgatggtttaactctgcgcaataccaatctggcag caaactttaatgggtcaaaataccgaaattataaatatgaattttaccaaaactga aaaattttaccggtctggtttgaattctataagctgctg GCGg TACCGG GTATCATCACCAGCAAACCAAAGCTGGA TAAAGGCTACAATAAGACCTATCGCAGCCGC AAATAT ASC AGCTGGTATCATCACCATCACCA TCACTAATAACTCGAG-3' (SEQ ID NO: 22) (ASC stands for <u>AGC</u> or <u>ACC</u> .)
PEP1- MKETWWETWWTEWSQPKKKRKVQFVN (AM) KQFNKYKDPVNGVDIAYIKIPNVGMQPV BTX-L KAFKIHNKIWIPIERDTFTNPEEGLNPP (C430S) - EAKQVPVSYDYSTYLSSTDNEKDNLYKGV BFGFRP- TKLFERIYSTDLGRMLLTSIVRGIPFWGGS His TIDTELKVIDTNCINVIQPDGYSRSEELNL VIIGPSADIIQFECKSFGHEVLNLTRNGYGS TQYIRFSPDFTFGFEESLEVDNPLLGAGK FATDPAVTLAHELIIHAGHRLYGIAINPNR VKVNTNAYYEMSGLEVSFEELRTFGGH DAKFIDSLQENEFRLYYYNKFKDIASLTN KAKSIVGTTASLQYMKNVFKEKYLSEDT SGKFSVDKLFKDKLYKMLTEIYTEDNFVK FFKVLNRKTYLNFDAVKINIVPKVNYTI YDGFNLRLNTLANFNGQNTTEINNMFNT KLNFTGLFEFYKLLSVRGIITSKTKSLDK GYNKYTRSRKYXSWYHHHHHH (SEQ ID NO: 23) (X stands for S or T.)	5'- catatgaaggaaacttggtgggaaacttggtggaccgaatggtctcaacca aagaagaagcgcaaggttcaatttggttaataaacaatttaattataaagatcc agtaaatggtgtcgacattgcttatatacaaaatccaaatgtaggccaaatgc aaccagtaaaagcttttaaaattcataaaaaatctgggttatccagaacgc gatacctttaccaatccggaagaaggtgatctgaatccaccaccagaagca aaacaagtccagttagctattatgatagcacctatctgagcaccgataatga aaaagataaattatctgaagggcggttaccaaactggttgagcgcatattatagca ctgatctgggtcgcatgctgctgaccagcatcgtagcggtatccattttg gggtggttagcaccatcgataccgaactgaaagtatttgataactaattgtatta atgtgatccaaccagatggttagctatcgagcgaagaactgaatctggtaa tcatcggtccgagcgctgatattatccagtttgaatgtaaaagattgggtcat gaagttctgaatctgaccctaatggttatggcagcaccacatcatcgctt tagcccagattttacctttggattgaggagagcctggaagtgtgataccaatc cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggca catgaactgatacatgctggccatcgctgtatggtatcgcaattaatccaa atcgcggtttttaagtaaataccaatgcctattatgaaatgagcggtctggaa gtaagctttgaggaaactgcgcacctttggtggtcatgatgcaaaagtattatcga tagcctgcaggaaaacgaatttcgctctgtattattataataagtttaagatat cgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc tgcaagtatatgaaaaatgtttttaagagaaaatctgctgtctgaagatacct ctggcaaattagcgtagataaaactgaaatttgataagctgtacaaaatgctg

TABLE 1-continued

Amino acid sequence	Nucleotide sequence
	accgagatttacaccgaggataaattttgttaagttttttaaagtactgaaccgc aaaacctatctgaatttttgataaagccgtattttaagatcaatatcgtaccgaa ggtaaattacaccatctatgatgggttttaattctgcgcaataccaatctggcag caaactttaatgggtcaaaataccgaaataataatatgaattttaccaaaactga aaaattttaccgggtctgtttgtaattctataagctgctg AGCgTACGCG GTATCATCACCAGCAAACCAAAAGCCTGGA TAAAGGCTACAATAAGACCTATCGCAGCCGC AAATAT ASC AGCTGGTATCATCACCATCACCA TCACTAATAACTCGAG-3' (SEQ ID NO: 24) (ASC stands for <u>AGC</u> or <u>ACC</u> .)
Linker	GGGG (SEQ ID NO: 25) GGTGGTGGTGGT (SEQ ID NO: 26)
Cleavage Peptide	CTGGTACCACGCGGTAGC (SEQ ID NO: 28)
BTX-L (C430G) MQFVNKQFNYKDPVNGVDIAYIKIPNVG QMQPVKAFKIHNKIWVIPERDTFTNPEEG DLNPPPEAKQVPVSYDSTYLSTDNEKDN YLKGVTKLFERIYSTDLGRMLLTSIVRGIP FWGGSTIDTELKVIDTNCINVIQPDGSYRS EELNLVIIGPSADIIQFECKSPGHEVLNLTR NGYGSTQYIRFSPDFTGFEESLEVDTNPL LGAGKFATDPAVTLAHELIHAGHRLYGIA INPNRVFKVNTNAYYEMSGLEVSFEELRT FGHDHAKFIDSLQENEFRLYYNKFKDIA STLNKAKSIVGTTASLQYMKNVFEKEYLL SEDTSGKESVDKLEKDKLYKMLTEIYTED NFVKFFKVLNRKTYLNFDAVKFKNIVPK VNYTIYDGFNLRLNTLANFNGQNTTEINN MNFTKLKNFTGLFEFYKLLGVRGIITSKT KSLDKGYNK (SEQ ID NO: 29)	5'- atgcaatttgttaataaacaatttaattataaagatccagtaaatgggtgtcgac attgcttataatcaaaattccaaatgtaggccaaatgcaaccagtaaaagcttt taaaattcataataaaaatctgggttattccagaacgcgataccattaccaatcc ggaagaagggtgatctgaatccaccaccagaagcaaaacaagttccagtta gctattatgatagcacctatctgagcaccgataaatgaaaagataattatctg aagggcgttaccaaaactggttgagcgcatttatagcactgatctgggtcgca tgctgctgaccagcatcgtacgcggatcccattttggggtggtagcaccat cgataccgaactgaaagtatttgatactaatgttattaatgtgatccaaccag atggtagctatcgcagcgaagaactgaatctggtaaatcatcgcgtccgagcg ctgatattatccagtttgaatgtaaaagctttgggtcatgaagtcttgaatctga ccgtaaatgggttatggcagacccaatacattcgctttagcccagattttacc tttgggttttgaggagagcctggaagtgtataccaatccgctgctgggtgcag gcaaatttgctaccgatccagcagtaacctggcacatgaactgatacatg ctggccatcgctgtatgggtatcgcaattaatccaaatcgctttttaaagta aataccaatgcctattatgaaatgagcggctctggaagtaagctttgaggaa tgccacaccttttggtggtcatgatgcaaaagtttatcgatagcctgcaggaaaa cgaatttcgctgtattattataataagtttaaagatatcgcaagcaccctgaa taaagctaaaagcatcgtaggtaccaccgctagcctgcagtatatgaaaaa tgtttttaaagagaaatattctgctgtctgaagatcccttggcaaatttagcgt agataaactgaaatttgataagctgtacaaaatgctgaccgagatttacacc gaggtataattttgttaagtttttaaagactgaaccgcaaaacctattctgaat ttgataaagccgtatttaagatcaaatatcgtaaccgaaggtaaattacaccatc tatgatggatttaattctgcgcaataccaatctggcagcaaacctttaatggtea aataaccgaaattataataatgaattttaccaaactgaaaaattttaccgctct gtttgaattctataagctgctg GCG ctacgcgggtatcatcaccagcaaaa ccaaaagcctggataaaggctacaataag-3' (SEQ ID NO: 30)
BTX-L (C430A) MQFVNKQFNYKDPVNGVDIAYIKIPNVG QMQPVKAFKIHNKIWVIPERDTFTNPEEG DLNPPPEAKQVPVSYDSTYLSTDNEKDN YLKGVTKLFERIYSTDLGRMLLTSIVRGIP FWGGSTIDTELKVIDTNCINVIQPDGSYRS EELNLVIIGPSADIIQFECKSPGHEVLNLTR NGYGSTQYIRFSPDFTGFEESLEVDTNPL LGAGKFATDPAVTLAHELIHAGHRLYGIA INPNRVFKVNTNAYYEMSGLEVSFEELRT FGHDHAKFIDSLQENEFRLYYNKFKDIA STLNKAKSIVGTTASLQYMKNVFEKEYLL SEDTSGKFSVDKLEKDKLYKMLTEIYTED NFVKFFKVLNRKTYLNFDAVKFKNIVPK VNYTIYDGFNLRLNTLANFNGQNTTEINN MNFTKLKNFTGLFEFYKLLAVRGIITSKT SLDKGYNK (SEQ ID NO: 31)	5'- atgcaatttgttaataaacaatttaattataaagatccagtaaatgggtgtcgac attgcttataatcaaaattccaaatgtaggccaaatgcaaccagtaaaagcttt taaaattcataataaaaatctgggttattccagaacgcgataccataccaatcc ggaagaagggtgatctgaatccaccaccagaagcaaaacaagttccagtta gctattatgatagcacctatctgagcaccgataaatgaaaagataattatctg aagggcgttaccaaaactggttgagcgcatttatagcactgatctgggtcgca tgctgctgaccagcatcgtacgcggatcccattttggggtggtagcaccat cgataccgaactgaaagtatttgatactaatgttattaatgtgatccaaccag atggtagctatcgcagcgaagaactgaatctggtaaatcatcgcgtccgagcg ctgatattatccagtttgaatgtaaaagctttgggtcatgaagtcttgaatctga ccgtaaatgggttatggcagcaccacaatacattcgctttagcccagattttacc tttgggttttgaggagagcctggaagtgtataccaatccgctgctgggtgcag gcaaatttgctaccgatccagcagtaacctggcacatgaactgatacatg ctggccatcgctgtatgggtatcgcaattaatccaaatcgctttttaaagta aataccaatgcctattatgaaatgagcggctctggaagtaagctttgaggaa tgccacaccttttggtggtcatgatgcaaaagtttatcgatagcctgcaggaaaa cgaatttcgctgtattattataataagtttaaagatatcgcaagcaccctgaa taaagctaaaagcatcgtaggtaccaccgctagcctgcagtatatgaaaaa tgtttttaaagagaaatattctgctgtctgaagatcccttggcaaatttagcgt agataaactgaaatttgataagctgtacaaaatgctgaccgagatttacacc gaggtataattttgttaagttttttaaagactgaaccgcaaaacctattctgaat ttgataaagccgtatttaagatcaaatatcgtaaccgaaggtaaattacaccatc tatgatggatttaattctgcgcaataccaatctggcagcaaacctttaatggtea aataaccgaaattataataatgaattttaccaaactgaaaaattttaccggtct gtttgaattctataagctgctg GCG ctacgcgggtatcatcaccagcaaaa ccaaaagcctggataaaggctacaataag-3' (SEQ ID NO: 32)
BTX-L (C430S) MQFVNKQFNYKDPVNGVDIAYIKIPNVG QMQPVKAFKIHNKIWVIPERDTFTNPEEG DLNPPPEAKQVPVSYDSTYLSTDNEKDN YLKGVTKLFERIYSTDLGRMLLTSIVRGIP FWGGSTIDTELKVIDTNCINVIQPDGSYRS	5'- atgcaatttgttaataaacaatttaattataaagatccagtaaatgggtgtcgac attgcttataatcaaaattccaaatgtaggccaaatgcaaccagtaaaagcttt taaaattcataataaaaatctgggttattccagaacgcgataccataccaatcc ggaagaagggtgatctgaatccaccaccagaagcaaaacaagttccagtta

TABLE 1-continued

Amino acid sequence	Nucleotide sequence
EELNLVIIGPSADIIQFECKSPGHEVLNLTR NGYGSTQYIRFSPDFTFGFEESLEVDTNPL LGAGKFATDPAVTLAHELIHAGHRLYGIA INPNRVFKVNTNAYYEMSGLEVSFEELRT FGGHDAKFIDSLQENEFRLYYNKFIDIA STLNKAKSIVGTASLQYMKNVFKEKYL SEDTSKGFVSVDKLPDKLYKMLTEIYTD NFVKFFKVLNRKTYLNFDAVKFKNIVPK VNYTIYDGNLRLNTLANFNGQNTTEINN MNFTKLKNFTGLFEFYKLLSVRGIITSKT SLDKGYNK (SEQ ID NO: 33)	gctattatgatagcacctatctgagcaccgataatgaaaaagataattatctg aagggcggtaccacaaactggttgagcgcatattatagcactgatctgggtcgca tgctgctgaccagcatcgtaacgggtatcccattttggggtggtagcaccat cgataccgaactgaaagtattgataactaattgtattaatgtgatccaaccag atggtagctatcgagcgaagaactgaatctggtaaatcatcggtccgagcg ctgatattatccagttgaatgtaaaagctttgggtcatgaagtctgaatctga cccgtaatgggtatggcagcaccacaaatattcgctttagcccagattttacc tttgggttttgaggagagcctggaagtgtgataccaatccgctgctgggtgcag gcaaatttgctaccgatccagcagtaacctggcacatgaactgatacatg ctggccatcgctgtatggatcgcaattaatccaaatcgctttttaaagta aataccaatgacctattatgaaatgagcggtctggaagtaagctttgaggaa tggcgacctttgggtggtcatgatgcaaagtttatcgatagcctgcaggaaaa cgaaatttcgtctgtattattataaagttaaagatatcgcaagcaccctgaa taaagctaaaagcatcgtaggtaccaccgctagcctgcagtatatgaaaaa tgtttttaaagagaaatattctgctgtctgaagatacctctggcaaatttagcgt agataaactgaaatttgataagctgtacaaatgctgacccgagatttacacc gaggataaattttgtaagtttttaaagtactgaaccgcaaaacctatctgaat ttgataaagcgtatttaagatcaatatcgtaaccgaaggtaaattacaccatc tatgatggatgaatctgcgcaatccaatctggcagcaaacctttaatggta aaataccgaaattaataatgaattttaccacaaactgaaaaattttaccggtct gtttgaaattctataagctgctgAGCgtacgcggtatcatcaccagcaaaa ccaaaagcctggataaaggctacaataag-3' (SEQ ID NO: 34)

TABLE 2

Mutation	Sequence
C430G	GAATTCTATAAGCTGCTGGGCGTACGCGGTATCATCACCAGCAAAACCAAAG CCTGGATAAAGGCTACAATAAGCATCACCATCACCATCACTAATAACTCGAG (SEQ ID NO: 35)
C430A	GAATTCTATAAGCTGCTGGGCGTACGCGGTATCATCACCAGCAAAACCAAAG CCTGGATAAAGGCTACAATAAGCATCACCATCACCATCACTAATAACTCGAG (SEQ ID NO: 36)
C430S	GAATTCTATAAGCTGCTGAGCGTACGCGGTATCATCACCAGCAAAACCAAAG CCTGGATAAAGGCTACAATAAGCATCACCATCACCATCACTAATAACTCGAG (SEQ ID NO: 37)
C430G-BFGRP	GAATTCTATAAGCTGCTGGGCGTACGCGGTATCATCACCAGCAAAACCAAAG CCTGGATAAAGGCTACAATAAGACCTATCGCAGCCGCAAAATATASCAGCTGGT ATCATCACCATCACCATCACTAATAACTCGAG-3' (SEQ ID NO: 38) (ASC stands for AGC or ACC.)
C430A-BFGRP	GAATTCTATAAGCTGCTGGGCGTACGCGGTATCATCACCAGCAAAACCAAAG CCTGGATAAAGGCTACAATAAGACCTATCGCAGCCGCAAAATATASCAGCTGGT ATCATCACCATCACCATCACTAATAACTCGAG-3' (SEQ ID NO: 39) (ASC stands for AGC or ACC.)
C430S-BFGRP	GAATTCTATAAGCTGCTGAGCGTACGCGGTATCATCACCAGCAAAACCAAAG CCTGGATAAAGGCTACAATAAGACCTATCGCAGCCGCAAAATATASCAGCTGGT ATCATCACCATCACCATCACTAATAACTCGAG-3' (SEQ ID NO: 40) (ASC stands for AGC or ACC.)
Belt	GAATTCTATAAGCTGCTGTGTGTACGCGGTATCATCACCAGCAAAACCAAAGCCT GGATAAAGGCTACAATAAGGCGCTGAACGATCTGTGCCATCACCATCACCATCAC TAATAACTCGAG (SEQ ID NO: 41)

Example 2. Expression of Non-Toxic Protease in *E. coli*

Each of the expression vectors for *E. coli* constructed in Example 1 was transformed into an *E. coli* BL21 (DE3) strain as a host for gene expression to construct recombinant *E. coli* strains. LB culture medium was inoculated with the ampicillin-resistant colonies and culturing was performed with shaking at 37° C. When the absorbance at 600 nm reached 0.6 to 0.7, 0.5 mM IPTG was added to the culture medium, and then expression of the non-toxic protease was induced at 18 to 25° C. depending on the conditions. Next,

the cells were cultured for 6 to 12 hours, and recombinant *E. coli* cells were harvested by centrifugation (at 3,000 rpm for 20 minutes).

The harvested *E. coli* cells were suspended in a 25% sucrose solution (containing 50 mM Tris HCl, pH 7.8, 0.1 mM EDTA), and 0.2 mg/ml of lysozyme was added thereto, followed by incubation at 4° C. for 1.5 hours. Next, a 1.5-fold volume of lysis buffer (20 mM Tris HCl, pH7.8, 0.2 M NaCl, 1% DCA, 1.6% NP-40 (or 1% SDS), 2 mM EDTA, 0.5 mM DTT) was added to the suspension, and then the cells were lysed using ultrasound at 4° C. The protein was quantified and loaded on SDS-PAGE gel, followed by Coomassie blue staining.

As a result, as shown in FIG. 7, it could be confirmed that the wild-type and mutated non-toxic proteases were effectively expressed by induction with IPTG. As shown in FIG. 8, it was confirmed that the non-toxic protease appeared only in the total cell lysate and the pellet, suggesting that it was mostly present in an insoluble form. In addition, as a result of subjecting the wild-type non-toxic protease to SDS-PAGE under non-reducing conditions and reducing conditions, it could be confirmed that the wild-type non-toxic protease existed in the form of high-molecular-weight protease under the non-reducing conditions, suggesting that disulfide bonds affect the formation of aggregates of the wild-type non-toxic protease (FIG. 9). Meanwhile, when the wild-type and mutated non-toxic proteases were codon-optimized to be suitable for expression in *E. coli*, the expression levels thereof could be significantly increased (FIGS. 10 and 11).

Example 3. Purification of Non-Toxic Protease

To purify the non-toxic proteases, each of the recombinant *E. coli* strains was cultured according to the method of Example 2, and then the *E. coli* cells were harvested by centrifugation (at 3,000 rpm for 20 minutes). 1 g of the harvested cells were suspended in a cell suspending solution (50 mM Tris HCl, pH 7.8, 0.1 mM EDTA, 25% sucrose), and 0.2 mg/ml of lysozyme was added thereto, followed by incubation at 4° C. for 1.5 hours. Then, a 1.5-fold volume (3 ml) of lysis buffer (20 mM Tris HCl, pH 7.8, 0.2 M NaCl, 1% DCA, 1.6% NP-40 (or 1% SDS), 2 mM EDTA, 0.5 mM DTT) was added to the suspension, and the cells were disrupted using an ultrasonic disruptor until the viscosity of the cell lysate disappeared when viewed with the naked eye. The cell lysate was separated into a soluble fraction and an insoluble fraction by centrifugation at 12,000 rpm and 4° C. for 15 minutes.

As a result of observing the fractions by Coomassie blue staining or Western blotting after SDS-PAGE, it was confirmed that a significant portion of each of the wild-type and mutated non-toxic proteases was present in the insoluble fractions.

The non-toxic protease present in the soluble fraction was loaded onto a Ni²⁺-IDA affinity column (iminodiacetic acid). The column was washed with a 10-fold volume of binding buffer (50 mM potassium phosphate buffer (pH 8.0)+300 mM NaCl+1 mM PMSF+1 mM β-mercaptoethanol) and a 6-fold volume of washing buffer (50 mM potassium phosphate buffer (pH 8.0)+300 mM NaCl+1 mM PMSF+1 mM β-mercaptoethanol+10 mM imidazole), and then eluted stepwise with elution buffers (50 mM potassium phosphate buffer (pH 8.0)+300 mM NaCl+1 mM PMSF+1 mM β-mercaptoethanol) containing 200, 300 or 500 mM imidazole.

Meanwhile, after the soluble fraction was isolated, the non-toxic protease precipitate of the insoluble fraction was washed three times with 5 ml of the same volume as the supernatant (washing buffer, 2 to 4 M urea, 0.5% Triton X100, mM EDTA, 1 mM DTT) and the supernatant was removed. Meanwhile, after the soluble fraction was isolated, the non-toxic protease precipitate of the insoluble fraction was washed three times with 5 ml (the same volume as that of the supernatant) of washing buffer (2 to 4 M urea, 0.5% Triton X100, 1 mM EDTA, 1 mM DTT), and then the supernatant was removed. The recovered precipitate was denatured with 1 ml of urea solution (8 M urea, 20 mM Tris HCl, pH 7.8, 20 μM DTT), and then transferred to a dialysis membrane (molecular weight cutoff: 12,000, Sigma, cat no: D-0530, USA) and subjected to refolding by dialysis three

times with a volume 100 times the volume of the urea solution of dialysis buffer (0.08 mM NaCl, 20 mM Tris HCl, pH 7.8, 0.03% tween 20), and 10 to 20 μM ZnCl₂ was added thereto as needed. The resulting material was loaded onto a Ni²⁺-IDA affinity column (iminodiacetic acid), and the column was washed stepwise with a 10-fold volume of binding buffer (50 mM potassium phosphate buffer (pH 8.0)+300 mM NaCl+1 mM PMSF+1 mM β-mercaptoethanol) and a 6-fold volume of washing buffer (50 mM potassium phosphate buffer (pH 8.0)+300 mM NaCl+1 mM PMSF+1 mM β-mercaptoethanol+10 mM imidazole), and then eluted stepwise with elution buffers (50 mM potassium phosphate buffer (pH 8.0)+300 mM NaCl+1 mM PMSF+1 mM β-mercaptoethanol) containing each of 200, 300 and 500 mM imidazole. The fractions were combined and desalted with a PD-10 column (GE Healthcare, USA). Protein concentration was measured by BCA protein assay using bovine serum albumin as a standard.

As a result, as shown in FIG. 14, it could be confirmed that the wild-type and mutated non-toxic proteases and fusion non-toxic proteases were effectively purified.

Example 4. Analysis of Folding/Refolding Efficiencies of Non-Toxic Proteases

As a result of performing the purification and refolding process on the insoluble fraction in Example 3, as shown in FIG. 15, it was confirmed that, in the case of the wild-type protease, aggregation still appeared after the refolding performed by dialysis, but in the case of the mutated non-toxic protease or the mutated fusion non-toxic protease, aggregation hardly appeared after refolding.

Meanwhile, after centrifugation at 12,000 rpm for 15 minutes, the content (%) of the non-toxic protease in each of the soluble fraction and the insoluble fraction (pellet fraction, aggregation) was analyzed by Coomassie blue staining after SDS-PAGE. As a result, as shown in FIG. 16, it was confirmed that about 60 to 90% of each of the wild-type and mutated non-toxic proteases was present in the insoluble fraction, suggesting that each of the non-toxic proteases was mostly present in the insoluble fraction rather than the soluble fraction. However, the tendency of the non-toxic-protease to be recovered through the refolding process was completely different between the wild-type non-toxic protease and the mutated non-toxic protease. That is, the wild-type non-toxic-protease was mostly present in a precipitated inclusion body state even after the refolding process, and thus was impossible to recover, whereas almost all of the mutated non-toxic-protease could be refolded and recovered.

Example 5. Analysis of Enzymatic Activity of Non-Toxic Protease

In order to analyze the enzymatic activities of the non-toxic proteases and variants thereof according to the present invention, a substrate for analyzing the cleavage by the non-toxic protease was synthesized. As the substrate, the ANNEXIN V protein fused to the C-terminus of SNAP25 was used for expression (Table 3), and a nucleotide sequence encoding the SNAP25-ANNEXIN V fusion protein was cloned into a pET22b (+) expression vector which was then transformed into an *E. coli* BL21 (DE3) strain. The transformed *E. coli* strain was cultured with shaking in LB medium at 37° C., and when it reached an O.D₆₀₀ of 0.6 to 0.7, 0.5 to 1 mM IPTG was added thereto, and the strain was cultured for 6 hours to induce protein expression. After the

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cells were disrupted with an ultrasonic disruptor, only the soluble fraction (supernatant) was recovered and purification was performed using his-6 tag contained in SNAP25-ANNEXIN V in a manner similar to the method of Example 3. As a result, as shown in FIG. 17, it was confirmed that SNAP25-ANNEXIN V was effectively expressed and purified.

In order to confirm whether all of the non-toxic proteases purified according to the present invention retain the same enzymatic activity as the wild-type non-toxic protease, cleavage of the purified non-toxic proteases was tested using the purified SNAP25-ANNEXIN V as a substrate. To this end, the prepared SNAP25-ANNEXIN V and the non-toxic

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protease were mixed together at a ratio of 20:1 (w/w) (substrate (SNAP25-ANNEXIN V): enzyme (protease)) in an enzymatic reaction solution (100 mM HEPES pH 7.4, 1 mM NaCl, 20 μ M ZnCl₂, mM DTT 2) and subjected to enzymatic reaction at 37° C. for 5 hours. As a result of analyzing the samples after completion of the reaction, it was confirmed that all of the non-toxic proteases according to the present invention cleaved 50-kDa SNAP25-ANNEXIN V to form 36-kDa and 14-kDa protein bands. This demonstrated that the mutated non-toxic proteases with improved productivity maintained their enzyme activity without changes.

TABLE 3

	Amino acid sequence	Nucleotide sequence
His-SNAP25- ANNEXIN V	MGGSHHHHHENLYFQSGGKSLKSSD AYKKAAGNNQDGVVASQPARVVDERE QMAISGGFIRRVNDARENEMDENLEQV SGII GNLRHMLDMGNEIDTQNRQIDRIM EKADSNKTRIDEANQRATKMLGSGAQV LRGTVTDFPQFDERADAETLRKAMKGL GTDEESILTLTSSRNAQRQEISAAPKTLF GRDLLDDLKSELTKFEKLIVALKMPSR LYDAYELKHALKGAGTNEKVLTEIIASR TPEELRAIKQVYEEYEGSSLEDDVVGDT GYYQRLVLVLLQANRDPDAGIDEAQVE QDAQALFQAGELKWTDEEFITIFGTRS VSHLRKVEDKYMTISGFQIETIDRETSG NLEQLLLAVKSIKIPAYLAETLYYAMK GAGTDDHTLIRVMVSRSEIDLLENIRKEFR KNFATSLYSMIKGDTSGDYKALLLLCG EDD (SEQ ID NO: 42)	5'- catATGGGCGGCAGCCATCATCATCA TCATGAAAACCTGTATTTTCAGGCTCT GGCGGCAACAAGCTGAAATCTAGCGAT GCTTACAAAAAGCCTGGGCAATAATC AGGACGGCGTGGTGGCCAGCCAGCCTGC TCGTGTAGTGGACGAACGTGACAGATG GCCATCAGCGCGGCTTCATCCGTCGTG TAACCAATGATGCCCGTGAAATGAAAT GGATGAAAACCTGGAGCAGGTGAGCGG CATCATCGGCAACCTGCGTCACATGGCC CTGGATATGGGCAATGAGATCGATACCC AGAATCGTCAGATCGACCGTATCATGGA GAAGGCTGATTCCAAACAAACCCGTATC GATGAGGCCAACCAACGTGCAACCAAG ATGCTGGGCAGCGCGCACAGGTTCTGC GTGGCACTGTGACCGACTTCCCTGGCTT TGATGAGCGTGTGATGCAGAAACCCGTG CGTAAGGCTATGAAAGGCTGGGCACCG ATGAGGAGAGCATCCTGACCTGCTGAC CTCCCGTAGCAATGCTCAGCGTCAGGAA ATCTCTGCAGCTTTTAAGACCCCTGTTGG CCGTGATCTGCTGGATGACCTGAAATCC GAACTGACCGGCAAAATTTGAAAACTGA TCGTGGCTCTGATGAAACCTTCTCGTCTG TATGATGCTTATGAACGAAACATGCCC TGAAGGGCGCTGGCACCATGAAAAAG TACTGACCGAAATCATGCTTCTCGTAC CCCTGAAGAACTGCGTGCCATCAACAA GTTTATGAAGAAGATATGGCTCTAGCC TGGAAGATGACGTGGTGGGCGACACTTC TGGCTACTACCAGCGTATGCTGGTGGTT CTGCTGCAGGCTAACCGTGACCCGTGATG CTGGCATCGATGAAGCTCAAGTTGAACA AGATGCTCAGGCTCTGTTTCAGGCTGGC GAACTGAAATGGGGCACCGATGAAGAA AAGTTTATCACCATCTTTGGCACCCGTA GCGTGTCTCATCTGAGAAAGGTGTTGA CAAGTACATGACCATCTCTGGCTTTCAA ATCGAGGAAACCATCGACCGTGAGACTT CTGGCAATCTGGAGCAACTGCTGCTGGC TGTGTGAAATCTATCCGTAGCATCCCT GCCTACCTGGCAGAGACCCTGTATTATG CTATGAAGGGCGCTGGCACCGATGATCA TACCCTGATCCGTGTATGGTTTCCCGTA GCGAGATCGATCTGTTTAAACATCCGTAA GGAGTTTCGTAAGAATTTTGCCACCTCT CTGTATTCCATGATCAAGGGCGATACCT CTGGCGACTATAAGAAAGCTCTGCTGCT GCTGTGTGGCGAAGATGACTAActcgag-3' (SEQ ID NO: 43)

Example 6. EXPRESSION OF NON-TOXIC
PROTEASE IN YEAST

In another embodiment, to express a non-toxic protease in yeast, the sequence shown in Table 4 below was subcloned by restriction enzymes (BamHI and NotI) downstream of the AOX1 promoter of pPIC9. Meanwhile, his 6-tag was introduced for purification of the non-toxic protease, and a TEV cleavage enzyme peptide sequence was added in order to remove his 6-tag after purification, thereby constructing a vector for expression in *Pichia pastoris*.

A *Pichia pastoris* strain was transformed with the vector and cultured in histidine-deficient medium. Then, the formed colonies were cultured in the methanol-assimilating yeast medium BM (buffered minimal medium: 100 mM potassium phosphate, pH 6.0 1.34% yeast nitrogen base,

Scientific, USA). Meanwhile, since a non-toxic protease with an increased molecular weight due to glycosylation is expressed in yeast, the non-target protease was aglycosylated by reaction with the enzyme PNGaseF (cat no. P0704S, New England Bio Lab USA), which cuts sugar chains from the protein, according to the manufacturer's instructions. Then, the non-target protease was subjected to SDS-PAGE and then to Western blotting with a botulinum light-chain antibody.

As a result, as shown in FIG. 19, it was confirmed that the non-toxic protease was effectively expressed in and purified from the yeast, and a multi-band, non-toxic protease with increased molecular weight due to glycosylation was identified in the yeast, and a single band of the non-toxic protease was observed after aglycosylation.

TABLE 4

	Amino acid sequence	Nucleotide sequence
SMF - (AM) BTX - L-PEP1 - TEV-His	MRFPSIFTAVLFAASSALAAPVNTTDD ETAQIPAEAVIGYSDLEGDFDVAVLPPFS NSTNNGLLFINTTIASTAAKEEGVSLEK RQFVNKQFNKYKDPVNGVDIAYIKIPNV GQMOPVKAFKIHNKIWIPIERDFTNP EEGDLNPPPEAKQVPVSYDSTYLSTD NEKDNYLKGVTKLFERIYSTDLGRMLL TSIVRGIPFWGGSTIDTELKVIDTNCINV IQPDGYSRSEELNLVIIGPSADIIQFECKS FGHEVLNLTRNGYSTQYIRFSPDFTGEG FEESLEVDTNPLLGAAGFATDPAVTLA HELIHAGHRLYGIAINPNRVFKVNTNA YYEMSGLEVSFEELRTFGGHDAKFIDS LQENEFRLYYNKFKDIASTLNKAKSI VGTASLQYMKNVFKEKYLLEDSTSG KFSVDKLFKFKLYKMLTEIYEDNFVK FFKVLNRKTYLNFDAVFKINIVPKVN YTIYDGFNLRLNTLANFNGQNTENN MNFVKLKNFTGLFEPYKLLCVRGIIITSK TKSLDKGYNKKETWWETWWTEW SQPKKKRKVENLYFQSNHHHHHH (SEQ ID NO: 44)	5'- atgagatttccatctatttttactgcagttttgtttgcagcatctttcgca ttggcagcaccagtttaacactactactgaagatgaaactgcacaaattcc agcagaagcagttatttggttactctgatttggaaaggtgattttgatgttg ctgttttggcatttttctaactctactaataacggtttgtgtttattaata ctactattgcattctattgcagcaaaaggaaggtgtttctttggaaaaaa gacagtttgtttaacaagcaattcaattacaagatccagtttaaggtgttg acattgcttatattaaaattccaaacgttggtgcagatgcagccagttaaa gctttcaagatccataaacaatttgggttatcccagagagagacactt tcactaatccagaggaaggtgatttgaaccaccaccagaagcaaaa caggttccagtttagttattatgacagtaacttatctttccactgataacgag aaagataactatttgaaggtgttactaagctgtttgaaagaatttacag tactgacttgggaagaatgttgctgacttcaattgttagaggaattccat tttgggttggaatctactattgacactgaattgaaagttatcgataactatt gtattacgttatccaaccagacggttctctacagatctgaggagcttaa cttggttattatttggtccatccgctgacatttcaatttgagtgtaagtc attcggacatgaggttttgaacctgactcgtgaatggttgcgttccactca atatattagattttccccagattttacttttgggtttcgaggaaagtttg gaggttgatactaaccatttgctgggtgcaggttaagtttgcactgatcca gcagttactcttgctcatgagctgactccatgcaggttcataagattgtagt attgcaattaatccaaacggtgtttttaaagtttaactaatgcttatatt gaaatgtcaggtttggaagttttcttgcagaggttgagaactttcgaggag acatgatgctaagttcattgactctttgcaggaaaacaggttccgtctgt actactataacaagttcaaggacatcgcatctactttgaacaaggcaa agtcatttgggtactactgcttccactgcaatatatgaagaacggttttca aggtaagtaacttgggttgcaggagataactagtggttaattctctgttga caattgaggttcgacaaattgtacaaaatgctgactgagatctatact gaagacaacttcgttaagttttttaaagttctgaacagaagacttatttg aactttgataaggctgtttttaaagattaacattgttccaaaggttaactat actatttatgacggtttcaatctgagaacactaatcttgcagctaatctc aatggtcagaatactgagattaataatgaacttcaactaagttgaaaaat tttactggtttgtttgaattttacaagttgtgtgtgttcgtggaattatt acttccaagactaaatctttggataaggggttacaacaaaaaggaaacttg tggaagacttggtggactgaatggtctcaaccaagaagaagagaa aggttgagaacctg tacttccaatccaatcatcaccatcaccatcacta a-3' (SEQ ID NO: 45)
SMF	MRFPSIFTAVLFAASSALAAPVNTTDD ETAQIPAEAVIGYSDLEGDFDVAVLPPFS NSTNNGLLFINTTIASTAAKEEGVSLEK R (SEQ ID NO: 46)	5'- atgagatttccatctatttttactgcagttttgtttgcagcatctttcgca ttggcagcaccagtttaacactactactgaagatgaaactgcacaaattcc agcagaagcagttatttggttactctgatttggaaaggtgattttgatgttg ctgttttggcatttttctaactctactaataacggtttgtgtttattaata ctactattgcattctattgcagcaaaaggaaggtgtttctttggaaaaaa a-3' (SEQ ID NO: 47)
TEV	ENLYFQSN (SEQ ID NO: 48)	5'-gagaacctgtacttccaatccaat-3' (SEQ ID NO: 49)

4x10⁻⁵% biotin) by supplying glycerol (1%) as a carbon source, and then the carbon source was replaced with methanol (0.5%), thus inducing expression of the non-toxic protease. The culture medium was recovered, and the expressed non-toxic protease was purified therefrom using His-Tag and subjected to Western blotting with a botulinum light-chain antibody (cat no. PA5-48053 (ThermoFisher

Example 7. Massive Fermentation and Purification
of Non-Toxic Protease

It was attempted to ferment and purify large amounts of the recombinant *E. coli* strains constructed in Example 2. To this end, the stock solution of each recombinant strain was inoculated into LB medium supplemented with ampicillin

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and was cultured with shaking overnight at 37° C. and 200 rpm (seed culture). 3 to 4 L of LB medium supplemented with 50 µg/ml of ampicillin was placed in a 7-L jar fermenter, and 50 ppm of an Antifoam B emulsion was added thereto. Then, 30 mL of the strain cultured overnight was inoculated into the medium and fermented at 37° C. at an impeller speed of 250 rpm under an air pressure corresponding to 2 L/min of filtered air. When the strain reached an OD₆₀₀ of 0.3, the internal temperature of the fermenter was lowered with a cooler of the fermenter and the strain was cultured. When the strain reached an OD₆₀₀ of 0.6 to 0.7, 0.5 mM IPTG was added and the strain was cultured overnight (16 to 18 hours) at 18° C.

200 ml of lysis buffer (50 mM Tris-HCl (pH 8.0), 150 mM NaCl, 10% glycerol) was added to the cell pellet (about 5 g by wet weight) of 1L of the fermentation culture medium, and the cells were completely suspended. A buffer for cell disruption (1 mM PMSF, 1 mM β-ME (beta-mercaptoethanol), 0.1% Triton-X 100) was added to the cell suspension, and the cells were sonicated five times using a Sonics Vibra-cell sonicator for 4 min at pulse on/off of 4 sec/8 sec and at power of 65%. The cell lysate was divided into a supernatant and a cell lysate precipitate by centrifugation at 4° C. at 12,000 rpm for 30 minutes.

Meanwhile, a Ni-IDA column (Workbeads™ 40 Ni-IDA, 40-650-001, Bio-Works, Sweden) was packed and the column was equilibrated with a 10-fold volume of equilibration buffer (50 mM Tris-HCl (pH 8.0), 150 mM NaCl, 10% glycerol, 0.1% TritonX-100, 1 mM PMSF, 1 mM β-ME). Ni-IDA resin was added to the centrifuged supernatant containing the soluble protein to induce a binding reaction with his-tag of the non-toxic protease at 4° C. for 2 hours, and was loaded into the column. The column was washed with a 20-fold volume of washing buffer A (50 mM Tris-HCl (pH 8.0), 150 mM NaCl, 10% glycerol, 1 mM β-ME), and then washed with a 20-fold volume of washing buffer (50 mM Tris-HCl (pH 8.0), 150 mM NaCl, 50 mM imidazole, 10% glycerol, 1 mM β-ME). The column was eluted stepwise with 1.5-fold volumes of elution buffers (100/200/300/500/700 mM imidazole, 50 mM Tris-HCl (pH 8.0), 150 mM NaCl, 10% glycerol, 1 mM β-ME). EDTA was added to each eluted fraction to a final concentration of 1 mM. The purity of each eluted fraction was analyzed by SDS-PAGE, and the high-purity eluted fractions obtained using the elution buffers containing 500 to 700 mM imidazole were combined, transferred to a dialysis membrane (molecular weight cut-off: 10-30 kDa), and dialyzed using PBS buffer to remove other chemical components, and the purified non-toxic protease was recovered. For cryopreservation, the non-toxic protease was dialyzed with PBS buffer containing 10% glycerol (FIG. 20).

As a result, as shown in FIGS. 21 to 26, it was confirmed that the wild-type non-toxic protease was recovered from the soluble fraction at a concentration of 3 to 6 mg/L, the non-toxic protease obtained by substituting the amino acid cysteine at position 430 of the wild-type non-toxic protease with serine was recovered from the soluble fraction at a concentration of 8 to 12 mg/L, and the non-toxic protease obtained by substituting the amino acid cysteine at position 430 of the wild-type non-toxic protease with alanine was recovered from the soluble fraction at a concentration of 15 to 20 mg/L, whereas the non-toxic protease obtained by substituting the amino acid cysteine at position 430 of the

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wild-type non-toxic protease with glycine was recovered from the soluble fraction at a concentration of 40 to 60 mg/L, suggesting that cysteine-to-glycine substitution at position 430 significantly increased the recovery of the non-toxic protease from the soluble fraction. It was believed that the reason why the recovery of the non-toxic protease from the soluble fraction can be enhanced as described is because cysteine-to-glycine substitution at position 430 of the non-toxic protease inhibited the formation of a bisulfide bond, the amount of the *E. coli* strain was increased through culturing at 37° C., the metabolism of the *E. coli* strain was slowed by lowering the culture temperature to 18° C. when an appropriate O.D. was reached, and expression of the protein was induced relatively slowly by adding IPTG at a decreased concentration of 0.5 mM, whereby the refolding efficiency of the non-toxic protease was increased. Meanwhile, in the case in which cysteine at position 430 was substituted with alanine or serine, a non-specific disulfide bond could be induced, and thus the efficiency of production of the non-toxic proteases from the soluble fraction was lower than that in the case in which cysteine at position 430 was substituted with glycine.

This effect of increasing the efficiency of expression purification of the mutated non-toxic protease in and from the soluble protein was higher than that for the wild-type non-toxic protease, even when the BFGFRP domain of the fibroblastic growth factor was fused to the mutated non-toxic protease. Thus, it could be seen that, even when another peptide is fused to the non-toxic peptide to the mutated non-toxic protease, the efficiency of expression and efficiency of purification of the mutated non-toxic protease is maintained. Specifically, the fusion non-toxic protease obtained by substituting cysteine at position of the non-toxic protease with glycine and fusing BFGFRP to the non-toxic protease was recovered at a concentration of 15 to 80 mg/L.

Example 8. Examination of Single Dose Toxicity of Non-Toxic Proteases

In order to confirm whether the toxicity of the non-toxic protease to be used in the present invention is reduced, the single dose toxicity thereof was evaluated in mice. An experiment was conducted using the wild-type non-toxic protease represented by SEQ ID NO: 1 of the present invention, and a "test for single dose administration of intraperitoneal administration of test substance in mice" was performed by the animal testing institution KPC (212-9, Opo-ro, Opo-eup, Gwangju-si, Gyeonggi-do, Korea).

As experimental animals, ICR mice were purchased from ORIENTBIO Inc. (Korea). The experiment was carried out using 35 6-week-old male mice, and animals that were not abnormal in appearance were brought into the breeding area and acclimatized in the animal room where the test was conducted, for 7 days. After 7 days of quarantine and acclimatization, healthy animals were selected by checking their health and suitability for testing. For drinking water, tap water was sterilized with a filter water sterilizer, irradiated with UV light, and freely fed using a drinking water bottle (250 mL) made of polycarbonate. The animals were allowed to freely access the rodent feed for laboratory animals provided by Cargill Agri Purina, Inc.

TABLE 5

Test group	Drug	Dose (mg/kg)	Volume	Number of animals	Administration route and frequency	Observation period
G1	Vehicle	n/a	0.25 mL	Five mice per group	Intraperitoneal injection, single	4 days after administration
G2	Test substance	1 µg/mouse	0.25 mL			
G4	Test substance	10 µg/mouse	0.25 mL			
G4	Test substance	40 µg/mouse	0.25 mL			
G5	Test substance	100 µg/mouse	0.25 mL			
G6	Test substance	110 µg/mouse	0.25 mL			
G7	Test substance	120 µg/mouse	0.25 mL			

As shown in Table 5 above, as the test substance, the wild-type non-toxic protease from which the heavy chain has been removed was purified according to the present invention and intraperitoneally injected into ICR mice (average body weight 34 g, n=5 for each concentration) at a dose of 1 to 120 µg/mouse. As a result, as shown in Table 6, it was confirmed that all the mice survived. Here, the dose (120 µg/mouse) for test group G7 corresponds to about 3.6 mg/kg mouse body weight. It is known that the median lethal dose (LD50) of natural botulinum toxin type A is about 0.3 ng/kg in rodent mice, and about 1 ng/kg in humans, which corresponds to a lethal dose of 1 µg for a 70-kg adult (Annu. Rev. Microbiol. 1999, 53:551-75, Eric A. Johnson, CLOSTRIDIAL TOXINS AS THERAPEUTIC AGENTS: Benefits of Nature's Most Toxic Proteins). Therefore, it was confirmed that the toxicity of the wild-type non-toxic protease used in the present invention decreased by 1.2×10^7 times compared to the reported median lethal dose (LD50) (0.3 ng/kg) of the natural botulinum toxin type A in mice, and that the mutated non-toxic protease of the present invention also had a significantly reduced median lethal dose (LD50) in mice, which is similar to that of the wild-type non-toxic protease.

TABLE 6

Summary: Incidence of mortality and daily observations									
Group	Sex	N	Mortality	Clinical observation	Days				
					1	2	3	4	5
G1 (Vehicle)	Male	5	0%	No clinical sign	5	5	5	5	5
G2 (1 µg/mouse)	Male	5	0%	No clinical sign	5	5	5	5	5
G3 (10 µg/mouse)	Male	5	0%	No clinical sign	5	5	5	5	5

TABLE 6-continued

Summary: Incidence of mortality and daily observations									
					Clinical	Days			
					observation	1	2	3	4
25	Group	Sex	N	Mortality					
	G4 (50 µg/mouse)	Male	5	0%	No clinical sign	5	5	5	5
	G5 (100 µg/mouse)	Male	5	0%	No clinical sign	5	5	5	5
	G6 (110 µg/mouse)	Male	5	0%	No clinical sign	5	5	5	5
	G7 (120 µg/mouse)	Male	5	0%	No clinical sign	5	5	5	5

Although the present invention has been described above with reference to the embodiments, it is to be understood that the present invention is not necessarily limited to the embodiments, and various modifications are possible without departing from the scope and spirit of the present invention. Accordingly, the scope of the present invention should be construed to include all embodiments falling within the scope of the claims appended hereto.

INDUSTRIAL APPLICABILITY

The present invention has advantages in that it is possible to inhibit aggregation into inclusion bodies, which generally occurs in the process of producing wild-type non-toxic protease using a recombinant microorganism, and thus it is possible to obtain active mutated non-toxic proteases from both a soluble fraction and an insoluble fraction, which are formed by disrupting the recombinant microorganism, even by a simple dialysis/refolding process alone, and accordingly, it is possible to produce non-toxic proteases in high yield.

SEQUENCE LIST FREE TEXT

An electronic file is attached.

SEQUENCE LISTING

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic Botulinum toxin Light Chain

<400> SEQUENCE: 1

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          20          25          30

Val  Lys  Ala  Phe  Lys  Ile  His  Asn  Lys  Ile  Trp  Val  Ile  Pro  Glu  Arg
          35          40          45

Asp  Thr  Phe  Thr  Asn  Pro  Glu  Glu  Gly  Asp  Leu  Asn  Pro  Pro  Pro  Glu
 50          55          60

Ala  Lys  Gln  Val  Pro  Val  Ser  Tyr  Tyr  Asp  Ser  Thr  Tyr  Leu  Ser  Thr
 65          70          75          80

Asp  Asn  Glu  Lys  Asp  Asn  Tyr  Leu  Lys  Gly  Val  Thr  Lys  Leu  Phe  Glu
          85          90          95

Arg  Ile  Tyr  Ser  Thr  Asp  Leu  Gly  Arg  Met  Leu  Leu  Thr  Ser  Ile  Val
          100          105          110

Arg  Gly  Ile  Pro  Phe  Trp  Gly  Gly  Ser  Thr  Ile  Asp  Thr  Glu  Leu  Lys
          115          120          125

Val  Ile  Asp  Thr  Asn  Cys  Ile  Asn  Val  Ile  Gln  Pro  Asp  Gly  Ser  Tyr
          130          135          140

Arg  Ser  Glu  Glu  Leu  Asn  Leu  Val  Ile  Ile  Gly  Pro  Ser  Ala  Asp  Ile
          145          150          155          160

Ile  Gln  Phe  Glu  Cys  Lys  Ser  Phe  Gly  His  Glu  Val  Leu  Asn  Leu  Thr
          165          170          175

Arg  Asn  Gly  Tyr  Gly  Ser  Thr  Gln  Tyr  Ile  Arg  Phe  Ser  Pro  Asp  Phe
          180          185          190

Thr  Phe  Gly  Phe  Glu  Glu  Ser  Leu  Glu  Val  Asp  Thr  Asn  Pro  Leu  Leu
          195          200          205

Gly  Ala  Gly  Lys  Phe  Ala  Thr  Asp  Pro  Ala  Val  Thr  Leu  Ala  His  Glu
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Leu  Ile  His  Ala  Gly  His  Arg  Leu  Tyr  Gly  Ile  Ala  Ile  Asn  Pro  Asn
          225          230          235          240

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          245          250          255

Glu  Val  Ser  Phe  Glu  Glu  Leu  Arg  Thr  Phe  Gly  Gly  His  Asp  Ala  Lys
          260          265          270

Phe  Ile  Asp  Ser  Leu  Gln  Glu  Asn  Glu  Phe  Arg  Leu  Tyr  Tyr  Tyr  Asn
          275          280          285

Lys  Phe  Lys  Asp  Ile  Ala  Ser  Thr  Leu  Asn  Lys  Ala  Lys  Ser  Ile  Val
          290          295          300

Gly  Thr  Thr  Ala  Ser  Leu  Gln  Tyr  Met  Lys  Asn  Val  Phe  Lys  Glu  Lys
          305          310          315          320

Tyr  Leu  Leu  Ser  Glu  Asp  Thr  Ser  Gly  Lys  Phe  Ser  Val  Asp  Lys  Leu
          325          330          335

Lys  Phe  Asp  Lys  Leu  Tyr  Lys  Met  Leu  Thr  Glu  Ile  Tyr  Thr  Glu  Asp
          340          345          350

Asn  Phe  Val  Lys  Phe  Phe  Lys  Val  Leu  Asn  Arg  Lys  Thr  Tyr  Leu  Asn
          355          360          365

Phe  Asp  Lys  Ala  Val  Phe  Lys  Ile  Asn  Ile  Val  Pro  Lys  Val  Asn  Tyr
          370          375          380

Thr  Ile  Tyr  Asp  Gly  Phe  Asn  Leu  Arg  Asn  Thr  Asn  Leu  Ala  Ala  Asn
          385          390          395          400

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-continued

Phe Asn Gly Gln Asn Thr Glu Ile Asn Asn Met Asn Phe Thr Lys Leu
 405 410 415

Lys Asn Phe Thr Gly Leu Phe Glu Phe Tyr Lys Leu Leu Cys Val Arg
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 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Botulinum toxin Light Chain

<400> SEQUENCE: 2

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gaatttcgtc tatattatta taataagttt aaagatatag caagtacact taataaagct      900
aaatcaatag taggtactac tgcctcatta cagtatatga aaaatgtttt taaagagaaa      960
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Lys Lys Lys Arg Lys Val

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Gly	Gln	Met	Gln	Pro	Val	Lys 55	Ala	Phe	Lys	Ile	His 60	Asn	Lys	Ile	Trp
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Thr	Lys	Leu	Phe 115	Glu	Arg	Ile	Tyr	Ser 120	Thr	Asp	Leu	Gly 125	Arg	Met	Leu
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Pro	Ser	Ala	Asp 180	Ile	Ile	Gln	Phe	Glu	Cys 185	Lys	Ser	Phe	Gly 190	His	Glu
Val	Leu	Asn	Leu 195	Thr	Arg	Asn	Gly 200	Tyr	Gly	Ser	Thr	Gln 205	Tyr	Ile	Arg
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Thr	Leu	Ala	His 245	Glu	Leu	Ile	His	Ala	Gly 250	His	Arg	Leu	Tyr	Gly	Ile 255
Ala	Ile	Asn	Pro 260	Asn	Arg	Val	Phe	Lys	Val 265	Asn	Thr	Asn	Ala	Tyr	Tyr
Glu	Met	Ser	Gly 275	Leu	Glu	Val	Ser	Phe	Glu 280	Glu	Leu	Arg 285	Thr	Phe	Gly
Gly 290	His	Asp	Ala	Lys	Phe	Ile 295	Asp	Ser	Leu	Gln	Glu 300	Asn	Glu	Phe	Arg
Leu 305	Tyr	Tyr	Tyr	Asn 310	Lys	Phe	Lys	Asp	Ile 315	Ala	Ser	Thr	Leu	Asn	Lys 320
Ala	Lys	Ser	Ile 325	Val	Gly	Thr	Thr	Ala	Ser 330	Leu	Gln	Tyr	Met	Lys	Asn 335
Val	Phe	Lys	Glu 340	Lys	Tyr	Leu	Leu	Ser	Glu 345	Asp	Thr	Ser	Gly	Lys	Phe
Ser	Val	Asp 355	Lys	Leu	Lys	Phe	Asp 360	Lys	Leu	Tyr	Lys	Met 365	Leu	Thr	Glu
Ile 370	Tyr	Thr	Glu	Asp	Asn	Phe 375	Val	Lys	Phe	Phe	Lys 380	Val	Leu	Asn	Arg
Lys 385	Thr	Tyr	Leu	Asn 390	Phe	Asp	Lys	Ala	Val 395	Phe	Lys	Ile	Asn	Ile	Val 400
Pro	Lys	Val	Asn 405	Tyr	Thr	Ile	Tyr	Asp	Gly 410	Phe	Asn	Leu	Arg	Asn	Thr

-continued

Asn Leu Ala Ala Asn Phe Asn Gly Gln Asn Thr Glu Ile Asn Asn Met
420 425 430

Asn Phe Thr Lys Leu Lys Asn Phe Thr Gly Leu Phe Glu Phe Tyr Lys
435 440 445

Leu Leu Cys Val Arg Gly Ile Ile Thr Ser Lys Thr Lys Ser Leu Asp
450 455 460

Lys Gly Tyr Asn Lys His His His His His His
465 470 475

<210> SEQ ID NO 10
<211> LENGTH: 1428
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: PEP1-(-M)BTX-L-His

<400> SEQUENCE: 10

```
atgaaggaaa cttggtggga aacttggtgg actgaatggt ctcaaccaa gaagaagaga    60
aagggtcaat ttgttaataa acaatttaac tataaagatc ctgtaaatgg tgttgatatt    120
gcttatataa aaattccaaa tgtaggacaa atgcaaccag taaaagcttt taaaattcat    180
aataaaatat ggggtatttc agaaagagat acatttacaa atcctgaaga aggagattta    240
aatccaccac cagaagcaaa acaagttcca gtttcattat atgattcaac atatttaagt    300
acagataatg aaaagataa ttatttaaag ggagttacaa aattatttga gagaatttat    360
tcaactgatac ttggaagaat gttgttaaca tcaatagtaa ggggaatacc attttggggt    420
ggaagtacaa tagatacaga attaaaagtt attgatacta attgtattaa tgtgatacaa    480
ccagatggta gttatagatc agaagaactt aatctagtaa taataggacc ctgagctgat    540
attatacagt ttgaatgtaa aagcttttga catgaagttt tgaatcttac gcgaaatggt    600
tatggctcta ctcaatacat tagatttagc ccagatttta catttggttt tgaggagtca    660
cttgaagttg atacaatatc tcttttaggt gcaggcaaat ttgctacaga tccagcagta    720
acattagcac atgaacttat acatgctgga catagattat atggaatagc aattaatcca    780
aataggggtt ttaaagtaaa tactaatgcc tattatgaaa tgagtgggtt agaagtaagc    840
tttaggaac  ttagaacatt tgggggacat gatgcaaatg ttatagatag ttacaggaa    900
aacgaatttc gtctatatta ttataataag tttaaagata tagcaagtac acttaataaa    960
gctaaatcaa tagtaggtac tactgcttca ttacagtata tgaaaaatgt ttttaaagag   1020
aaatatctcc tatctgaaga tacatctgga aaattttcgg tagataaatt aaaatttgat   1080
aagttataca aaatgttaac agagatttac acagaggata attttgttaa gttttttaa   1140
gtacttaaca gaaaaacata tttgaatttt gataaagccg tatttaagat aaatatagta   1200
cctaaggtaa attacacaat atatgatgga tttaatttaa gaaatacaaa tttagcagca   1260
aactttaatg gtcaaaatac agaaattaat aatatgaatt ttactaaact aaaaaatttt   1320
actggattgt ttgaatttta taagttgcta tgtgtaagag ggataataac ttctaaaact   1380
aatcattag  ataaaggata caataagcat caccatcacc atcactaa                   1428
```

<210> SEQ ID NO 11
<211> LENGTH: 475
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: PEP1-(-M)BTX-L-His (E.coli codon optimization)

<400> SEQUENCE: 11

-continued

Asn Leu Ala Ala Asn Phe Asn Gly Gln Asn Thr Glu Ile Asn Asn Met
420 425 430

Asn Phe Thr Lys Leu Lys Asn Phe Thr Gly Leu Phe Glu Phe Tyr Lys
435 440 445

Leu Leu Cys Val Arg Gly Ile Ile Thr Ser Lys Thr Lys Ser Leu Asp
450 455 460

Lys Gly Tyr Asn Lys His His His His His His
465 470 475

<210> SEQ ID NO 12

<211> LENGTH: 1440

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: gene cassette for cloning PEP1-(-M)BTX-L-His
(E.coli codon optimization)

<400> SEQUENCE: 12

```

catatgaagg aaacttggtg ggaaacttgg tggaccgaat ggtctcaacc aaagaagaag      60
cgcaagggttc aatttggttaa taaacaattt aattataaag atccagtaaa tgggtgctgac    120
attgcttata tcaaaattcc aaatgtaggc caaatgcaac cagtaaaagc ttttaaatt      180
cataataaaa tctgggttat tccagaacgc gataccttta ccaatccgga agaaggtgat      240
ctgaatccac caccagaagc aaaacaagtt ccagttagct attatgatag cacctatctg      300
agcaccgata atgaaaaaga taattatctg aagggcggtta ccaaactggt tgagcgcatt      360
tatagcactg atctgggtcg catgctgctg accagcatcg tacgcggtat cccatttttg      420
ggtggttagca ccacogatac cgaactgaaa gttattgata ctaattgtat taatgtgatc      480
caaccagatg gtagctatcg cagcgaagaa ctgaatctgg taatcctcgg tccgagcgct      540
gatattatcc agtttgaatg taaaagcttt ggtcatgaag ttctgaatct gacccgtaat      600
ggttatggca gcacccaata cattcgcttt agcccagatt ttacctttgg ttttgaggag      660
agcctggaag ttgataccaa tccgctgctg ggtgcaggca aatttgctac cgatccagca      720
gtaaccctgg cacatgaact gatacatgct ggccatcgcc tgtatggtat cgcaattaat      780
ccaaatcgcg tttttaaagt aaataccaat gcctattatg aaatgagcgg tctggaagta      840
agctttgagg aactgcgcac ctttgggtgt catgatgcaa agtttatcga tagcctgcag      900
gaaaacgaat ttcgtctgta ttattataat aagtttaaag atatcgcaag caccctgaat      960
aaagctaaaa gcacgttagg taccaccgct agcctgcagt atatgaaaaa tgttttttaa    1020
gagaaatata tgctgtctga agatacctct ggcaaaatta gcgtagataa actgaaattt    1080
gataagctgt acaaaatgct gaccgagatt tacaccgagg ataattttgt taagtttttt    1140
aaagtactga accgcaaaac ctatctgaat tttgataaag ccgtatttaa gatcaatata    1200
gtaccgaagg taaattacac catctatgat ggttttaata tgcgcaatac caatctggca    1260
gcaaacttta atgggtcaaaa taccgaaatt aataatatga attttaccaa actgaaaaat    1320
tttaccggtc tgtttgtaatt ctataagctg ctgtgtgtac gcggtatcat caccagcaaa    1380
acaaaaagcc tggataaagg ctacaataag catcaccatc accatcacta ataactcgag    1440

```

<210> SEQ ID NO 13

<211> LENGTH: 475

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: PEP1-(-M)BTX-L((C430G))-His (E.coli codon
optimization)

-continued

<400> SEQUENCE: 13

```

Met Lys Glu Thr Trp Trp Glu Thr Trp Trp Thr Glu Trp Ser Gln Pro
1      5      10      15

Lys Lys Lys Arg Lys Val Gln Phe Val Asn Lys Gln Phe Asn Tyr Lys
20      25      30

Asp Pro Val Asn Gly Val Asp Ile Ala Tyr Ile Lys Ile Pro Asn Val
35      40      45

Gly Gln Met Gln Pro Val Lys Ala Phe Lys Ile His Asn Lys Ile Trp
50      55      60

Val Ile Pro Glu Arg Asp Thr Phe Thr Asn Pro Glu Glu Gly Asp Leu
65      70      75      80

Asn Pro Pro Pro Glu Ala Lys Gln Val Pro Val Ser Tyr Tyr Asp Ser
85      90      95

Thr Tyr Leu Ser Thr Asp Asn Glu Lys Asp Asn Tyr Leu Lys Gly Val
100     105     110

Thr Lys Leu Phe Glu Arg Ile Tyr Ser Thr Asp Leu Gly Arg Met Leu
115     120     125

Leu Thr Ser Ile Val Arg Gly Ile Pro Phe Trp Gly Gly Ser Thr Ile
130     135     140

Asp Thr Glu Leu Lys Val Ile Asp Thr Asn Cys Ile Asn Val Ile Gln
145     150     155     160

Pro Asp Gly Ser Tyr Arg Ser Glu Glu Leu Asn Leu Val Ile Ile Gly
165     170     175

Pro Ser Ala Asp Ile Ile Gln Phe Glu Cys Lys Ser Phe Gly His Glu
180     185     190

Val Leu Asn Leu Thr Arg Asn Gly Tyr Gly Ser Thr Gln Tyr Ile Arg
195     200     205

Phe Ser Pro Asp Phe Thr Phe Gly Phe Glu Glu Ser Leu Glu Val Asp
210     215     220

Thr Asn Pro Leu Leu Gly Ala Gly Lys Phe Ala Thr Asp Pro Ala Val
225     230     235     240

Thr Leu Ala His Glu Leu Ile His Ala Gly His Arg Leu Tyr Gly Ile
245     250     255

Ala Ile Asn Pro Asn Arg Val Phe Lys Val Asn Thr Asn Ala Tyr Tyr
260     265     270

Glu Met Ser Gly Leu Glu Val Ser Phe Glu Glu Leu Arg Thr Phe Gly
275     280     285

Gly His Asp Ala Lys Phe Ile Asp Ser Leu Gln Glu Asn Glu Phe Arg
290     295     300

Leu Tyr Tyr Tyr Asn Lys Phe Lys Asp Ile Ala Ser Thr Leu Asn Lys
305     310     315     320

Ala Lys Ser Ile Val Gly Thr Thr Ala Ser Leu Gln Tyr Met Lys Asn
325     330     335

Val Phe Lys Glu Lys Tyr Leu Leu Ser Glu Asp Thr Ser Gly Lys Phe
340     345     350

Ser Val Asp Lys Leu Lys Phe Asp Lys Leu Tyr Lys Met Leu Thr Glu
355     360     365

Ile Tyr Thr Glu Asp Asn Phe Val Lys Phe Phe Lys Val Leu Asn Arg
370     375     380

Lys Thr Tyr Leu Asn Phe Asp Lys Ala Val Phe Lys Ile Asn Ile Val
385     390     395     400

Pro Lys Val Asn Tyr Thr Ile Tyr Asp Gly Phe Asn Leu Arg Asn Thr

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405	410	415	
Asn Leu Ala Ala Asn Phe Asn Gly Gln Asn Thr Glu Ile Asn Asn Met			
420	425	430	
Asn Phe Thr Lys Leu Lys Asn Phe Thr Gly Leu Phe Glu Phe Tyr Lys			
435	440	445	
Leu Leu Gly Val Arg Gly Ile Ile Thr Ser Lys Thr Lys Ser Leu Asp			
450	455	460	
Lys Gly Tyr Asn Lys His His His His His His			
465	470	475	

<210> SEQ ID NO 14
 <211> LENGTH: 1440
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: gene casstte for cloning PEP1-(-M)BTX-
 L((C430G))-His (E.coli codon optimization)

<400> SEQUENCE: 14

catatgaagg aaacttggtg ggaacttg tggaccgaat ggtctcaacc aaagaagaag	60
cgcaaggttc aatttgtaa taaacaattt aattataaag atccagtaaa tgggtgcgac	120
attgcttata tcaaaattcc aaatgtaggc caaatgcaac cagtaaaagc ttttaaaatt	180
cataataaaa tctgggttat tccagaacgc gataccttta ccaatccgga agaaggtgat	240
ctgaatccac caccagaagc aaaacaagtt ccagttagct attatgatag caccatctcg	300
agcaccgata atgaaaaaga taattatctg aaggcggtta ccaaactggt tgagcgcat	360
tatagcactg atctgggtcg catgctgctg accagcatcg tacgcggtat cccatcttgg	420
ggtggtagca ccatcgatgc cgaactgaaa gttattgata ctaattgtat taatgtgatc	480
caaccagatg gtatgctatc gagcgaagaa ctgaatctgg taatcctcgg tccgagcgct	540
gatattatcc agtttgaatg taaaagcttt ggtcatgaag ttctgaatct gaccctgaat	600
ggttatggca gcaccaata cattcgcttt agcccagatt ttacotttgg ttttgaggag	660
agcctggaag ttgataccaa tccgctgctg ggtgcaggca aatttgctac cgatccagca	720
gtaaccctgg cacatgaact gatacatgct ggccatcgcc tgtatggtat cgcaattaat	780
ccaaatcgcg tttttaaaag aaataccaat gcctattatg aaatgagcgg tctggaagta	840
agctttgagg aactgcgcac ctttggtggt catgatgcaa agtttatcga tagcctgcag	900
gaaaacgaat ttcgtctgta ttattataat aagtttaaa atacgcaag caccctgaat	960
aaagctaaaa gcatcgtagg taccaccgct agcctgcagt atatgaaaa tgtttttaa	1020
gagaaatata tgctgtctga agatacctct ggcaaattha gcgtagataa actgaaattt	1080
gataagctgt acaaaatgct gaccgagatt tacaccgagg ataattttgt taagttttt	1140
aaagtactga accgcaaac ctatctgaat ttgataaaag ccgtatttaa gatcaatata	1200
gtaccgaagg taaattacac catctatgat ggttttaata tgcgcaatac caatctggca	1260
gcaaacttta atggtcaaaa taccgaaatt aataatatga attttaccaa actgaaaaat	1320
tttaccggtc tgtttgaatt ctataagctg ctgggcgtac gcggtatcat caccagcaaa	1380
accaaagccc tggataaagg ctacaataag catcaccatc accatcacta ataactcgag	1440

<210> SEQ ID NO 15
 <211> LENGTH: 475
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: PEP1-(-M)BTX-L(C430A)-His (E.coli codon optimization)

<400> SEQUENCE: 15

```

Met Lys Glu Thr Trp Trp Glu Thr Trp Trp Thr Glu Trp Ser Gln Pro
1          5          10          15

Lys Lys Lys Arg Lys Val Gln Phe Val Asn Lys Gln Phe Asn Tyr Lys
20          25          30

Asp Pro Val Asn Gly Val Asp Ile Ala Tyr Ile Lys Ile Pro Asn Val
35          40          45

Gly Gln Met Gln Pro Val Lys Ala Phe Lys Ile His Asn Lys Ile Trp
50          55          60

Val Ile Pro Glu Arg Asp Thr Phe Thr Asn Pro Glu Glu Gly Asp Leu
65          70          75          80

Asn Pro Pro Pro Glu Ala Lys Gln Val Pro Val Ser Tyr Tyr Asp Ser
85          90          95

Thr Tyr Leu Ser Thr Asp Asn Glu Lys Asp Asn Tyr Leu Lys Gly Val
100         105         110

Thr Lys Leu Phe Glu Arg Ile Tyr Ser Thr Asp Leu Gly Arg Met Leu
115         120         125

Leu Thr Ser Ile Val Arg Gly Ile Pro Phe Trp Gly Gly Ser Thr Ile
130         135         140

Asp Thr Glu Leu Lys Val Ile Asp Thr Asn Cys Ile Asn Val Ile Gln
145         150         155         160

Pro Asp Gly Ser Tyr Arg Ser Glu Glu Leu Asn Leu Val Ile Ile Gly
165         170         175

Pro Ser Ala Asp Ile Ile Gln Phe Glu Cys Lys Ser Phe Gly His Glu
180         185         190

Val Leu Asn Leu Thr Arg Asn Gly Tyr Gly Ser Thr Gln Tyr Ile Arg
195         200         205

Phe Ser Pro Asp Phe Thr Phe Gly Phe Glu Glu Ser Leu Glu Val Asp
210         215         220

Thr Asn Pro Leu Leu Gly Ala Gly Lys Phe Ala Thr Asp Pro Ala Val
225         230         235         240

Thr Leu Ala His Glu Leu Ile His Ala Gly His Arg Leu Tyr Gly Ile
245         250         255

Ala Ile Asn Pro Asn Arg Val Phe Lys Val Asn Thr Asn Ala Tyr Tyr
260         265         270

Glu Met Ser Gly Leu Glu Val Ser Phe Glu Glu Leu Arg Thr Phe Gly
275         280         285

Gly His Asp Ala Lys Phe Ile Asp Ser Leu Gln Glu Asn Glu Phe Arg
290         295         300

Leu Tyr Tyr Tyr Asn Lys Phe Lys Asp Ile Ala Ser Thr Leu Asn Lys
305         310         315         320

Ala Lys Ser Ile Val Gly Thr Thr Ala Ser Leu Gln Tyr Met Lys Asn
325         330         335

Val Phe Lys Glu Lys Tyr Leu Leu Ser Glu Asp Thr Ser Gly Lys Phe
340         345         350

Ser Val Asp Lys Leu Lys Phe Asp Lys Leu Tyr Lys Met Leu Thr Glu
355         360         365

Ile Tyr Thr Glu Asp Asn Phe Val Lys Phe Phe Lys Val Leu Asn Arg
370         375         380

Lys Thr Tyr Leu Asn Phe Asp Lys Ala Val Phe Lys Ile Asn Ile Val
385         390         395         400

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-continued

Pro Lys Val Asn Tyr Thr Ile Tyr Asp Gly Phe Asn Leu Arg Asn Thr
 405 410 415

Asn Leu Ala Ala Asn Phe Asn Gly Gln Asn Thr Glu Ile Asn Asn Met
 420 425 430

Asn Phe Thr Lys Leu Lys Asn Phe Thr Gly Leu Phe Glu Phe Tyr Lys
 435 440 445

Leu Leu Ala Val Arg Gly Ile Ile Thr Ser Lys Thr Lys Ser Leu Asp
 450 455 460

Lys Gly Tyr Asn Lys His His His His His His
 465 470 475

<210> SEQ ID NO 16

<211> LENGTH: 1440

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: gene casstte for cloning PEP1-(-M)BTX-L(C430A) -
 His (E.coli codon optimization)

<400> SEQUENCE: 16

```

catatgaagg aaacttggtg ggaaacttgg tggaccgaat ggtctcaacc aaagaagaag      60
cgcaagggtc aatttgtaa taaacaattt aattataaag atccagtaaa tgggtgcgac      120
attgcttata tcaaaattcc aaatgtaggc caaatgcaac cagtaaaagc ttttaaaatt      180
cataataaaa tctgggttat tccagaacgc gataccttta ccaatccgga agaaggtgat      240
ctgaatccac caccagaagc aaaacaagtt ccagttagct attatgatag cacctatctg      300
agcaccgata atgaaaaaga taattatctg aaggcgctta ccaaactggt tgagcgcatt      360
tatagcactg atctgggtcg catgctgctg accagcatcg tacgcggtat cccattttgg      420
ggtggtagca ccacgcgata cgaactgaaa gttattgata ctaattgtat taatgtgata      480
caaccagatg gtatgctatc cagcgaagaa ctgaatctgg taatcatcgg tccgagcgct      540
gatattatcc agtttgaatg taaaagcttt ggtcatgaag ttctgaatct gaccgcgaat      600
ggttatggca gcacccaata cattcgcttt agcccagatt ttacctttgg ttttgaggag      660
agcctggaag ttgataccaa tccgctgctg ggtgcaggca aatttgctac cgatccagca      720
gtaaccctgg cacatgaact gatacatgct ggccatcgcc tgtatggtat cgcaattaat      780
ccaaatcgcg tttttaaagt aaataccaat gcctattatg aaatgagcgg tctggaagta      840
agctttgagg aactgcgcac ctttggtggt catgatgcaa agtttatcga tagcctgcag      900
gaaaacgaat ttcgtctgta ttattataat aagttaaag atatcgcaag caccctgaat      960
aaagctaaaa gcacgcgtagg taccaccgct agcctgcagt atatgaaaaa tgtttttaa      1020
gagaaatata tgctgtctga agatacctct ggcaaattta gcgtagataa actgaaattt      1080
gataagctgt acaaaatgct gaccgagatt tacaccgagg ataattttgt taagtttttt      1140
aaagtactga accgcaaaac ctatctgaat ttgataaag ccgtatttaa gatcaatata      1200
gtaccgaagg taaattacac catctatgat ggttttaata tgcgcaatac caatctggca      1260
gcaaaacttta atggtcaaaa taccgaaatt aataatatga attttaccaa actgaaaaat      1320
tttaccggtc tgtttgaatt ctataagctg ctgagcgtac gcggtatcat caccagcaaa      1380
accaaaagcc tggataaagg ctacaataag catcaccatc accatcacta ataactcgag      1440

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<210> SEQ ID NO 17

<211> LENGTH: 475

<212> TYPE: PRT

-continued

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: PEP1-(-M)BTX-L(C430S)-His

<400> SEQUENCE: 17

```

Met Lys Glu Thr Trp Trp Glu Thr Trp Trp Thr Glu Trp Ser Gln Pro
1          5          10          15

Lys Lys Lys Arg Lys Val Gln Phe Val Asn Lys Gln Phe Asn Tyr Lys
20          25          30

Asp Pro Val Asn Gly Val Asp Ile Ala Tyr Ile Lys Ile Pro Asn Val
35          40          45

Gly Gln Met Gln Pro Val Lys Ala Phe Lys Ile His Asn Lys Ile Trp
50          55          60

Val Ile Pro Glu Arg Asp Thr Phe Thr Asn Pro Glu Glu Gly Asp Leu
65          70          75          80

Asn Pro Pro Pro Glu Ala Lys Gln Val Pro Val Ser Tyr Tyr Asp Ser
85          90          95

Thr Tyr Leu Ser Thr Asp Asn Glu Lys Asp Asn Tyr Leu Lys Gly Val
100         105         110

Thr Lys Leu Phe Glu Arg Ile Tyr Ser Thr Asp Leu Gly Arg Met Leu
115         120         125

Leu Thr Ser Ile Val Arg Gly Ile Pro Phe Trp Gly Gly Ser Thr Ile
130         135         140

Asp Thr Glu Leu Lys Val Ile Asp Thr Asn Cys Ile Asn Val Ile Gln
145         150         155         160

Pro Asp Gly Ser Tyr Arg Ser Glu Glu Leu Asn Leu Val Ile Ile Gly
165         170         175

Pro Ser Ala Asp Ile Ile Gln Phe Glu Cys Lys Ser Phe Gly His Glu
180         185         190

Val Leu Asn Leu Thr Arg Asn Gly Tyr Gly Ser Thr Gln Tyr Ile Arg
195         200         205

Phe Ser Pro Asp Phe Thr Phe Gly Phe Glu Glu Ser Leu Glu Val Asp
210         215         220

Thr Asn Pro Leu Leu Gly Ala Gly Lys Phe Ala Thr Asp Pro Ala Val
225         230         235         240

Thr Leu Ala His Glu Leu Ile His Ala Gly His Arg Leu Tyr Gly Ile
245         250         255

Ala Ile Asn Pro Asn Arg Val Phe Lys Val Asn Thr Asn Ala Tyr Tyr
260         265         270

Glu Met Ser Gly Leu Glu Val Ser Phe Glu Glu Leu Arg Thr Phe Gly
275         280         285

Gly His Asp Ala Lys Phe Ile Asp Ser Leu Gln Glu Asn Glu Phe Arg
290         295         300

Leu Tyr Tyr Tyr Asn Lys Phe Lys Asp Ile Ala Ser Thr Leu Asn Lys
305         310         315         320

Ala Lys Ser Ile Val Gly Thr Thr Ala Ser Leu Gln Tyr Met Lys Asn
325         330         335

Val Phe Lys Glu Lys Tyr Leu Leu Ser Glu Asp Thr Ser Gly Lys Phe
340         345         350

Ser Val Asp Lys Leu Lys Phe Asp Lys Leu Tyr Lys Met Leu Thr Glu
355         360         365

Ile Tyr Thr Glu Asp Asn Phe Val Lys Phe Phe Lys Val Leu Asn Arg
370         375         380

Lys Thr Tyr Leu Asn Phe Asp Lys Ala Val Phe Lys Ile Asn Ile Val

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-continued

385	390	395	400
Pro Lys Val Asn Tyr Thr Ile Tyr Asp Gly Phe Asn Leu Arg Asn Thr	405	410	415
Asn Leu Ala Ala Asn Phe Asn Gly Gln Asn Thr Glu Ile Asn Asn Met	420	425	430
Asn Phe Thr Lys Leu Lys Asn Phe Thr Gly Leu Phe Glu Phe Tyr Lys	435	440	445
Leu Leu Ser Val Arg Gly Ile Ile Thr Ser Lys Thr Lys Ser Leu Asp	450	455	460
Lys Gly Tyr Asn Lys His His His His His His	465	470	475

<210> SEQ ID NO 18
 <211> LENGTH: 1440
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: gene cassette for cloning PEP1-(-M)BTX-
 L(C430S)-His

<400> SEQUENCE: 18

catatgaagg aaacttggtg ggaaacttgg tggaccgaat ggtctcaacc aaagaagaag	60
cgcaagggtc aatttggttaa taaacaattt aattataaag atccagtaaa tgggtgtcgac	120
attgcttata tcaaaattcc aaatgtaggc caaatgcaac cagtaaaagc ttttaaaatt	180
cataataaaa tctgggttat tccagaacgc gataccttta ccaatccgga agaaggtgat	240
ctgaatccac caccagaagc aaaacaagtt ccagttagct attatgatag cacctatctg	300
agcaccgata atgaaaaaga taattatctg aaggggcgta ccaaactggt tgagcgcatt	360
tatagcactg atctgggtcg catgctgctg accagcatcg tacgcggtat cccattttgg	420
ggtggttagca ccatcgatag cgaactgaaa gttattgata ctaattgtat taatgtgatc	480
caaccagatg gtatgctatc gagcgaagaa ctgaatctgg taatcctcgg tccgagcgct	540
gatattatcc agtttgaatg taaaagcttt ggtcatgaag ttctgaatct gaccctgaat	600
ggttatggca gcaccaata cattcgcttt agcccagatt ttacctttgg ttttgaggag	660
agcctggaag ttgataccaa tccgctgctg ggtgcaggca aatttgctac cgatccagca	720
gtaacctctg cacatgaact gatacatgct ggccatcgcc tgtatggtat cgcaattaat	780
ccaaatcgcg tttttaaagt aaataccaat gcctattatg aaatgagcgg tctggaagta	840
agctttgagg aactgcgcac ctttggtggt catgatgcaa agtttatcga tagcctgcag	900
gaaaacgaat ttcgtctgta ttattataat aagtttaaag atatcgcaag caccctgaat	960
aaagctaaaa gcatcgtagg taccaccgct agcctgcagt atatgaaaaa tgttttttaa	1020
gagaaatato tgctgtctga agatacctct ggcaaaattta gcgtagataa actgaaattt	1080
gataagctgt acaaaatgct gaccgagatt tacaccgagg ataattttgt taagtttttt	1140
aaagtactga accgcaaaac ctatctgaat ttgataaaag ccgtatttaa gatcaatata	1200
gtaccgaagg taaattacac catctatgat ggttttaata tgcgcaatac caatctggca	1260
gcaaacttta atgggtcaaaa taccgaaatt aataatatga attttaccaa actgaaaaat	1320
tttaccggtc tgtttgaatt ctataagctg ctgagcgtac gcggtatcat caccagcaaa	1380
acaaaaagcc tggataaagg ctacaataag catcaccatc accatcacta ataactcgag	1440

<210> SEQ ID NO 19
 <211> LENGTH: 486

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: PEP1-(-M)BTX-L(C430G)-BFGFRP-His
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (477)..(477)
<223> OTHER INFORMATION: X is S or T

<400> SEQUENCE: 19

Met Lys Glu Thr Trp Trp Glu Thr Trp Trp Thr Glu Trp Ser Gln Pro
1          5          10          15

Lys Lys Lys Arg Lys Val Gln Phe Val Asn Lys Gln Phe Asn Tyr Lys
20          25          30

Asp Pro Val Asn Gly Val Asp Ile Ala Tyr Ile Lys Ile Pro Asn Val
35          40          45

Gly Gln Met Gln Pro Val Lys Ala Phe Lys Ile His Asn Lys Ile Trp
50          55          60

Val Ile Pro Glu Arg Asp Thr Phe Thr Asn Pro Glu Glu Gly Asp Leu
65          70          75          80

Asn Pro Pro Pro Glu Ala Lys Gln Val Pro Val Ser Tyr Tyr Asp Ser
85          90          95

Thr Tyr Leu Ser Thr Asp Asn Glu Lys Asp Asn Tyr Leu Lys Gly Val
100         105         110

Thr Lys Leu Phe Glu Arg Ile Tyr Ser Thr Asp Leu Gly Arg Met Leu
115         120         125

Leu Thr Ser Ile Val Arg Gly Ile Pro Phe Trp Gly Gly Ser Thr Ile
130         135         140

Asp Thr Glu Leu Lys Val Ile Asp Thr Asn Cys Ile Asn Val Ile Gln
145         150         155         160

Pro Asp Gly Ser Tyr Arg Ser Glu Glu Leu Asn Leu Val Ile Ile Gly
165         170         175

Pro Ser Ala Asp Ile Ile Gln Phe Glu Cys Lys Ser Phe Gly His Glu
180         185         190

Val Leu Asn Leu Thr Arg Asn Gly Tyr Gly Ser Thr Gln Tyr Ile Arg
195         200         205

Phe Ser Pro Asp Phe Thr Phe Gly Phe Glu Glu Ser Leu Glu Val Asp
210         215         220

Thr Asn Pro Leu Leu Gly Ala Gly Lys Phe Ala Thr Asp Pro Ala Val
225         230         235         240

Thr Leu Ala His Glu Leu Ile His Ala Gly His Arg Leu Tyr Gly Ile
245         250         255

Ala Ile Asn Pro Asn Arg Val Phe Lys Val Asn Thr Asn Ala Tyr Tyr
260         265         270

Glu Met Ser Gly Leu Glu Val Ser Phe Glu Glu Leu Arg Thr Phe Gly
275         280         285

Gly His Asp Ala Lys Phe Ile Asp Ser Leu Gln Glu Asn Glu Phe Arg
290         295         300

Leu Tyr Tyr Tyr Asn Lys Phe Lys Asp Ile Ala Ser Thr Leu Asn Lys
305         310         315         320

Ala Lys Ser Ile Val Gly Thr Thr Ala Ser Leu Gln Tyr Met Lys Asn
325         330         335

Val Phe Lys Glu Lys Tyr Leu Leu Ser Glu Asp Thr Ser Gly Lys Phe
340         345         350

Ser Val Asp Lys Leu Lys Phe Asp Lys Leu Tyr Lys Met Leu Thr Glu
355         360         365

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Ile Tyr Thr Glu Asp Asn Phe Val Lys Phe Phe Lys Val Leu Asn Arg
 370 375 380

Lys Thr Tyr Leu Asn Phe Asp Lys Ala Val Phe Lys Ile Asn Ile Val
 385 390 395 400

Pro Lys Val Asn Tyr Thr Ile Tyr Asp Gly Phe Asn Leu Arg Asn Thr
 405 410 415

Asn Leu Ala Ala Asn Phe Asn Gly Gln Asn Thr Glu Ile Asn Asn Met
 420 425 430

Asn Phe Thr Lys Leu Lys Asn Phe Thr Gly Leu Phe Glu Phe Tyr Lys
 435 440 445

Leu Leu Gly Val Arg Gly Ile Ile Thr Ser Lys Thr Lys Ser Leu Asp
 450 455 460

Lys Gly Tyr Asn Lys Thr Tyr Arg Ser Arg Lys Tyr Xaa Ser Trp Tyr
 465 470 475 480

His His His His His His
 485

<210> SEQ ID NO 20
 <211> LENGTH: 1473
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: gene cassette for cloning PEP1-(-M)BTX-
 L(C430G)-BFGFRP-His

<400> SEQUENCE: 20

catatgaagg aaacttggtg ggaaacttgg tggaccgaat ggtctcaacc aaagaagaag	60
cgcaagggtc aatttggtta taaacaattt aattataaag atccagtaaa tgggtgtcgac	120
attgcttata tcaaaattcc aaatgtaggc caaatgcaac cagtaaaagc ttttaaaatt	180
cataataaaa tctgggttat tccagaacgc gataccttta ccaatccgga agaaggtgat	240
ctgaatccac caccagaagc aaaacaagtt ccagttagct attatgatag cacctatctg	300
agcaccgata atgaaaaaga taattatctg aagggcggtta ccaaactggt tgagcgcatt	360
tatagcactg atctgggctg catgctgctg accagcatcg tacgcggtat cccattttgg	420
ggtggttagc ccacgcgata cgaactgaaa gttattgata ctaattgtat taatgtgatac	480
caaccagatg gtatgctatg cagcgaagaa ctgaatctgg taatcatcgg tccgagcgcct	540
gatattatcc agtttgaatg taaaagcttt ggtcatgaag ttctgaatct gaccgcgaat	600
ggttatggca gcacccaata cattcgcttt agcccagatt ttacctttgg ttttgaggag	660
agcctggaag ttgataccaa tccgctgctg ggtgcaggca aatttgctac cgatccagca	720
gtaaccctgg cacatgaact gatacatgct ggccatcgcc tgtatggtat cgcaattaat	780
ccaaatcgcg tttttaaagt aaataccaat gcctattatg aaatgagcgg tctggaagta	840
agctttgagg aactgcgcac ctttgggtgt catgatgcaa agtttatcga tagcctgcag	900
gaaaacgaat ttcgtctgta ttattataat aagtttaaag atatcgcaag caccctgaat	960
aaagctaaaa gcacgttagg taccaccgct agcctgcagt atatgaaaaa tgttttttaa	1020
gagaaatata tgctgtctga agatacctct ggcaaattha gcgtagataa actgaaattt	1080
gataagctgt acaaaatgct gaccgagatt tacaccgagg ataattttgt taagtttttt	1140
aaagtactga accgcaaaac ctatctgaat tttgataaag ccgtatttaa gatcaatatac	1200
gtaccgaagg taaattacac catctatgat ggttttaata tcgcgaatac caatctggca	1260
gcaaacttta atggtcaaaa taccgaaatt aataatatga attttaccaa actgaaaaat	1320

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tttacccggtc tgtttgaatt ctataagctg ctgggcgtac gcggtatcat caccagcaaa 1380
accaaaaagcc tggataaagg ctacaataag acctatcgca gccgcaaata tascagctgg 1440
tatcatcacc atcaccatca ctaataactc gag 1473

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<210> SEQ ID NO 21
<211> LENGTH: 486
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: PEP1-(-M)BTX-L(C430A)-BFGFRP-His
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (477)..(477)
<223> OTHER INFORMATION: X is S or T

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<400> SEQUENCE: 21

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Met Lys Glu Thr Trp Trp Glu Thr Trp Trp Thr Glu Trp Ser Gln Pro
1          5          10          15

Lys Lys Lys Arg Lys Val Gln Phe Val Asn Lys Gln Phe Asn Tyr Lys
20          25          30

Asp Pro Val Asn Gly Val Asp Ile Ala Tyr Ile Lys Ile Pro Asn Val
35          40          45

Gly Gln Met Gln Pro Val Lys Ala Phe Lys Ile His Asn Lys Ile Trp
50          55          60

Val Ile Pro Glu Arg Asp Thr Phe Thr Asn Pro Glu Glu Gly Asp Leu
65          70          75          80

Asn Pro Pro Pro Glu Ala Lys Gln Val Pro Val Ser Tyr Tyr Asp Ser
85          90          95

Thr Tyr Leu Ser Thr Asp Asn Glu Lys Asp Asn Tyr Leu Lys Gly Val
100         105         110

Thr Lys Leu Phe Glu Arg Ile Tyr Ser Thr Asp Leu Gly Arg Met Leu
115         120         125

Leu Thr Ser Ile Val Arg Gly Ile Pro Phe Trp Gly Gly Ser Thr Ile
130         135         140

Asp Thr Glu Leu Lys Val Ile Asp Thr Asn Cys Ile Asn Val Ile Gln
145         150         155         160

Pro Asp Gly Ser Tyr Arg Ser Glu Glu Leu Asn Leu Val Ile Ile Gly
165         170         175

Pro Ser Ala Asp Ile Ile Gln Phe Glu Cys Lys Ser Phe Gly His Glu
180         185         190

Val Leu Asn Leu Thr Arg Asn Gly Tyr Gly Ser Thr Gln Tyr Ile Arg
195         200         205

Phe Ser Pro Asp Phe Thr Phe Gly Phe Glu Glu Ser Leu Glu Val Asp
210         215         220

Thr Asn Pro Leu Leu Gly Ala Gly Lys Phe Ala Thr Asp Pro Ala Val
225         230         235         240

Thr Leu Ala His Glu Leu Ile His Ala Gly His Arg Leu Tyr Gly Ile
245         250         255

Ala Ile Asn Pro Asn Arg Val Phe Lys Val Asn Thr Asn Ala Tyr Tyr
260         265         270

Glu Met Ser Gly Leu Glu Val Ser Phe Glu Glu Leu Arg Thr Phe Gly
275         280         285

Gly His Asp Ala Lys Phe Ile Asp Ser Leu Gln Glu Asn Glu Phe Arg
290         295         300

Leu Tyr Tyr Tyr Asn Lys Phe Lys Asp Ile Ala Ser Thr Leu Asn Lys

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305	310	315	320
Ala Lys Ser Ile Val Gly Thr Thr Ala Ser Leu Gln Tyr Met Lys Asn			
	325	330	335
Val Phe Lys Glu Lys Tyr Leu Leu Ser Glu Asp Thr Ser Gly Lys Phe			
	340	345	350
Ser Val Asp Lys Leu Lys Phe Asp Lys Leu Tyr Lys Met Leu Thr Glu			
	355	360	365
Ile Tyr Thr Glu Asp Asn Phe Val Lys Phe Phe Lys Val Leu Asn Arg			
	370	375	380
Lys Thr Tyr Leu Asn Phe Asp Lys Ala Val Phe Lys Ile Asn Ile Val			
	385	390	395
Pro Lys Val Asn Tyr Thr Ile Tyr Asp Gly Phe Asn Leu Arg Asn Thr			
	405	410	415
Asn Leu Ala Ala Asn Phe Asn Gly Gln Asn Thr Glu Ile Asn Asn Met			
	420	425	430
Asn Phe Thr Lys Leu Lys Asn Phe Thr Gly Leu Phe Glu Phe Tyr Lys			
	435	440	445
Leu Leu Ala Val Arg Gly Ile Ile Thr Ser Lys Thr Lys Ser Leu Asp			
	450	455	460
Lys Gly Tyr Asn Lys Thr Tyr Arg Ser Arg Lys Tyr Xaa Ser Trp Tyr			
	465	470	475
His His His His His His			
	485		

<210> SEQ ID NO 22
 <211> LENGTH: 1473
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: gene cassette for cloning PEP1-(-M)BTX-
 L(C430A)-BFGFRP-His

<400> SEQUENCE: 22

catatgaagg aaacttggtg ggaaacttgg tggaccgaat ggtctcaacc aaagaagaag	60
cgcaagggtc aatttggttaa taaacaattt aattataaag atccagtaaa tgggtgctgac	120
attgcttata tcaaaattcc aaatgtaggc caaatgcaac cagtaaaagc ttttaaaatt	180
cataataaaa tctgggttat tccagaacgc gataccttta ccaatccgga agaaggtgat	240
ctgaatccac caccagaagc aaaacaagtt ccagttagct attatgatag cacctatctg	300
agcaccgata atgaaaaaga taattatctg aagggcgtta ccaaactggt tgagcgcatt	360
tatagcactg atctgggtcg catgctgctg accagcatcg tacgcggtat cccattttgg	420
ggtggttagca ccatcgatag cgaactgaaa gttattgata ctaattgtat taatgtgatc	480
caaccagatg gtatgctatcg cagcgaagaa ctgaatctgg taatcctcgg tccgagcgct	540
gatattatcc agtttgaatg taaaagcttt ggtcatgaag ttctgaatct gaccctgaat	600
ggttatggca gcaccaata cattcgcttt agccagatt ttacctttgg ttttgaggag	660
agcctggaag ttgataccaa tccgctgctg ggtgcaggca aatttgctac cgatccagca	720
gtaaccctgg cacatgaact gatacatgct ggccatcgcc tgtatggtat cgcaattaat	780
ccaaatcgcg tttttaaagt aaataccaat gcctattatg aaatgagcgg tctggaagta	840
agctttgagg aactgcgcac ctttgggtgt catgatgcaa agtttatoga tagcctgcag	900
gaaaacgaat ttcgtctgta ttattataat aagtttaaag atatcgcaag caccctgaat	960
aaagctaaaa gcacgttagg taccaccgct agcctgcagt atatgaaaaa tgtttttaa	1020

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gagaaatattc tgcgtgtctga agatacctctt ggcaaattta gcgtagataa actgaaattt 1080
gataagctgt acaaaatgct gaccgagatt tacaccgagg ataattttgt taagtttttt 1140
aaagtactga accgcaaaac ctatctgaat ttgataaag ccgtatttaa gatcaatattc 1200
gtaccgaagg taaattacac catctatgat ggttttaac tgcgcaatac caatctggca 1260
gcaaacttta atgggtcaaaa taccgaaatt aataatatga attttaccaa actgaaaaat 1320
tttaccggto tgtttgaatt ctataagctg ctggcgggtac gcggtatcat caccagcaaa 1380
accaaagcc tggataaagg ctacaataag acctatcgca gccgcaaata tascagctgg 1440
tatcatcacc atcaccatca ctaataactc gag 1473

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<210> SEQ ID NO 23
<211> LENGTH: 486
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: PEP1- (-M)BTX-L(C430S)-BFGFRP-His
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (477)..(477)
<223> OTHER INFORMATION: X is S or T

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<400> SEQUENCE: 23

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Met Lys Glu Thr Trp Trp Glu Thr Trp Trp Thr Glu Trp Ser Gln Pro
1           5           10           15
Lys Lys Lys Arg Lys Val Gln Phe Val Asn Lys Gln Phe Asn Tyr Lys
20          25          30
Asp Pro Val Asn Gly Val Asp Ile Ala Tyr Ile Lys Ile Pro Asn Val
35          40          45
Gly Gln Met Gln Pro Val Lys Ala Phe Lys Ile His Asn Lys Ile Trp
50          55          60
Val Ile Pro Glu Arg Asp Thr Phe Thr Asn Pro Glu Glu Gly Asp Leu
65          70          75          80
Asn Pro Pro Pro Glu Ala Lys Gln Val Pro Val Ser Tyr Tyr Asp Ser
85          90          95
Thr Tyr Leu Ser Thr Asp Asn Glu Lys Asp Asn Tyr Leu Lys Gly Val
100         105         110
Thr Lys Leu Phe Glu Arg Ile Tyr Ser Thr Asp Leu Gly Arg Met Leu
115         120         125
Leu Thr Ser Ile Val Arg Gly Ile Pro Phe Trp Gly Gly Ser Thr Ile
130         135         140
Asp Thr Glu Leu Lys Val Ile Asp Thr Asn Cys Ile Asn Val Ile Gln
145         150         155         160
Pro Asp Gly Ser Tyr Arg Ser Glu Glu Leu Asn Leu Val Ile Ile Gly
165         170         175
Pro Ser Ala Asp Ile Ile Gln Phe Glu Cys Lys Ser Phe Gly His Glu
180         185         190
Val Leu Asn Leu Thr Arg Asn Gly Tyr Gly Ser Thr Gln Tyr Ile Arg
195         200         205
Phe Ser Pro Asp Phe Thr Phe Gly Phe Glu Glu Ser Leu Glu Val Asp
210         215         220
Thr Asn Pro Leu Leu Gly Ala Gly Lys Phe Ala Thr Asp Pro Ala Val
225         230         235         240
Thr Leu Ala His Glu Leu Ile His Ala Gly His Arg Leu Tyr Gly Ile
245         250         255

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Ala Ile Asn Pro Asn Arg Val Phe Lys Val Asn Thr Asn Ala Tyr Tyr
 260 265 270

Glu Met Ser Gly Leu Glu Val Ser Phe Glu Glu Leu Arg Thr Phe Gly
 275 280 285

Gly His Asp Ala Lys Phe Ile Asp Ser Leu Gln Glu Asn Glu Phe Arg
 290 295 300

Leu Tyr Tyr Tyr Asn Lys Phe Lys Asp Ile Ala Ser Thr Leu Asn Lys
 305 310 315 320

Ala Lys Ser Ile Val Gly Thr Thr Ala Ser Leu Gln Tyr Met Lys Asn
 325 330 335

Val Phe Lys Glu Lys Tyr Leu Leu Ser Glu Asp Thr Ser Gly Lys Phe
 340 345 350

Ser Val Asp Lys Leu Lys Phe Asp Lys Leu Tyr Lys Met Leu Thr Glu
 355 360 365

Ile Tyr Thr Glu Asp Asn Phe Val Lys Phe Phe Lys Val Leu Asn Arg
 370 375 380

Lys Thr Tyr Leu Asn Phe Asp Lys Ala Val Phe Lys Ile Asn Ile Val
 385 390 395 400

Pro Lys Val Asn Tyr Thr Ile Tyr Asp Gly Phe Asn Leu Arg Asn Thr
 405 410 415

Asn Leu Ala Ala Asn Phe Asn Gly Gln Asn Thr Glu Ile Asn Asn Met
 420 425 430

Asn Phe Thr Lys Leu Lys Asn Phe Thr Gly Leu Phe Glu Phe Tyr Lys
 435 440 445

Leu Leu Ser Val Arg Gly Ile Ile Thr Ser Lys Thr Lys Ser Leu Asp
 450 455 460

Lys Gly Tyr Asn Lys Thr Tyr Arg Ser Arg Lys Tyr Xaa Ser Trp Tyr
 465 470 475 480

His His His His His His
 485

<210> SEQ ID NO 24

<211> LENGTH: 1473

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: gene cassette for cloning PEP1-(-M)BTX-L(C430S)-BFGFRP-His

<400> SEQUENCE: 24

catatgaagg aaacttggtg ggaaacttgg tggaccgaat ggtctcaacc aaagaagaag	60
cgcaagggttc aatttggttaa taaacaattt aattataaag atccagtaaa tgggtgtcgac	120
attgcttata tcaaaattcc aaatgtaggc caaatgcaac cagtaaaagc ttttaaaatt	180
cataataaaa tctgggttat tccagaacgc gataccttta ccaatccgga agaaggtgat	240
ctgaatccac caccagaagc aaaacaagtt ccagtttagct attatgatag cacctatctg	300
agcaccgata atgaaaaaga taattatctg aagggcgtta ccaaactggt tgagcgcatt	360
tatagcaactg atctgggtcg catgctgctg accagcatcg tacgcggtat cccatcttgg	420
ggtggttagca ccacgatac cgaactgaaa gttattgata ctaattgtat taatgtgatc	480
caaccagatg gtatgtatcg cagcgaagaa ctgaatctgg taatcatcgg tccgagcgct	540
gatattatcc agtttgaatg taaaagcttt ggtcatgaag ttctgaatct gaccgcgaat	600
ggttatggca gcaccaata cattcgcttt agcccagatt ttacctttgg ttttgaggag	660
agcctggaag ttgataccaa tccgctgctg ggtgcaggca aatttgcctac cgatccagca	720

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gtaaccctgg cacatgaact gatacatgct ggccatcgcc tgtatggat cgcaattaat 780
ccaaatcgcg tttttaagt aaataccaat gcctattatg aaatgagcgg tctggaagta 840
agctttgagg aactgcgcac ctttggtggt catgatgcaa agtttatcga tagcctgcag 900
gaaaacgaat ttcgtctgta ttattataat aagtttaaag atatcgcaag caccctgaat 960
aaagctaaaa gcatcgtagg taccaccgct agcctgcagt atatgaaaaa tgtttttaa 1020
gagaaatato tgctgtctga agatacctct ggcaaattha gcgtagataa actgaaattt 1080
gataagctgt acaaaatgct gaccgagatt tacacgagg ataattttgt taagttttt 1140
aaagtactga accgcaaac ctatctgaat ttgataaag ccgtatttaa gatcaatata 1200
gtaccgaagg taaattacac catctatgat ggttttaata tgcgcaatac caatctggca 1260
gcaaacttha atggtcaaaa taccgaaatt aataatatga attttaccaa actgaaaaa 1320
tttaccggtc tgtttgaatt ctataagctg ctgagcgtac gcggtatcat caccagcaaa 1380
acaaaagcc tggataaagg ctacaataag acctatcgca gccgcaata tascagctgg 1440
tatcatcacc atcaccatca ctaataactc gag 1473

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<210> SEQ ID NO 25
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker

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<400> SEQUENCE: 25

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Gly Gly Gly Gly
1

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<210> SEQ ID NO 26
<211> LENGTH: 12
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker

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<400> SEQUENCE: 26

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```

ggtggtggtg gt 12

```

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<210> SEQ ID NO 27
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Cleavage Peptide

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<400> SEQUENCE: 27

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Leu Val Pro Arg Gly Ser
1 5

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<210> SEQ ID NO 28
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Cleavage Peptide

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<400> SEQUENCE: 28

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ctggtaccac gcggtagc 18

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<210> SEQ ID NO 29

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<211> LENGTH: 448
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: BTX-L(C430G)

<400> SEQUENCE: 29

Met Gln Phe Val Asn Lys Gln Phe Asn Tyr Lys Asp Pro Val Asn Gly
1          5          10          15

Val Asp Ile Ala Tyr Ile Lys Ile Pro Asn Val Gly Gln Met Gln Pro
20          25          30

Val Lys Ala Phe Lys Ile His Asn Lys Ile Trp Val Ile Pro Glu Arg
35          40          45

Asp Thr Phe Thr Asn Pro Glu Glu Gly Asp Leu Asn Pro Pro Pro Glu
50          55          60

Ala Lys Gln Val Pro Val Ser Tyr Tyr Asp Ser Thr Tyr Leu Ser Thr
65          70          75          80

Asp Asn Glu Lys Asp Asn Tyr Leu Lys Gly Val Thr Lys Leu Phe Glu
85          90          95

Arg Ile Tyr Ser Thr Asp Leu Gly Arg Met Leu Leu Thr Ser Ile Val
100         105         110

Arg Gly Ile Pro Phe Trp Gly Gly Ser Thr Ile Asp Thr Glu Leu Lys
115         120         125

Val Ile Asp Thr Asn Cys Ile Asn Val Ile Gln Pro Asp Gly Ser Tyr
130         135         140

Arg Ser Glu Glu Leu Asn Leu Val Ile Ile Gly Pro Ser Ala Asp Ile
145         150         155         160

Ile Gln Phe Glu Cys Lys Ser Phe Gly His Glu Val Leu Asn Leu Thr
165         170         175

Arg Asn Gly Tyr Gly Ser Thr Gln Tyr Ile Arg Phe Ser Pro Asp Phe
180         185         190

Thr Phe Gly Phe Glu Glu Ser Leu Glu Val Asp Thr Asn Pro Leu Leu
195         200         205

Gly Ala Gly Lys Phe Ala Thr Asp Pro Ala Val Thr Leu Ala His Glu
210         215         220

Leu Ile His Ala Gly His Arg Leu Tyr Gly Ile Ala Ile Asn Pro Asn
225         230         235         240

Arg Val Phe Lys Val Asn Thr Asn Ala Tyr Tyr Glu Met Ser Gly Leu
245         250         255

Glu Val Ser Phe Glu Glu Leu Arg Thr Phe Gly Gly His Asp Ala Lys
260         265         270

Phe Ile Asp Ser Leu Gln Glu Asn Glu Phe Arg Leu Tyr Tyr Tyr Asn
275         280         285

Lys Phe Lys Asp Ile Ala Ser Thr Leu Asn Lys Ala Lys Ser Ile Val
290         295         300

Gly Thr Thr Ala Ser Leu Gln Tyr Met Lys Asn Val Phe Lys Glu Lys
305         310         315         320

Tyr Leu Leu Ser Glu Asp Thr Ser Gly Lys Phe Ser Val Asp Lys Leu
325         330         335

Lys Phe Asp Lys Leu Tyr Lys Met Leu Thr Glu Ile Tyr Thr Glu Asp
340         345         350

Asn Phe Val Lys Phe Phe Lys Val Leu Asn Arg Lys Thr Tyr Leu Asn
355         360         365

Phe Asp Lys Ala Val Phe Lys Ile Asn Ile Val Pro Lys Val Asn Tyr
370         375         380

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Thr Ile Tyr Asp Gly Phe Asn Leu Arg Asn Thr Asn Leu Ala Ala Asn
385 390 395 400

Phe Asn Gly Gln Asn Thr Glu Ile Asn Asn Met Asn Phe Thr Lys Leu
405 410 415

Lys Asn Phe Thr Gly Leu Phe Glu Phe Tyr Lys Leu Leu Gly Val Arg
420 425 430

Gly Ile Ile Thr Ser Lys Thr Lys Ser Leu Asp Lys Gly Tyr Asn Lys
435 440 445

<210> SEQ ID NO 30

<211> LENGTH: 1344

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: BTX-L(C430G)

<400> SEQUENCE: 30

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atgcaatttg ttaataaaca atttaattat aaagatccag taaatggtgt cgacattgct 60
tatatcaaaa ttccaaatgt aggccaaatg caaccagtaa aagcttttaa aattcataat 120
aaaatctggg ttattccaga acgcgatacc ttaccaatc cggaagaagg tgatctgaat 180
ccaccaccag aagcaaaaca agttccagtt agctattatg atagcaccta tctgagcacc 240
gataatgaaa aagataatta tctgaagggc gttaccaaac tgtttgagcg catttatagc 300
actgatctgg gtcgcatgct gctgaccagc atcgtagcgc gtatccatt ttggggtggt 360
agcaccatcg ataccgaact gaaagtattt gatactaatt gtattaatgt gatccaacca 420
gatggtagct atcgagcga agaactgaat ctggtaatca tcggtccgag cgctgatatt 480
atccagtttg aatgtaaaag ctttggatcat gaagttctga atctgaccg taatggttat 540
ggcagcacc ccaacattcg ctttagccca gattttacct ttggttttga ggagagcctg 600
gaagttgata ccaatccgct gctgggtgca ggcaaatgtg ctaccgatcc agcagtaacc 660
ctggcacatg aactgatata tgctggccat cgctgtatg gtatcgcaat taatccaaat 720
cgcgttttta aagtaaac caatgcctat tatgaaatga gcggtctgga agtaagcttt 780
gaggaactgc gcaccttgg tggtcatgat gcaaagtta tcgatagcct gcaggaaaac 840
gaatttcgct tgtattatta taataagttt aaagatatcg caagcaccct gaataaagct 900
aaaagcatcg taggtaccac cgctagcctg cagtatatga aaaatgtttt taaagagaaa 960
tatctgctgt ctgaagatac ctctggcaaa tttagcgtag ataaactgaa atttgataag 1020
ctgtacaaaa tgctgaccga gatttacacc gaggataatt ttgttaagtt ttttaaagta 1080
ctgaaccgca aaacctatct gaattttgat aaagccgtat ttaagatcaa tatcgtagcg 1140
aaggtaaatt acaccatcta tgatggtttt aatctgcgca ataccaatct ggcagcaaac 1200
tttaatggtc aaaataccga aattaataat atgaatttta ccaaactgaa aaattttacc 1260
ggtctggttg aattctataa gctgctgggc gtacgcggtg tcatcaccag caaaacaaaa 1320
agcctggata aaggctacaa taag 1344
```

<210> SEQ ID NO 31

<211> LENGTH: 448

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: BTX-L(C430A)

<400> SEQUENCE: 31

Met 1	Gln	Phe	Val	Asn 5	Lys	Gln	Phe	Asn	Tyr 10	Lys	Asp	Pro	Val	Asn 15	Gly
Val	Asp	Ile	Ala 20	Tyr	Ile	Lys	Ile	Pro 25	Asn	Val	Gly	Gln	Met 30	Gln	Pro
Val	Lys	Ala 35	Phe	Lys	Ile	His	Asn 40	Lys	Ile	Trp	Val	Ile 45	Pro	Glu	Arg
Asp	Thr 50	Phe	Thr	Asn	Pro	Glu 55	Glu	Gly	Asp	Leu	Asn 60	Pro	Pro	Pro	Glu
Ala 65	Lys	Gln	Val	Pro	Val 70	Ser	Tyr	Tyr	Asp	Ser 75	Thr	Tyr	Leu	Ser	Thr 80
Asp	Asn	Glu	Lys	Asp 85	Asn	Tyr	Leu	Lys	Gly 90	Val	Thr	Lys	Leu	Phe 95	Glu
Arg	Ile	Tyr	Ser 100	Thr	Asp	Leu	Gly	Arg	Met 105	Leu	Leu	Thr	Ser 110	Ile	Val
Arg	Gly 115	Ile	Pro	Phe	Trp	Gly	Gly 120	Ser	Thr	Ile	Asp	Thr 125	Glu	Leu	Lys
Val 130	Ile	Asp	Thr	Asn	Cys	Ile 135	Asn	Val	Ile	Gln	Pro 140	Asp	Gly	Ser	Tyr
Arg 145	Ser	Glu	Glu	Leu	Asn 150	Leu	Val	Ile	Ile	Gly	Pro 155	Ser	Ala	Asp	Ile 160
Ile	Gln	Phe	Glu	Cys 165	Lys	Ser	Phe	Gly	His 170	Glu	Val	Leu	Asn	Leu 175	Thr
Arg	Asn	Gly 180	Tyr	Gly	Ser	Thr	Gln	Tyr 185	Ile	Arg	Phe	Ser 190	Pro	Asp	Phe
Thr	Phe 195	Gly	Phe	Glu	Glu	Ser	Leu 200	Glu	Val	Asp	Thr	Asn 205	Pro	Leu	Leu
Gly 210	Ala	Gly	Lys	Phe	Ala	Thr 215	Asp	Pro	Ala	Val	Thr 220	Leu	Ala	His	Glu
Leu 225	Ile	His	Ala	Gly 230	His	Arg	Leu	Tyr	Gly 235	Ile	Ala	Ile	Asn	Pro	Asn 240
Arg	Val	Phe	Lys	Val 245	Asn	Thr	Asn	Ala	Tyr 250	Tyr	Glu	Met	Ser	Gly 255	Leu
Glu	Val	Ser	Phe 260	Glu	Glu	Leu	Arg	Thr 265	Phe	Gly	Gly	His 270	Asp	Ala	Lys
Phe	Ile 275	Asp	Ser	Leu	Gln	Glu	Asn 280	Glu	Phe	Arg	Leu	Tyr 285	Tyr	Tyr	Asn
Lys 290	Phe	Lys	Asp	Ile	Ala	Ser 295	Thr	Leu	Asn	Lys	Ala 300	Lys	Ser	Ile	Val
Gly 305	Thr	Thr	Ala	Ser 310	Leu	Gln	Tyr	Met	Lys 315	Asn	Val	Phe	Lys	Glu	Lys 320
Tyr	Leu	Leu	Ser	Glu 325	Asp	Thr	Ser	Gly	Lys 330	Phe	Ser	Val	Asp	Lys 335	Leu
Lys	Phe 340	Asp	Lys	Leu	Tyr	Lys	Met 345	Leu	Thr	Glu	Ile	Tyr 350	Thr	Glu	Asp
Asn	Phe 355	Val	Lys	Phe	Phe	Lys	Val 360	Leu	Asn	Arg	Lys	Thr 365	Tyr	Leu	Asn
Phe 370	Asp	Lys	Ala	Val	Phe	Lys 375	Ile	Asn	Ile	Val	Pro 380	Lys	Val	Asn	Tyr
Thr 385	Ile	Tyr	Asp	Gly 390	Phe	Asn	Leu	Arg	Asn 395	Thr	Asn	Leu	Ala	Ala	Asn 400
Phe	Asn	Gly	Gln	Asn 405	Thr	Glu	Ile	Asn	Asn 410	Met	Asn	Phe	Thr	Lys 415	Leu
Lys	Asn	Phe	Thr	Gly	Leu	Phe	Glu	Phe	Tyr	Lys	Leu	Leu	Ala	Val	Arg

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420	425	430	
Gly Ile Ile Thr Ser Lys Thr Lys Ser Leu Asp Lys Gly Tyr Asn Lys			
435	440	445	
 <210> SEQ ID NO 32			
<211> LENGTH: 1344			
<212> TYPE: DNA			
<213> ORGANISM: Artificial Sequence			
<220> FEATURE:			
<223> OTHER INFORMATION: BTX-L(C430A)			
 <400> SEQUENCE: 32			
atgcaatttg ttaataaaca atttaattat aaagatccag taaatggtgt cgacattgct		60	
tatatcaaaa ttccaaatgt aggccaaatg caaccagtaa aagcttttaa aattcataat		120	
aaaatctggg ttattccaga acgcgatacc ttaccaatc cggaagaagg tgatctgaat		180	
ccaccaccag aagcaaaaca agttccagtt agctattatg atagcaccta tctgagcacc		240	
gataatgaaa aagataatta tctgaagggc gttaccaaac tgtttgagcg catttatagc		300	
actgatctgg gtcgcatgct gctgaccagc atcgtacgag gtatccattt ttggggtggt		360	
agcaccatcg ataccgaact gaaagttatt gatactaatt gtattaatgt gatccaacca		420	
gatggtagct atcgacgaga agaactgaat ctggtaatca tcggtccgag cgctgatatt		480	
atccagtttg aatgtaaaag ctttggtcat gaagttctga atctgacccg taatggttat		540	
ggcagcacc aatacattcg ctttagacca gattttacct ttggttttga ggagagcctg		600	
gaagttgata ccaatccgct gctgggtgca ggcaaatgtg ctaccgatcc agcagtaacc		660	
ctggcacatg aactgataca tgcctggccat cgctgtatg gtatcgcaat taatccaaat		720	
cgcgttttta aagtaaatac caatgcctat tatgaaatga gcggtctgga agtaagcttt		780	
gaggaactgc gcaccttttg tggtcatgat gcaaagtta tcgatagcct gcaggaaaac		840	
gaatttcgtc tgtattatta taataagttt aaagatatcg caagcacct gaataaagct		900	
aaaagcatcg taggtaccac cgctagcctg cagtatatga aaaatgtttt taaagagaaa		960	
tatctgctgt ctgaagatac ctctggcaaa tttagcgtag ataaactgaa atttgataag		1020	
ctgtacaaaa tgctgaccga gatttacacc gaggataatt ttgttaagtt ttttaaagta		1080	
ctgaaccgca aaacctatct gaattttgat aaagccgtat ttaagatcaa tatcgtaccg		1140	
aaggtaaatt acaccatcta tgatggtttt aatctgcgca ataccaatct ggcagcaaac		1200	
tttaatggtc aaaataccga aattaataat atgaatttta ccaaactgaa aaattttacc		1260	
ggtctgtttg aattctataa gctgctggcg gtacgcggtg tcaccaccag caaaacaaaa		1320	
agcctggata aaggctacaa taag		1344	

<210> SEQ ID NO 33
 <211> LENGTH: 448
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: BTX-L(C430S)

<400> SEQUENCE: 33

Met Gln Phe Val Asn Lys Gln Phe Asn Tyr Lys Asp Pro Val Asn Gly			
1	5	10	15
 Val Asp Ile Ala Tyr Ile Lys Ile Pro Asn Val Gly Gln Met Gln Pro			
	20	25	30
 Val Lys Ala Phe Lys Ile His Asn Lys Ile Trp Val Ile Pro Glu Arg			
	35	40	45

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Asp	Thr	Phe	Thr	Asn	Pro	Glu	Glu	Gly	Asp	Leu	Asn	Pro	Pro	Pro	Glu
50						55					60				
Ala	Lys	Gln	Val	Pro	Val	Ser	Tyr	Tyr	Asp	Ser	Thr	Tyr	Leu	Ser	Thr
65					70					75					80
Asp	Asn	Glu	Lys	Asp	Asn	Tyr	Leu	Lys	Gly	Val	Thr	Lys	Leu	Phe	Glu
			85						90					95	
Arg	Ile	Tyr	Ser	Thr	Asp	Leu	Gly	Arg	Met	Leu	Leu	Thr	Ser	Ile	Val
			100					105						110	
Arg	Gly	Ile	Pro	Phe	Trp	Gly	Gly	Ser	Thr	Ile	Asp	Thr	Glu	Leu	Lys
		115					120					125			
Val	Ile	Asp	Thr	Asn	Cys	Ile	Asn	Val	Ile	Gln	Pro	Asp	Gly	Ser	Tyr
		130				135					140				
Arg	Ser	Glu	Glu	Leu	Asn	Leu	Val	Ile	Ile	Gly	Pro	Ser	Ala	Asp	Ile
145					150					155					160
Ile	Gln	Phe	Glu	Cys	Lys	Ser	Phe	Gly	His	Glu	Val	Leu	Asn	Leu	Thr
				165					170						175
Arg	Asn	Gly	Tyr	Gly	Ser	Thr	Gln	Tyr	Ile	Arg	Phe	Ser	Pro	Asp	Phe
			180					185						190	
Thr	Phe	Gly	Phe	Glu	Glu	Ser	Leu	Glu	Val	Asp	Thr	Asn	Pro	Leu	Leu
		195					200					205			
Gly	Ala	Gly	Lys	Phe	Ala	Thr	Asp	Pro	Ala	Val	Thr	Leu	Ala	His	Glu
	210					215					220				
Leu	Ile	His	Ala	Gly	His	Arg	Leu	Tyr	Gly	Ile	Ala	Ile	Asn	Pro	Asn
225					230					235					240
Arg	Val	Phe	Lys	Val	Asn	Thr	Asn	Ala	Tyr	Tyr	Glu	Met	Ser	Gly	Leu
			245						250					255	
Glu	Val	Ser	Phe	Glu	Glu	Leu	Arg	Thr	Phe	Gly	Gly	His	Asp	Ala	Lys
			260					265					270		
Phe	Ile	Asp	Ser	Leu	Gln	Glu	Asn	Glu	Phe	Arg	Leu	Tyr	Tyr	Tyr	Asn
		275					280					285			
Lys	Phe	Lys	Asp	Ile	Ala	Ser	Thr	Leu	Asn	Lys	Ala	Lys	Ser	Ile	Val
		290				295					300				
Gly	Thr	Thr	Ala	Ser	Leu	Gln	Tyr	Met	Lys	Asn	Val	Phe	Lys	Glu	Lys
305					310					315					320
Tyr	Leu	Leu	Ser	Glu	Asp	Thr	Ser	Gly	Lys	Phe	Ser	Val	Asp	Lys	Leu
			325						330					335	
Lys	Phe	Asp	Lys	Leu	Tyr	Lys	Met	Leu	Thr	Glu	Ile	Tyr	Thr	Glu	Asp
		340						345					350		
Asn	Phe	Val	Lys	Phe	Phe	Lys	Val	Leu	Asn	Arg	Lys	Thr	Tyr	Leu	Asn
		355					360					365			
Phe	Asp	Lys	Ala	Val	Phe	Lys	Ile	Asn	Ile	Val	Pro	Lys	Val	Asn	Tyr
	370					375					380				
Thr	Ile	Tyr	Asp	Gly	Phe	Asn	Leu	Arg	Asn	Thr	Asn	Leu	Ala	Ala	Asn
385					390					395					400
Phe	Asn	Gly	Gln	Asn	Thr	Glu	Ile	Asn	Asn	Met	Asn	Phe	Thr	Lys	Leu
			405						410					415	
Lys	Asn	Phe	Thr	Gly	Leu	Phe	Glu	Phe	Tyr	Lys	Leu	Leu	Ser	Val	Arg
			420					425					430		
Gly	Ile	Ile	Thr	Ser	Lys	Thr	Lys	Ser	Leu	Asp	Lys	Gly	Tyr	Asn	Lys
	435						440					445			

<210> SEQ ID NO 34

<211> LENGTH: 1344

-continued

<212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: BTX-L(C430S)

<400> SEQUENCE: 34

atgcaatttg ttaataaaca atttaattat aaagatccag taaatggtgt cgacattgct	60
tatatcaaaa ttccaaatgt aggccaaatg caaccagtaa aagcttttaa aattcataat	120
aaaatctggg ttattccaga acgcgatacc ttaccaatc cggaagaagg tgatctgaat	180
ccaccaccag aagcaaaaca agttccagtt agctattatg atagcaccta tctgagcacc	240
gataatgaaa aagataatta tctgaagggc gttaccaaac tggttgagcg cattttagc	300
actgatctgg gtcgcatgct gctgaccagc atcgtacgcg gtatcccatt ttggggtggt	360
agcaccatcg atacogaact gaaagttatt gatactaatt gtattaatgt gatccaacca	420
gatggtagct atcgcagcga agaactgaat ctggtaatca tcggtccgag cgctgatatt	480
atccagtttg aatgtaaag ctttggtcat gaagttctga atctgaccg taatggttat	540
ggcagcacc ccaacattcg ctttagccca gattttacct ttggttttga ggagagcctg	600
gaagttgata ccaatccgct gctgggtgca ggcaaatgtg ctaccgatcc agcagtaacc	660
ctggcacatg aactgatata tgctggccat cgcctgtatg gtatcgcaat taatccaaat	720
cgcgttttta aagtaatac caatgcctat tatgaaatga gcggtctgga agtaagcttt	780
gaggaaactgc gcaccttttg tggatcatgat gcaaagtta tcgatagcct gcaggaaaaac	840
gaatttcgct tgtattatta taataagttt aaagatatcg caagcaccct gaataaagct	900
aaaagcatcg taggtaccac cgctagcctg cagtatatga aaaatgtttt taaagagaaa	960
tatctgctgt ctgaagatac ctctggcaaa tttagcgtag ataaactgaa atttgataag	1020
ctgtacaaaa tgctgaccga gattttacacc gaggataatt ttgttaagtt ttttaaagta	1080
ctgaaccgca aaacctatct gaattttgat aaagccgtat ttaagatcaa tatcgtagcg	1140
aaggtaaatt acaccatcta tgatggtttt aatctgcgca ataccaatct ggcagcaaac	1200
tttaaatggtc aaaataccga aattaataat atgaatttta ccaaactgaa aaattttacc	1260
ggtctgtttg aattctataa gctgctgagc gtacgcggta tcataccag caaaacaaaa	1320
agcctggata aaggctacaa taag	1344

<210> SEQ ID NO 35
 <211> LENGTH: 105
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: gene cassette for cloning C430G

<400> SEQUENCE: 35

gaattctata agctgctggg cgtacgcggt atcatcacca gcaaaaccaa aagcctggat	60
aaaggctaca ataagcatca ccatcaccat cactaataac tcgag	105

<210> SEQ ID NO 36
 <211> LENGTH: 105
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: gene cassette for cloning C430A

<400> SEQUENCE: 36

gaattctata agctgctggc ggtacgcggt atcatcacca gcaaaaccaa aagcctggat	60
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aaaggctaca ataagcatca ccatcaccat cactaataac tcgag 105

<210> SEQ ID NO 37
 <211> LENGTH: 105
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: gene cassette for cloning C430S

<400> SEQUENCE: 37

gaattctata agctgctgag cgtacgcggt atcatcacca gcaaaaccaa aagcctggat 60

aaaggctaca ataagcatca ccatcaccat cactaataac tcgag 105

<210> SEQ ID NO 38
 <211> LENGTH: 138
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: gene cassette for cloning C430G-BFGRP

<400> SEQUENCE: 38

gaattctata agctgctggg cgtacgcggt atcatcacca gcaaaaccaa aagcctggat 60

aaaggctaca ataagaccta tcgcagccgc aaatatasca gctggtatca tcaccatcac 120

catcactaat aactcgag 138

<210> SEQ ID NO 39
 <211> LENGTH: 138
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: gene cassette for cloning C430A-BFGRP

<400> SEQUENCE: 39

gaattctata agctgctggc ggtacgcggt atcatcacca gcaaaaccaa aagcctggat 60

aaaggctaca ataagaccta tcgcagccgc aaatatasca gctggtatca tcaccatcac 120

catcactaat aactcgag 138

<210> SEQ ID NO 40
 <211> LENGTH: 138
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: gene cassette for cloning C430S-BFGRP

<400> SEQUENCE: 40

gaattctata agctgctgag cgtacgcggt atcatcacca gcaaaaccaa aagcctggat 60

aaaggctaca ataagaccta tcgcagccgc aaatatasca gctggtatca tcaccatcac 120

catcactaat aactcgag 138

<210> SEQ ID NO 41
 <211> LENGTH: 123
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: gene cassette for cloning belt'

<400> SEQUENCE: 41

gaattctata agctgctgtg tgtacgcggt atcatcacca gcaaaaccaa aagcctggat 60

aaaggctaca ataaggcgct gaacgatctg tgccatcacc atcaccatca ctaataactc 120

gag 123

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<210> SEQ ID NO 42
<211> LENGTH: 453
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: His-SNAP25-ANNEXIN V

<400> SEQUENCE: 42

Met Gly Gly Ser His His His His His His Glu Asn Leu Tyr Phe Gln
1      5      10      15
Gly Ser Gly Gly Asn Lys Leu Lys Ser Ser Asp Ala Tyr Lys Lys Ala
20     25     30
Trp Gly Asn Asn Gln Asp Gly Val Val Ala Ser Gln Pro Ala Arg Val
35     40     45
Val Asp Glu Arg Glu Gln Met Ala Ile Ser Gly Gly Phe Ile Arg Arg
50     55     60
Val Thr Asn Asp Ala Arg Glu Asn Glu Met Asp Glu Asn Leu Glu Gln
65     70     75     80
Val Ser Gly Ile Ile Gly Asn Leu Arg His Met Ala Leu Asp Met Gly
85     90     95
Asn Glu Ile Asp Thr Gln Asn Arg Gln Ile Asp Arg Ile Met Glu Lys
100    105    110
Ala Asp Ser Asn Lys Thr Arg Ile Asp Glu Ala Asn Gln Arg Ala Thr
115    120    125
Lys Met Leu Gly Ser Gly Ala Gln Val Leu Arg Gly Thr Val Thr Asp
130    135    140
Phe Pro Gly Phe Asp Glu Arg Ala Asp Ala Glu Thr Leu Arg Lys Ala
145    150    155    160
Met Lys Gly Leu Gly Thr Asp Glu Glu Ser Ile Leu Thr Leu Leu Thr
165    170    175
Ser Arg Ser Asn Ala Gln Arg Gln Glu Ile Ser Ala Ala Phe Lys Thr
180    185    190
Leu Phe Gly Arg Asp Leu Leu Asp Asp Leu Lys Ser Glu Leu Thr Gly
195    200    205
Lys Phe Glu Lys Leu Ile Val Ala Leu Met Lys Pro Ser Arg Leu Tyr
210    215    220
Asp Ala Tyr Glu Leu Lys His Ala Leu Lys Gly Ala Gly Thr Asn Glu
225    230    235    240
Lys Val Leu Thr Glu Ile Ile Ala Ser Arg Thr Pro Glu Glu Leu Arg
245    250    255
Ala Ile Lys Gln Val Tyr Glu Glu Glu Tyr Gly Ser Ser Leu Glu Asp
260    265    270
Asp Val Val Gly Asp Thr Ser Gly Tyr Tyr Gln Arg Met Leu Val Val
275    280    285
Leu Leu Gln Ala Asn Arg Asp Pro Asp Ala Gly Ile Asp Glu Ala Gln
290    295    300
Val Glu Gln Asp Ala Gln Ala Leu Phe Gln Ala Gly Glu Leu Lys Trp
305    310    315    320
Gly Thr Asp Glu Glu Lys Phe Ile Thr Ile Phe Gly Thr Arg Ser Val
325    330    335
Ser His Leu Arg Lys Val Phe Asp Lys Tyr Met Thr Ile Ser Gly Phe
340    345    350
Gln Ile Glu Glu Thr Ile Asp Arg Glu Thr Ser Gly Asn Leu Glu Gln
355    360    365

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Leu Leu Leu Ala Val Val Lys Ser Ile Arg Ser Ile Pro Ala Tyr Leu
370 375 380

Ala Glu Thr Leu Tyr Tyr Ala Met Lys Gly Ala Gly Thr Asp Asp His
385 390 395 400

Thr Leu Ile Arg Val Met Val Ser Arg Ser Glu Ile Asp Leu Phe Asn
405 410 415

Ile Arg Lys Glu Phe Arg Lys Asn Phe Ala Thr Ser Leu Tyr Ser Met
420 425 430

Ile Lys Gly Asp Thr Ser Gly Asp Tyr Lys Lys Ala Leu Leu Leu Leu
435 440 445

Cys Gly Glu Asp Asp
450

<210> SEQ ID NO 43

<211> LENGTH: 1371

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: His-SNAP25-ANNEXIN V

<400> SEQUENCE: 43

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catatgggcg gcagccatca tcatcatcat catgaaaacc tgtattttca gggctctggc      60
ggcaacaagc tgaatctag cgatgcttac aaaaagcct ggggcaataa tcaggacggc      120
gtggtggcca gccagcctgc tctgttagtg gacgaacgtg agcagatggc catcagcggc      180
ggcttcatcc gtcgtgtaac caatgatgcc cgtgaaaatg aaatggatga aaacctggag      240
cagggtgagcg gcacatcctg caacctgcgt cacatggccc tggatatggg caatgagatc      300
gatacccaga atcgtcagat cgaccgtatc atggagaagg ctgattccaa caaaaccgt      360
atcgatgagg ccaaccaacg tgcaaccaag atgctgggca gcggcgacaca ggttctgcgt      420
ggcactgtga ccgacttccc tggctttgat gagcgtgctg atgcagaaac cctgcgtaag      480
gctatgaaag gcctgggcac cgatgaggag agcatcctga ccctgctgac ctcccgtagc      540
aatgctcagc gtcaggaaat ctctgcagct tttaagacc tgtttggccg tgatctgctg      600
gatgacctga aatccgaact gaccggcaaa ttgaaaaac tgatcgtggc tctgatgaaa      660
ccttctcgtc tgtatgatgc ttatgaactg aaacatgccc tgaaggcgcg tggcaccaat      720
gaaaaagtac tgaccgaaat cattgcttct cgtaccctcg aagaactgcg tgccatcaaa      780
caagtttatg aagaagaata tggctctagc ctggaagatg acgtggtggg cgacacttct      840
ggctactacc agcgtatgct ggtggttctg ctgcaggcta accgtgacct tgatgctggc      900
atcgatgaag ctcaagtga acaagatgct caggctctgt ttcaggctgg cgaactgaaa      960
tggggcaccg atgaagaaaa gtttatcacc atctttggca cccgtagcgt gtctcatctg     1020
agaaagggtg ttgacaagta catgaccatc tctggtttc aaatcgagga aaccatcgac     1080
cgtgagactt ctggcaatct ggagcaactg ctgctggctg ttgtgaaatc tatccgtagc     1140
atccctgcct acctggcaga gacctgtat tatgctatga agggcgctgg caccgatgat     1200
cataccctga tccgtgtcat ggtttcccgt agcgagatcg atctgtttaa catccgtaag     1260
gagtttcgta agaattttgc cacctctctg tattccatga tcaagggcga tacctctggc     1320
gactataaga aagctctgct gctgctgtgt ggcgaagatg actaactcga g              1371

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<210> SEQ ID NO 44

<211> LENGTH: 567

<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: SMF-(-M)BTX-L-PEP1-TEV-His

<400> SEQUENCE: 44

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Met Arg Phe Pro Ser Ile Phe Thr Ala Val Leu Phe Ala Ala Ser Ser
 1           5           10           15

Ala Leu Ala Ala Pro Val Asn Thr Thr Thr Glu Asp Glu Thr Ala Gln
      20           25           30

Ile Pro Ala Glu Ala Val Ile Gly Tyr Ser Asp Leu Glu Gly Asp Phe
      35           40           45

Asp Val Ala Val Leu Pro Phe Ser Asn Ser Thr Asn Asn Gly Leu Leu
      50           55           60

Phe Ile Asn Thr Thr Ile Ala Ser Ile Ala Ala Lys Glu Glu Gly Val
      65           70           75           80

Ser Leu Glu Lys Arg Gln Phe Val Asn Lys Gln Phe Asn Tyr Lys Asp
      85           90           95

Pro Val Asn Gly Val Asp Ile Ala Tyr Ile Lys Ile Pro Asn Val Gly
      100          105          110

Gln Met Gln Pro Val Lys Ala Phe Lys Ile His Asn Lys Ile Trp Val
      115          120          125

Ile Pro Glu Arg Asp Thr Phe Thr Asn Pro Glu Glu Gly Asp Leu Asn
      130          135          140

Pro Pro Pro Glu Ala Lys Gln Val Pro Val Ser Tyr Tyr Asp Ser Thr
      145          150          155          160

Tyr Leu Ser Thr Asp Asn Glu Lys Asp Asn Tyr Leu Lys Gly Val Thr
      165          170          175

Lys Leu Phe Glu Arg Ile Tyr Ser Thr Asp Leu Gly Arg Met Leu Leu
      180          185          190

Thr Ser Ile Val Arg Gly Ile Pro Phe Trp Gly Gly Ser Thr Ile Asp
      195          200          205

Thr Glu Leu Lys Val Ile Asp Thr Asn Cys Ile Asn Val Ile Gln Pro
      210          215          220

Asp Gly Ser Tyr Arg Ser Glu Glu Leu Asn Leu Val Ile Ile Gly Pro
      225          230          235          240

Ser Ala Asp Ile Ile Gln Phe Glu Cys Lys Ser Phe Gly His Glu Val
      245          250          255

Leu Asn Leu Thr Arg Asn Gly Tyr Gly Ser Thr Gln Tyr Ile Arg Phe
      260          265          270

Ser Pro Asp Phe Thr Phe Gly Phe Glu Glu Ser Leu Glu Val Asp Thr
      275          280          285

Asn Pro Leu Leu Gly Ala Gly Lys Phe Ala Thr Asp Pro Ala Val Thr
      290          295          300

Leu Ala His Glu Leu Ile His Ala Gly His Arg Leu Tyr Gly Ile Ala
      305          310          315          320

Ile Asn Pro Asn Arg Val Phe Lys Val Asn Thr Asn Ala Tyr Tyr Glu
      325          330          335

Met Ser Gly Leu Glu Val Ser Phe Glu Glu Leu Arg Thr Phe Gly Gly
      340          345          350

His Asp Ala Lys Phe Ile Asp Ser Leu Gln Glu Asn Glu Phe Arg Leu
      355          360          365

Tyr Tyr Tyr Asn Lys Phe Lys Asp Ile Ala Ser Thr Leu Asn Lys Ala
      370          375          380

Lys Ser Ile Val Gly Thr Thr Ala Ser Leu Gln Tyr Met Lys Asn Val

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385	390	395	400
Phe Lys Glu Lys Tyr Leu Leu Ser Glu Asp Thr Ser Gly Lys Phe Ser			
	405	410	415
Val Asp Lys Leu Lys Phe Asp Lys Leu Tyr Lys Met Leu Thr Glu Ile			
	420	425	430
Tyr Thr Glu Asp Asn Phe Val Lys Phe Phe Lys Val Leu Asn Arg Lys			
	435	440	445
Thr Tyr Leu Asn Phe Asp Lys Ala Val Phe Lys Ile Asn Ile Val Pro			
	450	455	460
Lys Val Asn Tyr Thr Ile Tyr Asp Gly Phe Asn Leu Arg Asn Thr Asn			
	465	470	475
Leu Ala Ala Asn Phe Asn Gly Gln Asn Thr Glu Ile Asn Asn Met Asn			
	485	490	495
Phe Thr Lys Leu Lys Asn Phe Thr Gly Leu Phe Glu Phe Tyr Lys Leu			
	500	505	510
Leu Cys Val Arg Gly Ile Ile Thr Ser Lys Thr Lys Ser Leu Asp Lys			
	515	520	525
Gly Tyr Asn Lys Lys Glu Thr Trp Trp Glu Thr Trp Trp Thr Glu Trp			
	530	535	540
Ser Gln Pro Lys Lys Lys Arg Lys Val Glu Asn Leu Tyr Phe Gln Ser			
	545	550	555
Asn His His His His His His			
	565		

<210> SEQ ID NO 45
 <211> LENGTH: 1704
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: SMF-(-M)BTX-L-PEP1-TEV-His

<400> SEQUENCE: 45

atgagatttc catctatattt tactgcagtt ttgtttgcag catcttctgc attggcagca	60
ccagttaaca ctactactga agatgaaact gcacaaattc cagcagaagc agttattggt	120
tactctgatt tggaagggtga ttttgatggt gctgttttgc cattttctaa ctctactaat	180
aacggtttgt tgtttattaa tactactatt gcatctattg cagcaaagga agaagggtgtt	240
tctttggaaa aaagacagtt tgtaacaag caattcaatt acaaagatcc agttaatggt	300
gttgacattg cttatattaa aattccaaac gttggtcaga tgcagccagt taaagctttc	360
aagatccata acaaaatttg ggttatccca gagagagaca ctttactaa tccagaggaa	420
ggtgatttga acccaccacc agaagcaaaa cagggtccag ttagttatta tgacagtact	480
tatctttcca ctgataacga gaaagataac tatttgaaag gtgttactaa gctgtttgaa	540
agaatttaca gtactgactt gggaagaatg ttgctgactt caattgtag aggaattcca	600
ttttggggtg gatctactat tgacactgaa ttgaaagta tcgatactaa ttgtattaac	660
gttatccaac cagacgggtc ctacagatct gaggagctta acttggttat tattggtcca	720
tccgtgaca ttattcaatt tgagtgttaag tcattcggac atgaggtttt gaacctgact	780
cgtaatggtt atggttccac tcaatatatt agattttccc cagattttac ttttggtttc	840
gaggaaagt tggagggtga tactaacca ttgctgggtg caggtaagtt tgctactgat	900
ccagcagtta ctcttgctca tgagctgac catgcaggtc atagattgta tggatttgca	960
attaatccaa accgtgtttt taaagttaat actaatgctt attatgaaat gtcagggttg	1020

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gaagtttctt tcgaagagtt gagaactttc ggaggacatg atgctaagtt cattgactct 1080
ttgcaggaaa acgagttccg tctgtactac tataacaagt tcaaggacat cgcattctact 1140
ttgaacaagg caaagtcaat tgttggtact actgcttcac tgcaatatat gaagaacgtt 1200
ttcaaggaga agtacttggt gtccgaggat actagtggta aattctctgt tgacaaattg 1260
aagttcgaca aattgtacaa aatgctgact gagatctata ctgaagacaa cttcgttaag 1320
ttttttaaag ttctgaacag aaagacttat ttgaactttg ataaggctgt ttttaagatt 1380
aacattgttc caaagggtta ctatactatt tatgacgggt tcaatctgag aaacactaat 1440
cttcgagcta atttcaatgg tcagaatact gagattaata atatgaactt cactaagttg 1500
aaaaatttta ctggtttggt tgaattttac aagttgtgtg gtgttcgtgg aattattact 1560
tccaagacta aatctttgga taagggttac aacaaaaagg aaacttggtg ggaaacttgg 1620
tggactgaat ggtctcaacc aaagaagaag agaaaggttg agaacctgta cttccaatcc 1680
aatcatcacc atcaccatca ctaa 1704

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<210> SEQ ID NO 46
<211> LENGTH: 85
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: SMF

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<400> SEQUENCE: 46

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Met Arg Phe Pro Ser Ile Phe Thr Ala Val Leu Phe Ala Ala Ser Ser
1           5           10          15
Ala Leu Ala Ala Pro Val Asn Thr Thr Thr Glu Asp Glu Thr Ala Gln
20          25          30
Ile Pro Ala Glu Ala Val Ile Gly Tyr Ser Asp Leu Glu Gly Asp Phe
35          40          45
Asp Val Ala Val Leu Pro Phe Ser Asn Ser Thr Asn Asn Gly Leu Leu
50          55          60
Phe Ile Asn Thr Thr Ile Ala Ser Ile Ala Ala Lys Glu Glu Gly Val
65          70          75          80
Ser Leu Glu Lys Arg
85

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<210> SEQ ID NO 47
<211> LENGTH: 255
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: SMF

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<400> SEQUENCE: 47

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atgagatttc catctatttt tactgcagtt ttgtttgcag catcttctgc attggcagca 60
ccagttaaca ctactactga agatgaaact gcacaaattc cagcagaagc agttattggt 120
tactctgatt tggaagggtga ttttgatggt gctgttttgc cattttctaa ctctactaat 180
aacggtttgt tgtttattaa tactactatt gcattctattg cagcaaagga agaaggtggt 240
tctttgaaaa aaaga 255

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<210> SEQ ID NO 48
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TEV

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<400> SEQUENCE: 48

Glu Asn Leu Tyr Phe Gln Ser Asn
1 5

<210> SEQ ID NO 49

<211> LENGTH: 24

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: TEV

<400> SEQUENCE: 49

gagaacctgt acttccaatc caat

24

<210> SEQ ID NO 50

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Protease Cleavage Site

<400> SEQUENCE: 50

Asp Asp Asp Asp Lys
1 5

<210> SEQ ID NO 51

<211> LENGTH: 4

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Protease Cleavage Site

<400> SEQUENCE: 51

Ile Glu Gly Arg
1

<210> SEQ ID NO 52

<211> LENGTH: 4

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Protease Cleavage Site

<400> SEQUENCE: 52

Ile Asp Gly Arg
1

<210> SEQ ID NO 53

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Protease Cleavage Site

<400> SEQUENCE: 53

Glu Asn Leu Tyr Phe Gln Gly
1 5

<210> SEQ ID NO 54

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Protease Cleavage Site

<400> SEQUENCE: 54

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Leu Val Pro Arg Gly Ser
1 5

<210> SEQ ID NO 55
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Protease Cleavage Site

<400> SEQUENCE: 55

Leu Glu Val Leu Phe Gln Gly Pro
1 5

<210> SEQ ID NO 56
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Cell Penetrating Peptide(Protein-derived Penetration)

<400> SEQUENCE: 56

Arg Gln Ile Lys Ile Trp Phe Gln Asn Arg Arg Met Lys Trp Lys Lys
1 5 10 15

<210> SEQ ID NO 57
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Cell Penetrating Peptide(Tat peptide)

<400> SEQUENCE: 57

Gly Arg Lys Lys Arg Arg Gln Arg Arg Pro Pro Gln
1 5 10

<210> SEQ ID NO 58
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Cell Penetrating Peptide(pVEC)

<400> SEQUENCE: 58

Leu Leu Ile Ile Leu Arg Arg Arg Ile Arg Lys Gln Ala His Ala His
1 5 10 15

<210> SEQ ID NO 59
<211> LENGTH: 27
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Cell Penetrating Peptide(Transportan)

<400> SEQUENCE: 59

Gly Trp Thr Leu Asn Ser Ala Gly Tyr Leu Leu Gly Lys Ile Asn Leu
1 5 10 15

-continued

Lys Ala Leu Ala Ala Leu Ala Lys Lys Ile Leu
20 25

<210> SEQ ID NO 60
 <211> LENGTH: 27
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Cell Penetrating Peptide (MPG)
 <400> SEQUENCE: 60

Gly Ala Leu Phe Leu Gly Phe Leu Gly Ala Ala Gly Ser Thr Met Gly
1 5 10 15

Ala Trp Ser Gln Pro Lys Lys Lys Arg Lys Val
20 25

<210> SEQ ID NO 61
 <211> LENGTH: 21
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Cell Penetrating Peptide (Pep1)
 <400> SEQUENCE: 61

Lys Glu Thr Trp Trp Glu Thr Trp Trp Thr Glu Trp Ser Gln Pro Lys
1 5 10 15

Lys Lys Arg Lys Val
20

<210> SEQ ID NO 62
 <211> LENGTH: 18
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Cell Penetrating Peptide (MAP)
 <400> SEQUENCE: 62

Lys Leu Ala Leu Lys Leu Ala Leu Lys Ala Leu Lys Ala Ala Leu Lys
1 5 10 15

Leu Ala

<210> SEQ ID NO 63
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Cell Penetrating Peptide (R6W3)
 <400> SEQUENCE: 63

Arg Arg Trp Trp Arg Arg Trp Arg Arg
1 5

The invention claimed is:

1. A mutated non-toxic protease in which an amino acid cysteine (Cys) at position 430 of a non-toxic protease represented by the amino acid sequence of SEQ ID NO: 1 is substituted with an amino acid glycine (Gly).

2. A fusion non-toxic protease in which the mutated non-toxic protease of claim 1 is fused with any one or more peptides selected from the group consisting of:

- i) a cell penetrating peptide;
- ii) a belt domain fragment peptide; and
- iii) a cell targeting peptide.

3. The fusion non-toxic proteinase of claim 2, wherein the cell penetrating peptide is represented by the amino acid

55 sequence of SEQ ID NO: 3, the belt domain fragment peptide is represented by the amino acid sequence of SEQ ID NO: 5, and the cell targeting peptide is represented by the amino acid sequence of SEQ ID NO: 7.

60 4. The fusion non-toxic proteinase of claim 2, wherein the any one peptide is fused directly or via a linker to the non-toxic protease.

5. The fusion non-toxic proteinase of claim 4, wherein the linker is a peptide linker represented by (Gly) N wherein N is an integer ranging from 3 to 10.

65 6. The fusion non-toxic proteinase of claim 2, further comprising a cleavable peptide sequence by a protease for

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cleavage between a C-terminus of the non-toxic protease and an N-terminus of the peptide fused with the non-toxic protease.

7. The fusion non-toxic proteinase of claim 6, wherein the cleavable peptide sequence is represented by the amino acid sequence of SEQ ID NO: 27.

8. A mutant gene encoding the mutated non-toxic protease of claim 1.

9. The mutant gene of claim 8, which is represented by a nucleotide sequence of SEQ ID NO: 30.

10. A gene construct comprising:

the mutant gene of claim 9; and

any one or more nucleic acids selected from the group consisting of

i) a nucleic acid encoding a cell penetrating peptide;

ii) a nucleic acid encoding a belt domain fragment peptide; and

iii) a nucleic acid encoding a cell targeting peptide.

11. The gene construct of claim 10, wherein the nucleic acid encoding the cell penetrating peptide is represented by the nucleotide sequence of SEQ ID NO: 4, the nucleic acid encoding the belt domain fragment peptide is represented by the nucleotide sequence of SEQ ID NO: 6, and the nucleic acid encoding the cell targeting peptide is represented by the nucleotide sequence of SEQ ID NO: 8.

12. The gene construct of claim 10, further comprising a nucleic acid encoding a linker peptide between the mutant gene encoding the mutated non-toxic protease and the any one or more nucleic acids.

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13. The gene construct of claim 12, wherein the nucleic acid encoding the linker peptide is represented by the nucleotide sequence of SEQ ID NO: 26.

14. The gene construct of claim 10, further comprising a nucleic acid encoding a peptide sequence cleavable by a cleavage protease between the mutant gene encoding the mutated non-toxic protease and the one or more nucleic acids.

15. A pharmaceutical composition for treating Dystonia containing the non-toxic protease of claim 1 as an active ingredient.

16. The pharmaceutical composition of claim 15, wherein the Dystonia is selected from the group consisting of facial spasm, eyelid spasm, torticollis, blepharospasm, spasmodic torticollis, cervical dystonia, oromandibular dystonia, spasmodic dysphonia, migraine, anal pruritus, and hyperhidrosis.

17. The pharmaceutical composition of claim 16, which is for transdermal administration.

18. A hyaluronic acid microneedle patch containing the non-toxic protease of claim 1.

19. The hyaluronic acid microneedle patch of claim 18, which is for treatment or alleviation of a symptom selected from the group consisting of facial spasm, eyelid spasm, torticollis, blepharospasm, spasmodic torticollis, cervical dystonia, oromandibular dystonia, spasmodic dysphonia, migraine, anal pruritus, and hyperhidrosis.

* * * * *